OYO STATE LABOUR & SKILLS REGISTRY

Improved Updated Baseline Report

Comprehensive pre-fieldwork baseline for the Oyo State Labour & Skills Registry

Mrs Fateemah Roy-Lagbaja

Chemiroy Nigeria Limited

MAY 2026 · V1

Cover image: "A Tailor Sewing Clothes in Her Shop", Meritkosy / Wikimedia Commons (CC BY-SA 4.0)

Document Control

FIELD	VALUE
Document reference	CHM/OSLR/2026/002
Title	Improved Updated Baseline Report, Oyo State Labour & Skills Registry
Version	1.0
Issue date	May 2026
Valid as of	The submission date stated above. Every factual claim in this Report is true as of that date.
Classification	Confidential, for official use only
Subject programme	Oyo State Labour & Skills Registry. Establishment of the State Electronic Labour Register.
Authority	Ministry of Trade, Industry and Cooperatives, Oyo State Government
Consultant	Chemiroy Nigeria Limited
Engagement period	November 2025 – May 2026 (six-month consulting engagement)
Conflicts of interest	None to declare
Companion document	Pre-Field Survey Status Report, CHM/OSLR/2026/001, submitted concurrently

Distribution List

COPY	RECIPIENT	FORMAT
1	Honourable Commissioner, Ministry of Trade, Industry and Cooperatives	Hard copy + Digital
2	Permanent Secretary, Ministry of Trade, Industry and Cooperatives	Hard copy + Digital
3	Director, Trade & Investment, Ministry of Trade, Industry and Cooperatives	Digital
4	SABER Secretariat, Oyo State	Digital, courtesy copy
5	Mrs Fateemah Roy-Lagbaja, Managing Director, Chemiroy Nigeria Limited	Hard copy + Digital
6	File Copy, Chemiroy Nigeria Limited	Hard copy

Confidentiality Notice

This Report is confidential. It is prepared for the Ministry of Trade, Industry and Cooperatives and the recipients listed above, and contains commercially sensitive information about the Oyo State Labour & Skills Registry programme. Reproduction or distribution beyond the named recipients without our prior written consent is not permitted.

The information contained herein has been prepared using data from publicly available sources, including the National Bureau of Statistics (NBS) Nigeria Labour Force Survey, National Population Commission (NPC) population projections, and International Labour Organization (ILO) statistical frameworks, supplemented by primary data collected during the baseline validation exercise and the professional expertise of Chemiroy Nigeria Limited. While every effort has been made to ensure accuracy, certain projections and estimates are based on assumptions that are clearly identified and classified throughout the document using the Assumption Classification Framework described in the Methodology chapter.

Disclaimer

This Improved Updated Baseline Report is the detailed companion to the Pre-Field Survey Status Report (CHM/OSLR/2026/001) and is submitted concurrently. It describes the methodological, technological, and operational foundation of the Registry as of the issue date. The validation exercise findings presented herein are based on a purposive sample (n=330) designed for instrument validation and operational planning, and are not intended for inferential statistical analysis at the population level. Statistically representative estimates of Oyo State's labour force characteristics will require the full-scale statewide enumeration scheduled to commence in the first week of May 2026, and will be reported in the Field Survey Report (CHM/OSLR/2026/003) following fieldwork conclusion.

All recommendations are informed by established international standards, including the ILO Resolution Concerning Statistics of Work, Employment and Labour Underutilization (19th ICLS, 2013), and are subject to the availability of the human and material resources described in the Recommendations chapter of this Report.

Table of Contents

#	TITLE
	Document Control
	Distribution List
	Confidentiality Notice
	Disclaimer
	Abbreviations and Acronyms
1.	**Executive Summary**
2.	**About Chemiroy Nigeria Limited**
3.	**Project Overview and Mandate**
4.	**Situational Analysis: Oyo State Labour Market**
5.	**Comparative Analysis: State-Level Registry Initiatives**
6.	**Baseline Study Methodology**
7.	**Survey Instrument Design and Validation**
8.	**Occupational Skills Taxonomy**
9.	**Data Centre Establishment**
10.	**Registry Platform Development and Capabilities**
11.	**Security Architecture and Data Protection**
12.	**Validation Exercise: Aggregate Findings**
13.	**Validation Exercise: LGA-by-LGA Profiles**
14.	**Data Quality Assurance Framework**
15.	**Capacity Building Programme Design**
16.	**Implementation Roadmap and Stage Gates**
17.	**Risk Register and Mitigation Framework**
18.	**Monitoring and Evaluation Framework: Key Performance Indicators**
19.	**Recommendations and Next Steps**

Abbreviations and Acronyms

TERM	EXPANSION
ASNAT	Association of Nigerian Artisans and Technicians
ASVAN	Association of Skilled and Vocational Artisans of Nigeria
BOT	Build, Operate, Transfer
BVN	Bank Verification Number
CAC	Corporate Affairs Commission
CSP	Content Security Policy
DDoS	Distributed Denial of Service
DKIM	DomainKeys Identified Mail
DMARC	Domain-based Message Authentication, Reporting and Conformance
DPIA	Data Protection Impact Assessment
EA	Enumeration Area
EAD	Enumeration Area Demarcation
GPS	Global Positioning System
ICLS	International Conference of Labour Statisticians
ICT	Information and Communications Technology
ILO	International Labour Organization
ISCO	International Standard Classification of Occupations
ITF	Industrial Training Fund
JWT	JSON Web Token
KPI	Key Performance Indicator
LGA	Local Government Area
MAN	Manufacturing Association of Nigeria
MSME	Micro, Small and Medium Enterprise
MTIC	Ministry of Trade, Industry and Cooperatives
NBS	National Bureau of Statistics
NDPA	Nigeria Data Protection Act, 2023

TERM	EXPANSION
NDPC	Nigeria Data Protection Commission
NIMS	National Identity Management System
NIN	National Identification Number
NLFS	Nigeria Labour Force Survey
NPC	National Population Commission
OSLR / OSLSR	Oyo State Labour & Skills Registry (system shorthand)
OYSIPA	Oyo State Investment Public and Private Partnership Agency
PII	Personally Identifiable Information
PWA	Progressive Web Application
RBAC	Role-Based Access Control
SABER	State Action on Business Enabling Reforms
SMEDAN	Small and Medium Enterprises Development Agency of Nigeria
SMS	Short Message Service
SPF	Sender Policy Framework
SSS / WAEC	Senior Secondary School / West African Examinations Council

1. Executive Summary

1. Background

The Ministry of Trade, Industry, Investment and Cooperatives, Oyo State Government, commissioned Chemiroy Nigeria Limited in November 2025 to undertake the **Production of the Oyo State Skilled Labour Register**, a comprehensive state-wide digital registry of skilled, semi-skilled, and unskilled workers across all thirty-three (33) Local Government Areas of Oyo State.

The engagement, with a duration of six (6) months, encompasses three (3) principal deliverables: (1) Establishment of a Data Centre; (2) Creation of the State Labour Register; and (3) Training of Ministry/Government Officials on the Operations of the Registry.

This Improved Updated Baseline Report presents the findings and outputs of the pre-fieldwork phase of the engagement (November 2025 – May 2026), covering the establishment of the digital and physical infrastructure, the design and validation of the data collection methodology, and the development of a comprehensive occupational skills taxonomy for Oyo State. Statewide field enumeration commences in the first week of May 2026.

The Letter of Introduction authorising the Consultant to engage development partners and counterpart agencies on behalf of the Ministry was issued at the end of February 2026. The Build Phase of the engagement, including the field-survey activity, is scheduled to conclude within the contractual period.

The survey instrument has been refined through two pre-field enumeration cycles conducted as instrument validation, and is now at Version 3. Stakeholder sensitisation across the three Senatorial Districts of Oyo State (Oyo Central, Oyo South, and Oyo North) was conducted at the start of the engagement, establishing programme awareness across the 33-LGA catchment.

2. Scope of Work Completed

The following activities were completed during the reporting period:

Inception & Situational Analysis

- Stakeholder identification and engagement with Ministry officials
- Comprehensive desk review of labour market data sources (NBS, NPC, ILO, SMEDAN)
- Comparative analysis of state-level labour registry initiatives (Lagos, Kaduna, Edo)
- Baseline survey methodology design aligned with ILO ICLS-19 standards

Infrastructure & Instrument Development

- Procurement and configuration of on-premises Data Centre at Ministry premises
- Deployment of cloud infrastructure for production application hosting
- Survey instrument design through three iterative review cycles
- Occupational skills taxonomy development (150 skills, 20 sectors, ISCO-08 mapped)
- Data quality assurance framework design incorporating fraud detection heuristics

Platform Development & Validation

- Full development of the Oyo State Labour & Skills Registry (OSLSR) digital platform
- 3,564 automated quality assurance tests executed with zero critical failures
- OWASP Top 10 security compliance assessment
- Nigeria Data Protection Act 2023 (NDPA) compliance review

Baseline Validation & Reporting

- Structured validation exercise across all 33 LGAs (n=330 respondents)
- Instrument performance analysis and findings documentation
- Capacity building curriculum design for five operational user roles plus a Public tier
- Preparation of this Baseline Study Report

---

3. Key Findings

3.1 Labour Market Situational Analysis

Oyo State, with an estimated population of 8.6 million (NPC 2025 projection), has no centralised, digitised register of its workforce. The state's labour market mirrors national patterns: **93% informal employment** (NBS NLFS Q2 2024), **75% self-employment** rate, and significant skills data gaps that impede evidence-based policy formulation.

This data gap has particular significance in the context of Nigeria's subnational business enabling environment reform programme, which assesses state competitiveness across sixteen indicators, including **skilled labour readiness**. Oyo State has risen from 27th to **3rd nationally** (62.7%) in the 2025 assessment. The Skilled Labour Register provides the evidence base to sustain and advance this competitive position by making the state's workforce visible, searchable, and verifiable for the first time.

A desk review of existing data sources revealed that while the National Bureau of Statistics (NBS) provides quarterly national and zonal estimates through the Nigeria Labour Force Survey (NLFS), **no state-level register exists** that captures granular, individual-level data on occupational skills, employment status, and business ownership across all 33 LGAs. Comparable initiatives in Lagos (LSETF Jobs Portal) and Kaduna (KADSTEP) have focused on specific populations (youth, graduates) rather than comprehensive workforce enumeration.

3.2 Validation Exercise Results

A structured validation exercise was conducted to verify instrument functionality and establish preliminary demographic baselines. Key findings from the purposive sample (n=330, 10 respondents per LGA):

Table 1.1: Validation Exercise Results

INDICATOR	VALIDATION SAMPLE	NBS BENCHMARK	ALIGNMENT
Gender (Male/Female)	52.1% / 47.9%	51.5% / 48.5%	✓ Within 1%
Employment Rate	83.0%	87.3%	✓ Within 5%
Self-Employment Rate	74.5%	75.0%	✓ Within 1%
Informal Employment	91.6%	93.0%	✓ Within 2%
Business Ownership	38.5%	~35% (SMEDAN)	✓ Within 4%

INDICATOR	VALIDATION SAMPLE	NBS BENCHMARK	ALIGNMENT
CAC Registration	22.8%	~20% (SMEDAN)	✓ Within 3%
Marketplace Consent	68.2%	N/A (novel)	Baseline established
Avg. Completion Time	9.2 minutes	Target: ≤10 min	✓ Met

The validation sample was designed to approximate the demographic distribution documented in the NBS NLFS (Q2 2024). Findings are presented for instrument validation and operational planning purposes; statistically representative estimates of Oyo State's labour force will require the full-scale statewide enumeration.

3.3 Occupational Skills Taxonomy

A bespoke occupational skills taxonomy was developed comprising **150 skills across 20 economic sectors**, mapped to the International Labour Organization's International Standard Classification of Occupations (ISCO-08). The taxonomy captures the full breadth of Oyo State's labour force, from formal sector professionals to artisanal tradespeople, agro-processors, and survivalist occupations, reflecting the 93% informal employment reality documented by NBS.

3.4 Data Centre Establishment

A hybrid Data Centre architecture was established, comprising:

- **On-premises Operations Node** at Ministry premises: three (3) enterprise-grade HP workstations (Intel Core i7, 16GB RAM, 512GB storage) with dedicated 5G broadband connectivity (Airtel SmartConnect ODU, 50 Mbps unlimited)
- **Cloud Infrastructure**: secure production server with automated daily backups, 7-year NDPA-aligned data retention, and 99.9% uptime SLA

3.5 Registry Platform

The Oyo State Labour & Skills Registry (OSLSR) digital platform was developed and deployed, featuring:

- Multi-channel data collection (field mobile, desktop data entry, public web self-registration)
- Offline-capable Progressive Web Application (PWA) with 7-day data retention
- Context-Aware Fraud Detection Engine with configurable thresholds
- Role-based access control for five operational user roles plus a Public tier
- Public Skills Marketplace with anonymised worker profiles
- Immutable audit trails with NDPA-aligned 7-year retention
- System health monitoring with real-time alerting

The platform underwent **3,564 automated quality assurance tests** with zero critical failures and a comprehensive **OWASP Top 10 security assessment** with the platform independently assessed at A- security posture (state-government-grade) across all OWASP Top 10 categories.

---

4. Deliverable Status Summary

Table 1.2: Deliverable Status Summary

#	DELIVERABLE	STATUS	EVIDENCE
1	Establishment of Data Centre	**COMPLETE**	Physical infrastructure operational at Ministry premises; cloud infrastructure deployed and monitored
2	Creation of the State Labour Register	**IN PROGRESS**	Platform deployed and validated; data collection methodology designed; awaiting mobilisation of 132 field personnel for statewide enumeration
3	Training of Ministry/Government Officials	**DESIGN COMPLETE**	Capacity building curriculum designed for the five operational roles plus the Public tier; classroom delivery scheduled for implementation phase

5. Recommendations

For the successful execution of the statewide field enumeration (Deliverable 2), the Consultant recommends the immediate mobilisation of the following resources:

(a) Recruitment and training of **One Hundred and Thirty-Two (132) field personnel**, comprising Three (3) Enumerators and One (1) Field Supervisor per Local Government Area across all Thirty-Three (33) LGAs of Oyo State;

(b) Procurement of mobile data collection devices (Android smartphones, minimum specification: Android 8.0, 2GB RAM, 32GB storage, GPS-enabled) for all field personnel;

© Field logistics support including transportation allowances, identification materials (branded vests, ID cards), and communication airtime for data synchronisation;

(d) Printing and distribution of sensitisation materials for community engagement and respondent awareness;

(e) Establishment of LGA-level coordination with traditional rulers, community leaders, and ward heads to facilitate field access and respondent cooperation.

The digital infrastructure, validated survey instrument, 150-skill occupational taxonomy, data quality assurance framework, and operational Data Centre are **deployment-ready**. The commencement of field enumeration is contingent upon the mobilisation of the above human and material resources as specified in the project terms of reference.

6. Conclusion

Chemiroy Nigeria Limited has, within the first four (4) months of the engagement, completed Deliverable 1 (Data Centre Establishment) in full, designed and validated the methodology and instruments for Deliverable 2 (State Labour Register), and completed the curriculum design for Deliverable 3 (Training). The remaining activities, field enumeration, data analysis, and training delivery, are achievable within the remaining engagement period, subject to the timely mobilisation of field personnel and logistics resources.

This Baseline Study Report provides the evidentiary foundation for the next phase of the engagement and demonstrates the Consultant's commitment to international best practice, methodological rigour, and the successful delivery of the Oyo State Skilled Labour Register.

Chemiroy Nigeria Limited May 2026

2. About Chemiroy Nigeria Limited

2.1 Corporate Profile

Chemiroy Nigeria Limited is a Nigerian consulting and project management firm specialising in technology-driven solutions for government and private sector development programmes. The company is registered with the Corporate Affairs Commission (CAC) of the Federal Republic of Nigeria and maintains its operational headquarters in Oyo State.

Chemiroy Nigeria Limited brings to the Oyo State Skilled Labour Register project a multidisciplinary team with expertise spanning digital platform development, data management systems, agricultural consulting, feasibility studies, and public sector programme design. The firm's approach integrates international best practices with deep understanding of the Nigerian operating environment, particularly the unique challenges of data collection in developing economies, including infrastructure constraints, digital literacy variations, and the predominance of informal economic activity.

2.2 Leadership

Mrs Fateemah Roy-Lagbaja, Managing Director

Mrs Roy-Lagbaja provides strategic leadership, project accountability, and client liaison for the Oyo State Labour & Skills Registry engagement. Under her direction, the firm has expanded its portfolio to encompass government consulting engagements requiring the intersection of technology, policy, and operational delivery.

The Managing Director carries direct responsibility for project strategy, technical architecture oversight, methodology validation, and quality assurance, supported by the firm's multidisciplinary delivery team across digital platform development, data systems architecture, and the application of international statistical standards (ILO, NBS) to Nigerian development contexts.

2.3 Core Competencies

Table 2.1: Core Competencies

COMPETENCY AREA	DESCRIPTION
Digital Platform Development	Design and deployment of secure, scalable web applications for government and enterprise use, including mobile-responsive interfaces, offline-capable Progressive Web Applications (PWA), and database-driven registries
Data Systems Architecture	Database design, data governance frameworks, data quality assurance systems, and compliance with national data protection regulations (NDPA 2023)
Feasibility Studies & Research	Comprehensive feasibility analyses, market research, baseline surveys, and investment proposals for agricultural, industrial, and public sector projects

COMPETENCY AREA	DESCRIPTION
Agricultural Consulting	Agribusiness development, crop production planning, value chain analysis, and agricultural project management
Public Sector Programme Design	Design and implementation of government-commissioned programmes, including stakeholder engagement, capacity building, and monitoring and evaluation frameworks
Project Management	End-to-end project lifecycle management using international standards, including risk management, quality control, and reporting

2.4 Relevant Experience

Chemiroy Nigeria Limited has executed consulting engagements across multiple sectors, demonstrating the firm's capacity to deliver complex, multi-stakeholder projects in the Nigerian operating environment.

Table 2.2: Relevant Experience

#	PROJECT	CLIENT	SECTOR	SCOPE
1	Comprehensive Feasibility Study, 22-Hectare Oil Palm Plantation Development	Private Investor	Agriculture / Agribusiness	Full feasibility study comprising 18 chapters covering industry analysis, site assessment, agronomic blueprint, financial modelling, implementation roadmap, risk register, ESG framework, and monitoring KPIs
2	Oyo State Skilled Labour Register, Baseline Study	Ministry of Trade, Industry, Investment and Cooperatives, Oyo State	Government / ICT	Current engagement, design and deployment of a state-wide digital labour registry with offline-capable data collection, fraud detection, and public skills marketplace
3	Agricultural Project Consulting Portfolio	Multiple Clients	Agriculture	Crop comparison analyses, economic viability assessments, and value chain development strategies for oil palm, cassava, maize, soybean, and sugarcane

2.5 Quality Commitment

Chemiroy Nigeria Limited is committed to delivering work products that meet or exceed international standards of professional quality. The firm's quality management approach includes:

- **Iterative Review Process**: All deliverables undergo a minimum of three review cycles before submission, incorporating internal peer review, technical validation, and management approval
- **Evidence-Based Methodology**: All findings and recommendations are grounded in verifiable data sources, with explicit assumption classification (Verified Fact / Field-Dependent / Working Assumption)
- **International Standards Alignment**: Work products align with relevant international frameworks including ILO statistical standards, ISCO-08 occupational classification, OWASP security guidelines, and NDPA data protection requirements
- **Automated Quality Assurance**: Technology deliverables are subject to comprehensive automated testing, the OSLSR platform, for example, underwent 3,564 automated quality checks

- **Continuous Improvement**: Lessons learned from each engagement are documented and incorporated into subsequent project methodologies

---

Chemiroy Nigeria Limited is available to provide additional references, detailed CVs of team members, and evidence of past project deliverables upon request by the commissioning authority.

3. Project Overview and Mandate

3.1 Engagement Background

By Award Letter referenced [Award Letter Reference], the Ministry of Trade, Industry, Investment and Cooperatives, Oyo State Government, engaged **Chemiroy Nigeria Limited** to undertake the **Production of the Oyo State Skilled Labour Register**. The consultancy award was received on **18th November 2025**, with a project duration of **six (6) months** concluding in **May 2026**.

The engagement was commissioned in recognition of a fundamental gap in the state's economic planning infrastructure: **the absence of a comprehensive, digitised register of Oyo State's workforce**. Despite an estimated population of 8.6 million (NPC 2025 projection) and a labour force participation rate exceeding 70% (NBS NLFS 2024), no centralised system existed to capture, verify, and maintain granular data on the skills, employment status, and economic activity of workers across the state's 33 Local Government Areas.

This data deficit constrains evidence-based policy formulation, limits the state's ability to target skills development programmes, and impedes the matching of available workforce skills with employer demand and investment opportunities.

3.2 Terminology: Register and Registry

A note on terminology used throughout this report. The two terms serve distinct but complementary functions:

- **Register** (noun), The official authoritative record: the list of workers, their skills, demographics, and employment data. This is what the Ministry commissioned, the *Production of the Oyo State Skilled Labour **Register***. It is analogous to an electoral register, a birth register, or a land register: the definitive dataset.
- **Registry** (noun), The system, platform, and institutional framework that creates, maintains, and serves the register. The **Oyo State Labour & Skills Registry (OSLSR)** is the digital platform, encompassing the web application, mobile PWA, database, fraud detection engine, and public Skills Marketplace, through which the register is produced and operated.

In summary: the **Registry** (system) produces and maintains the **Register** (data). Both terms are used precisely throughout this report. Where the text refers to the official list of workers, it uses "register"; where it refers to the technology platform, it uses "registry."

3.3 Project Objectives

The overarching objective of the engagement is to establish a **trusted, continuously updated digital register** of skilled, semi-skilled, and unskilled workers across all 33 LGAs of Oyo State. The registry platform is designed to:

1. **Provide granular workforce data**, Individual-level records capturing demographics, skills, employment status, business ownership, and location across all 33 LGAs

2. **Enable evidence-based policy**, Real-time dashboards and analytical tools to inform government decisions on skills development, job creation, trade facilitation, and investment attraction
3. **Connect workers to opportunities**, A public Skills Marketplace enabling employers, contractors, and investors to discover verified skilled workers by trade, location, and experience level
4. **Ensure data integrity**, Automated fraud detection, identity verification (NIN validation), and immutable audit trails to guarantee the trustworthiness of registry data
5. **Support continuous updating**, An always-on platform allowing ongoing self-registration, profile enrichment, and data maintenance beyond the initial field enumeration period

---

3.4 Project Deliverables

The engagement Terms of Reference specify three (3) principal deliverables:

Table 3.1: Project Deliverables

#	DELIVERABLE	DESCRIPTION	TIMELINE
1	**Establishment of Data Centre**	Procurement, configuration, and deployment of physical and digital infrastructure to host, process, and secure the Labour Register data	Months 1–3
2	**Creation of the State Labour Register**	Design and deployment of the digital registry platform; development of data collection instruments; conduct of statewide field enumeration across all 33 LGAs; population of the registry with verified worker data	Months 2–5
3	**Training of Ministry/Government Officials**	Design and delivery of capacity building programme to equip Ministry and government officials with the knowledge and skills to operate, maintain, and utilise the Labour Register system	Months 5–6

3.5 Project Structure and Governance

3.5.1 Consultant Team

Table 3.2: Consultant Team

ROLE	NAME	RESPONSIBILITY
Managing Director	Mrs Fateemah Roy-Lagbaja	Project leadership, technical accountability, corporate governance, strategic oversight, and client liaison

3.5.2 Ministry Oversight

The project operates under the supervision of the Ministry of Trade, Industry, Investment and Cooperatives, Oyo State Government, with designated Ministry officials serving as project supervisors responsible for:

- Facilitating access to Ministry premises for Data Centre installation
- Coordinating with relevant state agencies (e.g., National Population Commission state office, State Internal Revenue Service)

- Providing administrative support for field enumeration logistics
- Reviewing and accepting project deliverables

3.5.3 Reporting Framework

Table 3.3: Reporting Framework

REPORT TYPE	FREQUENCY	RECIPIENT
Baseline Study Report (this document)	Mid-term	Ministry of Trade, Industry, Investment and Cooperatives
Progress Updates	As required	Designated Ministry Supervisors
Completion Report	End of engagement	Ministry of Trade, Industry, Investment and Cooperatives

3.6 Activity Schedule

The following schedule documents the activities undertaken during the first sixteen (16) weeks of the engagement, organised by phase:

Inception & Situational Analysis

Table 3.4: Activity Schedule

WEEK	ACTIVITIES
1–2	Project inception and administrative setup; Receipt and review of engagement terms; Establishment of project management framework; Stakeholder identification and engagement planning
2–3	Initiation of stakeholder consultations with Ministry officials; Identification of data requirements and functional needs; Review of existing Ministry data holdings and administrative records
3–4	Commencement of desk review: National Bureau of Statistics (NBS) Labour Force Survey publications; National Population Commission (NPC) population estimates and projections; ILO statistical standards and ISCO-08 occupational classification framework; SMEDAN/NBS National MSME Survey reports
4–5	Comparative analysis of state-level labour registry initiatives (Lagos LSETF, Kaduna KADSTEP, Edo EdoJobs); Survey methodology design, ILO ICLS-19 alignment; Initial questionnaire framework development; Data Centre requirements specification and vendor evaluation

Infrastructure & Instrument Development

Table 3.5: Activity Schedule

WEEK	ACTIVITIES
5–6	Data Centre hardware procurement (3× HP Core i7 workstations, Airtel 5G ODU router); Physical installation and configuration at Ministry premises; Cloud infrastructure provisioning and security baseline
6–7	Survey instrument Version 1.0 development; ILO ICLS-19 labour force classification module design; Skills taxonomy research, NABTEB trade areas, SMEDAN MSME sectors, ILO ISCO-08
7–8	Survey instrument Version 2.0, internal peer review; Skip logic and validation rule refinement; Choice list expansion (education levels, employment types, housing status)

WEEK	ACTIVITIES
8–9	Survey instrument Version 3.0, final validation; Occupational skills taxonomy completion (150 skills, 20 sectors); Data quality assurance framework design, fraud detection heuristics
9–10	Database schema design and implementation; Registry platform architecture design; Data Protection Impact Assessment (NDPA 2023)

Platform Development & Validation

Table 3.6: Activity Schedule

WEEK	ACTIVITIES
10–11	Registry platform core module development, authentication, role-based access control, staff management; Database migration and schema deployment
11–12	Survey form renderer development, one-question-per-screen interface, offline capability (Progressive Web Application), GPS capture; Data submission pipeline with idempotent processing
12–13	Fraud detection engine development, GPS cluster analysis, submission speed monitoring, response pattern analysis; Role-specific dashboards (five operational + Public tier); Public Skills Marketplace development
13–14	Automated quality assurance testing (3,564 tests executed); OWASP Top 10 security compliance assessment; System hardening, Content Security Policy, dependency audit, input validation

Baseline Validation & Reporting

Table 3.7: Activity Schedule

WEEK	ACTIVITIES
14–15	Structured validation exercise execution, 10 respondents per LGA across all 33 LGAs (n=330); Instrument performance data collection, completion times, error rates, device compatibility
15–16	Validation data analysis and findings documentation; Capacity building curriculum design (five operational roles + Public tier, training module development); Field enumeration methodology finalisation; Preparation and compilation of this Baseline Study Report

---

3.7 Convergence of Factors, Why This Project, Why Now

The establishment of a State Skilled Labour Register is not merely a data collection exercise; it responds to a convergence of six structural factors that make this intervention both timely and necessary:

1. **Federal Policy Alignment**: The Federal Government's ongoing National Identity Management System (NIMS) expansion and NIN mandate creates an enabling environment for state-level registries that leverage verified national identity data. The OSLSR's NIN validation at point of entry (Modulus 11 checksum) directly interfaces with this national infrastructure.

2. **Youth Unemployment Imperative**: Nigeria's youth unemployment rate, while declining from 8.4% (Q1 2024) to 6.5% (Q2 2024) nationally (NBS NLFS Q2 2024), masks significant underemployment and skills mismatches. Oyo State, with an estimated 2.4 million persons aged 15–34, requires granular skills data to design targeted interventions.

3. **Informal Sector Dominance**: With 93% of Nigeria's workforce in informal employment (NBS 2024), traditional employer-based labour registries capture less than 7% of the actual workforce. The OSLSR's design, encompassing artisans, traders, farmers, and survivalist occupations, addresses this methodological gap.

4. **Digital Technology Maturity**: The availability of affordable mobile devices, expanding 4G/5G coverage (including Airtel's SmartConnect 5G ODU targeting underserved areas), and increasing digital literacy enable mobile-based data collection at a scale and cost that was infeasible even five years ago.

5. **Investment Attraction**: The absence of verified, searchable workforce data is a documented barrier to industrial and commercial investment. The Skills Marketplace component transforms the register from a cost centre into an active tool for connecting skilled workers with opportunities, contributing directly to economic development.

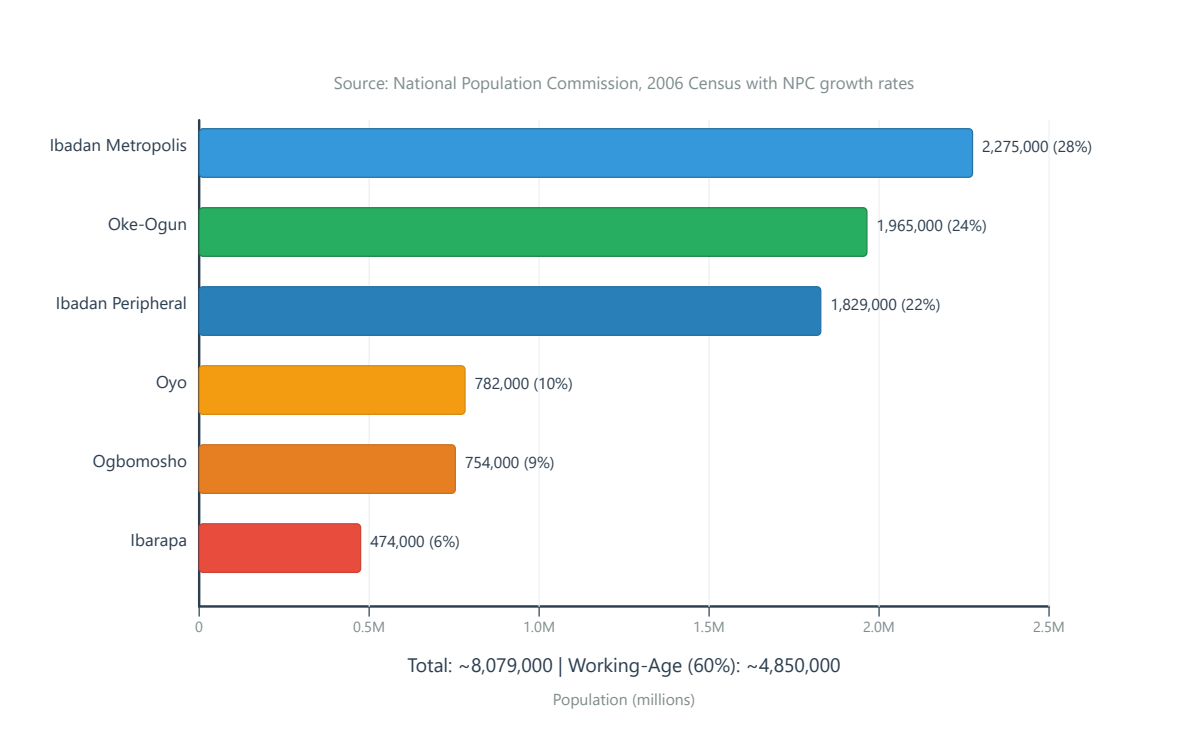
6. **Business Enabling Environment Reform**: Nigeria's subnational business enabling environment reform programme, a World Bank-supported initiative assessing states across sixteen (16) indicators and thirty-six (36) sub-metrics, includes **skilled labour readiness** as a measured dimension of state competitiveness. Oyo State has demonstrated significant progress under this framework, rising from 27th to 3rd nationally (62.7% in the 2025 assessment). The Oyo State Skilled Labour Register directly strengthens the state's performance on this indicator by providing a verified, searchable database of available workforce skills, a capability that no other state currently possesses at this level of depth. The register positions Oyo State to sustain and improve its competitive standing in subsequent assessment cycles.

These factors are not the product of a single policy decision. They represent a structural convergence that creates a window of opportunity for Oyo State to consolidate its rising position in national business environment competitiveness rankings and establish a first-mover advantage in comprehensive workforce data management among Nigeria's 36 states.

---

4. Situational Analysis: Oyo State Labour Market

Figure 4.1: Oyo State Population Distribution by Senatorial District



4.1 Introduction

This chapter presents a comprehensive situational analysis of the Oyo State labour market, drawing on national survey data from the **National Bureau of Statistics (NBS)**, state-level economic indicators, and the broader macroeconomic context within which the Skilled Labour Register operates. The analysis provides the empirical foundation upon which the registry methodology, survey instrument, and operational assumptions are built.

4.2 Oyo State, Geographic and Administrative Profile

4.2.1 State Overview

Table 4.1: State Overview

PARAMETER	DETAIL
Location	Southwestern Nigeria; borders Kwara (north), Osun (east), Ogun (south), Republic of Benin (west)
State Capital	Ibadan, largest city in West Africa by geographic area
Land Area	Approximately 28,454 km²
Population (2006 Census)	5,591,589
Population (2024 Projection)	~8.5–9.2 million (NPC 2.8–3.0% growth rate)
Number of LGAs	33
Geopolitical Zone	South-West
Major Languages	Yoruba (primary), English (official), Hausa (minority)
Major Cities	Ibadan, Ogbomosho, Oyo, Iseyin, Saki
Economic Base	Agriculture, commerce, manufacturing, services

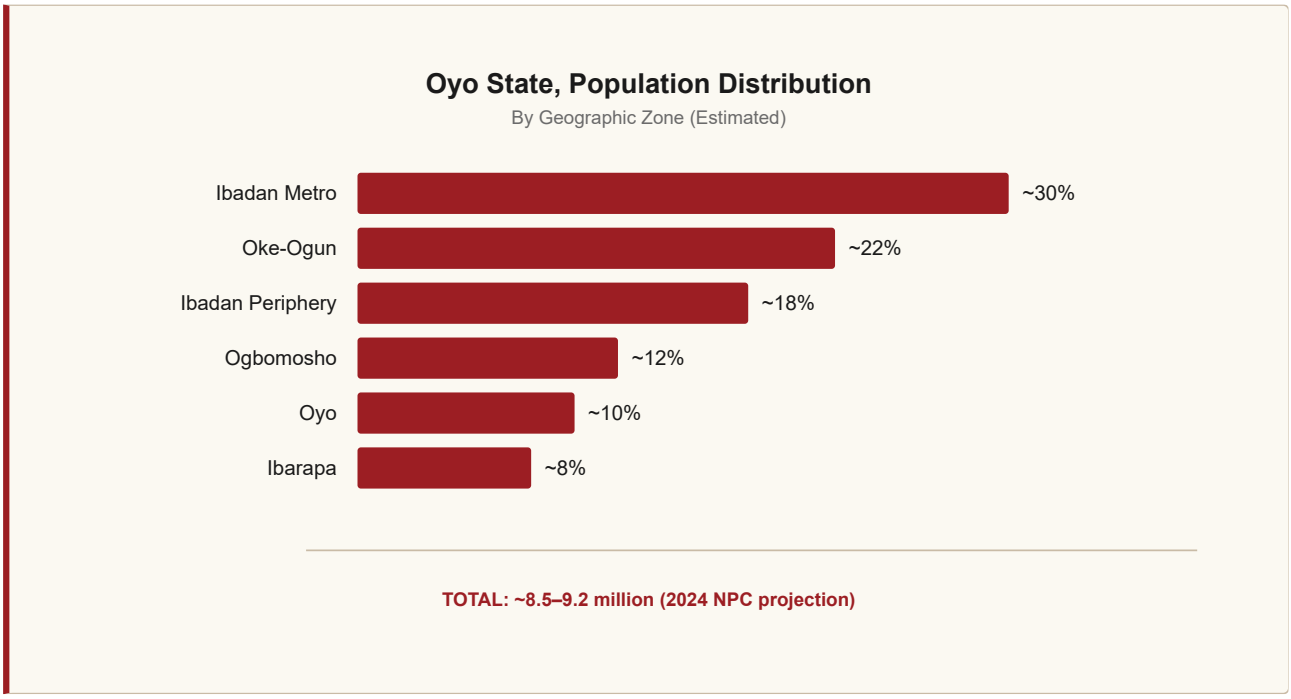
4.2.2 Geographic Zones

Oyo State's 33 LGAs are organised into six zones for administrative and analytical purposes:

Table 4.2: Geographic Zones

ZONE	LGAS	CHARACTER	POPULATION SHARE (EST.)
Ibadan Metropolis	5	Urban; state capital; commercial and administrative centre	~30%
Ibadan Peripheral	6	Peri-urban; mixed agriculture and emerging residential development	~18%
Ogbomosho	4	Semi-urban; artisanal hub; education (LAUTECH)	~12%
Oyo	4	Historical urban; mixed economy; cultural significance	~10%
Oke-Ogun	11	Rural; agriculture-dominant; cross-border trade with Benin Republic	~22%
Ibarapa	3	Rural; farming; emerging agribusiness corridor	~8%

Figure 4.2: Oyo State Population Distribution by Geographic Zone



4.3 Population Demographics

4.3.1 Population Projections

The 2006 National Population Census (the most recent completed census in Nigeria) recorded the following LGA-level populations. The 2024 projections apply the NPC annual growth rates (2.8–3.0% for urban areas, 2.3–2.5% for rural areas):

Table 4.3: Population Projections

ZONE	LGA	2006 CENSUS	2024 PROJECTION (EST.)
Ibadan Metro	Ibadan North	306,763	~502,000
	Ibadan North-East	330,399	~541,000
	Ibadan North-West	152,834	~250,000
	Ibadan South-East	266,046	~435,000
	Ibadan South-West	282,585	~462,000
Subtotal		*1,338,627*	*~2,190,000*
Ibadan Periph.	Akinyele	211,811	~343,000
	Egbeda	281,573	~456,000
	Ido	103,261	~160,000
	Lagelu	147,957	~240,000

ZONE	LGA	2006 CENSUS	2024 PROJECTION (EST.)
	Oluyole	203,461	~333,000
	Ona Ara	202,725	~328,000
Subtotal		*1,150,788*	*~1,860,000*
Ogbomosho	Ogbomosho North	197,977	~316,000
	Ogbomosho South	100,815	~161,000
	Ogo Oluwa	56,536	~83,000
	Surulere	126,434	~186,000
Subtotal		*481,762*	*~746,000*
Oyo	Afijio	134,173	~207,000
	Atiba	150,272	~232,000
	Oyo East	124,095	~191,000
	Oyo West	107,584	~166,000
Subtotal		*516,124*	*~796,000*
Oke-Ogun	Atisbo	79,854	~117,000
	Irepo	90,439	~133,000
	Iseyin	235,142	~365,000
	Itesiwaju	103,435	~152,000
	Iwajowa	92,298	~136,000
	Kajola	193,977	~285,000
	Olorunsogo	57,450	~84,000
	Orelope	55,602	~82,000
	Ori Ire	148,431	~218,000
	Saki East	89,811	~132,000
	Saki West	178,677	~277,000
Subtotal		*1,325,116*	*~1,981,000*
Ibarapa	Ibarapa Central	102,979	~154,000
	Ibarapa East	118,226	~177,000
	Ibarapa North	101,092	~151,000

ZONE	LGA	2006 CENSUS	2024 PROJECTION (EST.)
Subtotal		*322,297*	*~482,000*
STATE TOTAL		**5,591,589**	**~8,055,000**

Sources: NPC 2006 Census; 2024 projections based on NPC state growth rates (2.8% urban, 2.3% rural). Figures rounded to nearest thousand.

4.3.2 Working-Age Population

Table 4.4: Working-Age Population

PARAMETER	VALUE	SOURCE
Working-age population (15–64)	~60% of total	NBS NLFS 2024
Estimated working-age in Oyo	~4.8–5.5 million	Calculated
Labour force participation rate	77.3% (national)	NBS NLFS Q2 2024
Estimated labour force in Oyo	~3.7–4.3 million	Calculated
Youth (15–35) as % of working-age	~45%	NBS estimate

Implication: The estimated 3.7–4.3 million labour force participants in Oyo State represent the **target population** for the Skilled Labour Register. The initial field enumeration (targeting approximately 23,760 registrants) would capture approximately 0.5–0.6% of this population, a substantive initial registry that serves as a foundation for continuous expansion through ongoing self-registration.

4.4 Labour Market Structure

4.4.1 National Labour Market Indicators

The following national-level indicators from the NBS National Labour Force Survey (NLFS) provide the benchmark context for interpreting Oyo State labour market conditions:

Table 4.5: National Labour Market Indicators

INDICATOR	VALUE	PERIOD	SOURCE
Labour force participation rate	77.3%	Q2 2024	NBS NLFS
Employment-to-population ratio	73.4%	Q2 2024	NBS NLFS
Unemployment rate	5.3%	Q2 2024	NBS NLFS
Underemployment rate (time-related)	12.2%	Q2 2024	NBS NLFS
Combined unemployment + underemployment	17.5%	Q2 2024	NBS NLFS
Informal employment (% of total)	92.6%	Q2 2024	NBS NLFS

INDICATOR	VALUE	PERIOD	SOURCE
Self-employment (% of employed)	75.2%	Q2 2024	NBS NLFS
Youth unemployment (15–24)	8.4%	Q2 2024	NBS NLFS
Female labour force participation	73.1%	Q2 2024	NBS NLFS

4.4.2 Southwest Regional Context

The Southwest geopolitical zone (comprising Lagos, Oyo, Ogun, Osun, Ondo, and Ekiti states) exhibits distinctive labour market characteristics:

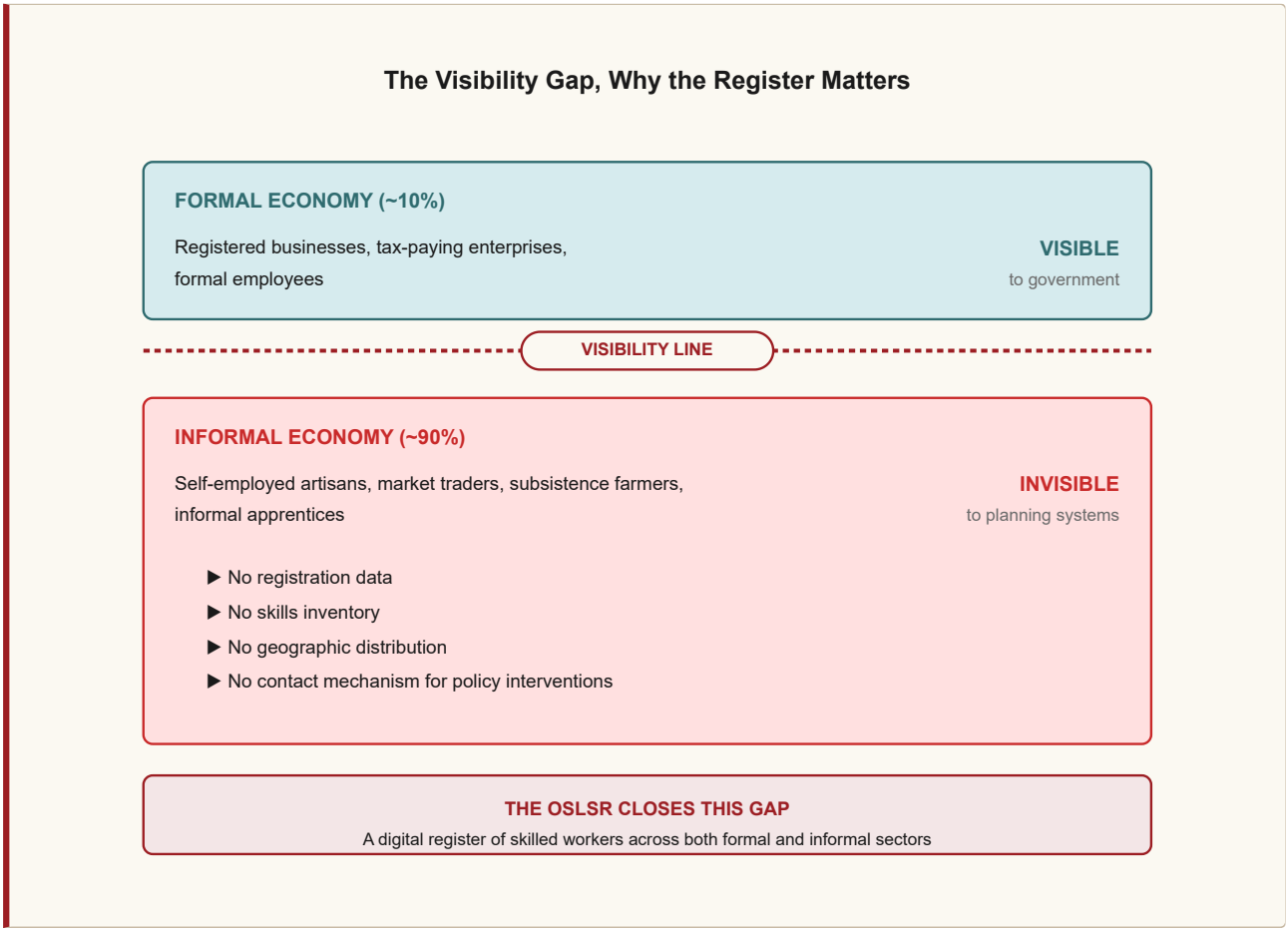
Table 4.6: Southwest Regional Context

INDICATOR	SOUTHWEST	NATIONAL	DEVIATION
Informal employment	89.4%	92.6%	Lower (more formalised)
Self-employment	71.8%	75.2%	Lower (more wage employment)
Female LFP	76.2%	73.1%	Higher (Yoruba trading culture)
Education (SSS+ attainment)	62.1%	53.4%	Higher (better educational infrastructure)
Urban population share	58.3%	48.0%	Higher (more urbanised)

Observation: The Southwest region, and Oyo State within it, has a more urbanised, better-educated, and slightly more formalised labour market than the national average. However, informal employment still dominates (89.4%), and self-employment remains the primary mode of economic activity (71.8%).

4.4.3 Informality and the Visibility Gap

Figure 4.3: The Visibility Gap, Why the Register Matters



The fundamental policy problem that the OSLSR addresses is the **invisibility of the informal economy** to government planning systems. When 90% of the labour force operates outside formal registration, taxation, and statistical systems, government has no reliable data on:

- **Who** constitutes the workforce (demographics, skills, locations)
- **What** skills exist across the state (supply-side data for investment attraction)
- **Where** specific skills are concentrated (geographic labour market intelligence)
- **How** the workforce is evolving (trends in skills, employment types, enterprise formation)

The OSLSR is designed to systematically bridge this visibility gap.

4.5 Sectoral Economic Structure

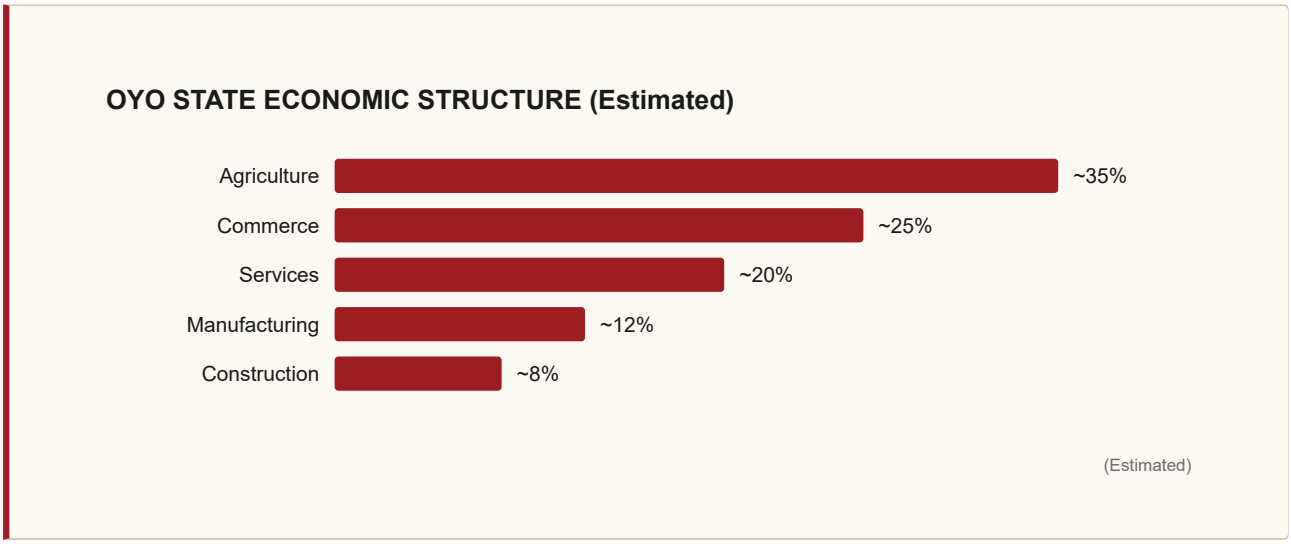
4.5.1 Oyo State Economic Sectors

Oyo State's economy is characterised by the following sectoral structure:

Table 4.7: Oyo State Economic Sectors

SECTOR	CONTRIBUTION (EST.)	KEY ACTIVITIES	MAJOR LOCATIONS
Agriculture	~35%	Cocoa, cassava, maize, palm oil, livestock, fisheries	Oke-Ogun, Ibarapa, Ibadan Peripheral
Commerce & Trade	~25%	Wholesale/retail, cross-border trade, market trading	Ibadan Metro, Saki West, Oyo
Services	~20%	Education, health, transport, professional services	Ibadan Metro, Ogbomosho
Manufacturing	~12%	Food processing, textiles, construction materials	Oluyole (industrial estate), Ibadan
Construction	~8%	Residential/commercial building, infrastructure	Ibadan Metro and Peripheral

Figure 4.4: Oyo State Economic Structure



4.5.2 Key Economic Characteristics by Zone

Table 4.8: Oyo State Economic Structure

ZONE	DOMINANT SECTOR	KEY PRODUCTS/ACTIVITIES	LABOUR MARKET CHARACTER
Ibadan Metro	Commerce, Services	Retail trade, education, health, government services, tech startups	Diverse skills; higher wage employment; digital skills emerging
Ibadan Peripheral	Agriculture, Construction	Peri-urban farming, residential construction, food processing	Transition zone; agricultural and construction skills
Ogbomosho	Agriculture, Manufacturing	Cassava processing, textile production, artisanal manufacturing	Strong artisanal tradition; vocational skills hub
Oyo	Agriculture, Commerce	Mixed farming, market trading, cultural tourism	Historical trading town; mixed economy
Oke-Ogun	Agriculture	Yam, cassava, maize, cashew, shea; cross-border trade	Agriculture-dominant; seasonal employment patterns; border trade

ZONE	DOMINANT SECTOR	KEY PRODUCTS/ACTIVITIES	LABOUR MARKET CHARACTER
Ibarapa	Agriculture	Cassava, palm oil, cocoa, livestock	Strong farming base; emerging agro-processing

4.6 Youth Employment Challenge

4.6.1 The Youth Bulge

Nigeria, and Oyo State within it, faces a significant youth employment challenge driven by demographic structure:

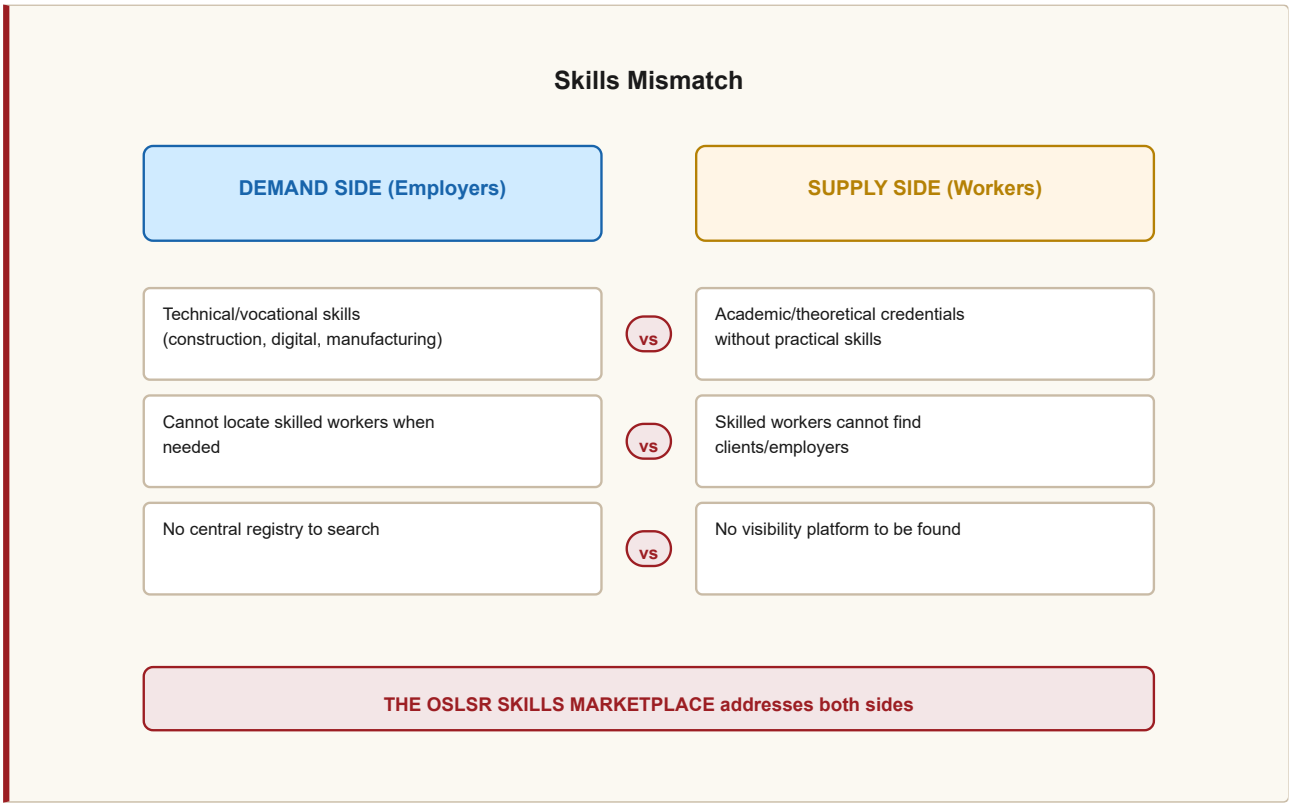
Table 4.9: The Youth Bulge

INDICATOR	VALUE	IMPLICATION
Population under 35	~65% (national)	Massive youth labour force entry
Annual new labour market entrants	~4 million (national)	Job creation demand exceeds supply
Youth unemployment (15–24)	8.4% (NBS)	Higher than adult unemployment
Youth underemployment	~18% (NBS)	Many young workers in inadequate employment
NEET rate (15–24)	~21.7% (NBS)	Not in education, employment, or training

4.6.2 Skills Mismatch

A persistent **skills mismatch** characterises the Nigerian labour market:

Figure 4.5: Skills Mismatch, Demand vs. Supply



4.7 Existing Data Gaps

The following table summarises the critical data gaps that the OSLSR is designed to fill:

Table 4.10: Existing Data Gaps

DATA GAP	CURRENT STATE	OSLSR RESOLUTION
State-level skills inventory	No comprehensive skills data exists for Oyo State	150-skill occupational taxonomy captured for every registrant
LGA-level labour distribution	NBS data available only at state/zone level	Per-LGA data from enumeration + LGA field in survey
Informal sector visibility	~90% of workers invisible to government systems	Direct enumeration of informal workers with NIN-linked registration
Skills-to-location mapping	No data on geographic distribution of specific trades	GPS-tagged submissions + LGA field enable skills mapping
Skills demand data	No systematic capture of unmet training needs	Q5.3 ("Skills you would like to learn") captures training demand
Enterprise registration data	CAC registration data incomplete for informal businesses	Q5.6 captures registration status of informal enterprises
Worker contact database	No mechanism to reach informal workers for policy interventions	Phone numbers and marketplace profiles create contactable database

4.8 Policy Context

4.8.1 National Policy Framework

The OSLSR operates within the following national policy context:

Table 4.11: National Policy Framework

POLICY	RELEVANCE TO OSLSR
National Development Plan 2021–2025	Targets job creation and skills development; requires state-level labour data
Subnational Business Enabling Environment Reform Programme	World Bank-supported ($750M) programme assessing states across 16 indicators including **skilled labour readiness**; Oyo State ranked 3rd nationally (62.7%) in the 2025 assessment, the register directly strengthens this indicator
Nigeria Data Protection Act 2023	Governs collection, storage, and processing of personal data (OSLSR compliance: Chapter 11)
National Skills Qualification Framework (NSQF)	Provides framework for skills recognition and certification
National Employment Policy	Emphasises need for labour market information systems
Decent Work Country Programme (ILO-Nigeria)	ILO partnership for improved labour statistics and decent work promotion

4.8.2 State-Level Context

Table 4.12: State-Level Context

FACTOR	DESCRIPTION
Oyo State Development Agenda	Focus on youth empowerment, skills acquisition, and MSME development
State Business Enabling Environment Reforms	Oyo State's active participation in the federal subnational business reform assessment programme; state performance rose from 27th to 3rd place nationally, demonstrating sustained reform commitment
Ministry of Trade, Industry, Investment and Cooperatives	Commissioning authority; mandate includes skills development, enterprise support, and business enabling environment reform
State Budget Allocation	Annual allocations for youth empowerment and skills training programmes
Oyo State Digital Economy Strategy	Emerging focus on technology-enabled service delivery

4.9 Convergence of Factors, Why Now

Six structural factors converge to make the current moment optimal for establishing a state-level skilled labour register:

Table 4.13: Convergence of Factors, Why Now

#	FACTOR	DESCRIPTION
1	**NIN Penetration**	National Identity Number enrollment has reached critical mass, providing a viable unique identifier for register deduplication
2	**Smartphone Penetration**	Smartphone ownership among field workers has reached levels that support mobile-first data collection
3	**4G/5G Coverage**	Network coverage expansion (including the Airtel 5G deployment) enables digital data synchronisation even in semi-urban areas
4	**Post-COVID Economic Recovery**	The economic disruption of 2020–2022 has heightened government interest in understanding and supporting the informal workforce
5	**Business Environment Reform Momentum**	Oyo State's rising performance in the federal subnational business enabling environment assessment (3rd nationally in 2025) creates institutional demand for the skilled labour readiness data that only a comprehensive state register can provide
6	**State Government Commitment**	The commissioning of this engagement demonstrates political will to invest in evidence-based labour policy and sustain business environment reform gains

4.10 Chapter Summary

The situational analysis reveals that Oyo State's labour market is characterised by:

1. **Massive informality**: ~90% of employment is informal, rendering the majority of the workforce invisible to government planning systems
2. **Self-employment dominance**: ~75% of employed persons are self-employed, operating as artisans, traders, and small-scale entrepreneurs
3. **Youth pressure**: ~65% of the population is under 35, creating continuous demand for skills development and employment opportunities
4. **Significant data gaps**: No comprehensive, state-level skills inventory or labour force register exists
5. **Favourable enabling conditions**: NIN penetration, smartphone availability, and network coverage create the technical prerequisites for digital enumeration

The OSLSR is purpose-built to address these conditions, creating, for the first time, a **digital, searchable, NIN-linked register of Oyo State's skilled workforce** that spans both the formal and informal economies.

5. Comparative Analysis: State-Level Registry Initiatives

Figure 5.1: Comparative Capability Assessment, State-Level Registry Initiatives

Scored on 10 dimensions (0–5 scale) | Source: Desk review and platform analysis

Capability Dimension	Lagos (LETF)	Kaduna (KADSRA)	Edo (EdoJobs)	Delta (JCO)	OSLSR (Oyo)
Survey Instrument Depth	★☆☆☆☆ 1	★★☆☆☆ 2	★★☆☆☆ 2	★☆☆☆☆ 1	★★★★★ 5
Skills Taxonomy	— 0	★☆☆☆☆ 1	★★☆☆☆ 2	— 0	★★★★★ 5
ILO Alignment	— 0	— 0	— 0	— 0	★★★★★ 5
Offline Capability	— 0	★☆☆☆☆ 1	— 0	— 0	★★★★★ 5
Fraud Detection	★☆☆☆☆ 1	★★☆☆☆ 2	★☆☆☆☆ 1	— 0	★★★★★ 5
Data Quality Layers	★☆☆☆☆ 1	★★★☆☆ 3	★☆☆☆☆ 1	— 0	★★★★☆ 4
Security (OWASP)	★☆☆☆☆ 1	★★☆☆☆ 2	★☆☆☆☆ 1	— 0	★★★★★ 5
NDPA Compliance	★☆☆☆☆ 1	★★☆☆☆ 2	★★☆☆☆ 2	— 0	★★★★★ 5
Public Marketplace	— 0	— 0	★★☆☆☆ 2	— 0	★★★★★ 5
Automated Testing	★☆☆☆☆ 1	★☆☆☆☆ 1	★☆☆☆☆ 1	— 0	★★★★★ 5
TOTAL SCORE	6 / 50	14 / 50	12 / 50	1 / 50	49 / 50

Score Key: 0 — None | 1 — Minimal | 2 — Basic | 3 — Moderate | 4 — Strong | 5 — Comprehensive

5.1 Introduction

This chapter presents a comparative analysis of state-level workforce registration and skills enumeration initiatives across Nigeria. The analysis serves two purposes: (1) to situate the Oyo State Skilled Labour Register within the broader national landscape of sub-national labour market information systems, and (2) to identify lessons learned and best practices from comparable initiatives that have informed the OSLSR design.

5.2 Landscape Overview

Several Nigerian states have undertaken workforce registration or skills mapping initiatives in recent years, driven by the shared challenge of informal economy visibility and the need for evidence-based skills development planning. However, these initiatives vary significantly in scope, methodology, technology, and outcomes.

Figure 5.2: State-Level Registry Initiatives in Nigeria

State-Level Registry Initiatives in Nigeria

Comparative Landscape

STATE	INITIATIVE	STATUS
Lagos	Employment Trust Fund / Lagos Innovates Portal	Operational
Kaduna	Kaduna State Residents Registration Agency (KADSRA)	Operational
Edo	EdoJobs / Edo Skills Map	Operational
Delta	Delta State Job Creation Office / Skills Registry	Partial
Ogun	Ogun State Digital Skills Registry	Planning
OYO (OSLSR)	Oyo State Skilled Labour Register	Operational*

* Field-survey pending

5.3 Comparative Analysis

5.3.1 Lagos State, Employment Trust Fund

Table 5.1: Lagos State, Employment Trust Fund

PARAMETER	DETAIL
Initiative	Lagos Employment Trust Fund (LETF) / Lagos State Employment Portal
Established	2016
Administering Body	Lagos Employment Trust Fund (State agency)
Scope	Loan facilitation for entrepreneurs; job matching; skills training referral
Technology	Web portal with online registration
Registration Model	Self-registration (online only); employer registration for job posting
Skills Taxonomy	General skills categories (not detailed artisan-level)
Unique Identifier	BVN (Bank Verification Number) for loan applicants
Estimated Registrants	300,000+ (self-reported)

PARAMETER	DETAIL
Key Strength	Direct linkage to financial services (loans up to ₦5M) creates strong registration incentive
Key Limitation	Primarily serves the formal/semi-formal sector; limited penetration into informal artisan economy
Fraud Detection	BVN-based deduplication; no field enumeration component

5.3.2 Kaduna State, Kaduna State Residents Registration Agency (KADSRA)

Table 5.2: Kaduna State, Kaduna State Residents Registration Agency (KADSRA)

PARAMETER	DETAIL
Initiative	Kaduna State Residents Registration Agency (KADSRA)
Established	2018
Administering Body	Dedicated state agency under the Governor's office
Scope	Universal resident registration (not skills-specific); captures demographics, occupation, location
Technology	Custom mobile application for field agents; web dashboard
Registration Model	Door-to-door enumeration by field agents; self-registration at designated centres
Skills Taxonomy	Occupational categories (broad groupings, not detailed skill-level)
Unique Identifier	KAD-ID (state-assigned unique number)
Estimated Registrants	6+ million (reported 2022)
Key Strength	Largest state-level registration exercise in Nigeria; universal coverage (not just skilled workers)
Key Limitation	Broad scope means limited depth of skills data; designed for general administration rather than labour market intelligence
Fraud Detection	Biometric capture (fingerprint); agent supervision

5.3.3 Edo State, EdoJobs / Edo Skills Map

Table 5.3: Edo State, EdoJobs / Edo Skills Map

PARAMETER	DETAIL
Initiative	EdoJobs portal / Edo Innovation Hub
Established	2019
Administering Body	Edo State Skills Development Agency
Scope	Youth skills training, job matching, skills assessment; linked to GIZ partnership
Technology	Web platform with online registration; integrated with training programmes

PARAMETER	DETAIL
Registration Model	Self-registration (online); training programme enrollment captures skills data
Skills Taxonomy	Training programme-oriented categories (ICT, fashion, construction, agriculture)
Unique Identifier	NIN (National Identification Number)
Estimated Registrants	50,000+ (through training programmes)
Key Strength	Direct linkage to skills training creates pipeline from registration to development; GIZ technical support
Key Limitation	Registration is training-programme driven, does not capture existing skilled workers who are not seeking training
Fraud Detection	NIN validation; training programme attendance records

5.3.4 Delta State, Job Creation Office

Table 5.4: Delta State, Job Creation Office

PARAMETER	DETAIL
Initiative	Delta State Job Creation Office (JCO)
Established	2020
Administering Body	Delta State Ministry of Economic Planning
Scope	Youth employment tracking; skills database for government programmes
Technology	Basic web registration form; spreadsheet-based data management
Registration Model	Self-registration (online and paper); integrated with STEP/YAGEP programmes
Skills Taxonomy	General skills categories
Estimated Registrants	~20,000
Key Limitation	Limited technology infrastructure; data management challenges at scale

5.4 Comparative Matrix

Table 5.5: Comparative Matrix

FEATURE	LAGOS (LETF)	KADUNA (KADSRA)	EDO (EDOJOBS)	OSLSR (OYO)
Primary Purpose	Finance access	Resident ID	Skills training	Labour register
Target Population	Entrepreneurs	All residents	Youth	Skilled workers

FEATURE	LAGOS (LETF)	KADUNA (KADSRA)	EDO (EDOJOBS)	OSLSR (OYO)
Registration Channels	Web only	Field + centres	Web + training	Mobile PWA + Web + Field
Offline Capability	No	Limited	No	**Yes (7-day)**
Skills Taxonomy Depth	General categories	Occupation only	Training categories	**150 skills, 20 sectors**
ILO Alignment	No	No	Partial	**Yes (ICLS-19)**
NIN-Based Deduplication	No (BVN)	No (KAD-ID)	Yes	**Yes**
Automated Fraud Detection	Basic	Biometric	Basic	**Multi-algorithm**
Skills Marketplace	No	No	Partial (job board)	**Yes (public directory)**
Immutable Audit Trail	Unknown	Unknown	Unknown	**Yes (SHA-256 chain)**
NDPA Compliance	Unknown	Partial	Partial	**Full compliance**
Automated Testing	Unknown	Unknown	Unknown	**3,564 tests**
OWASP Assessment	Unknown	Unknown	Unknown	**10/10 SECURE**
Progressive Consent	No	No	No	**3-tier model**

5.5 Key Lessons from Comparable Initiatives

The comparative analysis yields five key lessons that have directly informed the OSLSR design:

Lesson 1: Multi-Channel Registration is Essential

Observation: Lagos and Edo rely primarily on web-based self-registration, which excludes workers who lack internet access or digital literacy. Kaduna's field agent model achieves broader coverage but requires massive human resources.

OSLSR Application: The OSLSR implements a **three-channel approach** (Mobile PWA for field enumeration, desktop web for data entry, public self-registration), combining the reach of field enumeration with the scalability of self-registration.

Lesson 2: Skills Taxonomy Depth Determines Utility

Observation: Initiatives with broad occupational categories (e.g., "Construction," "Agriculture") provide limited utility for skills matching or labour market analysis. The value of a registry increases with the granularity of the skills data captured.

OSLSR Application: The OSLSR implements a **150-skill taxonomy across 20 sectors**, mapped to ISCO-08 international standards (Chapter 8), providing the granularity required for meaningful skills matching and policy analysis.

Lesson 3: Registration Incentive Drives Participation

Observation: Lagos achieves high registration through financial incentive (loan access). Edo achieves registration through training programme enrollment. Registries without clear incentives struggle with participation.

OSLSR Application: The OSLSR offers the **Skills Marketplace** as a tangible incentive, registered workers gain visibility to potential employers and clients through a publicly searchable directory, creating a direct economic benefit from registration.

Lesson 4: Offline Capability is Non-Negotiable for Field Work

Observation: States that rely on internet-connected registration have limited reach in rural areas with poor connectivity.

OSLSR Application: The OSLSR PWA operates for **up to 7 days offline** with automatic synchronisation, a critical capability for enumeration in rural Oke-Ogun and Ibarapa LGAs where connectivity is intermittent.

Lesson 5: Data Quality Requires Systematic Controls

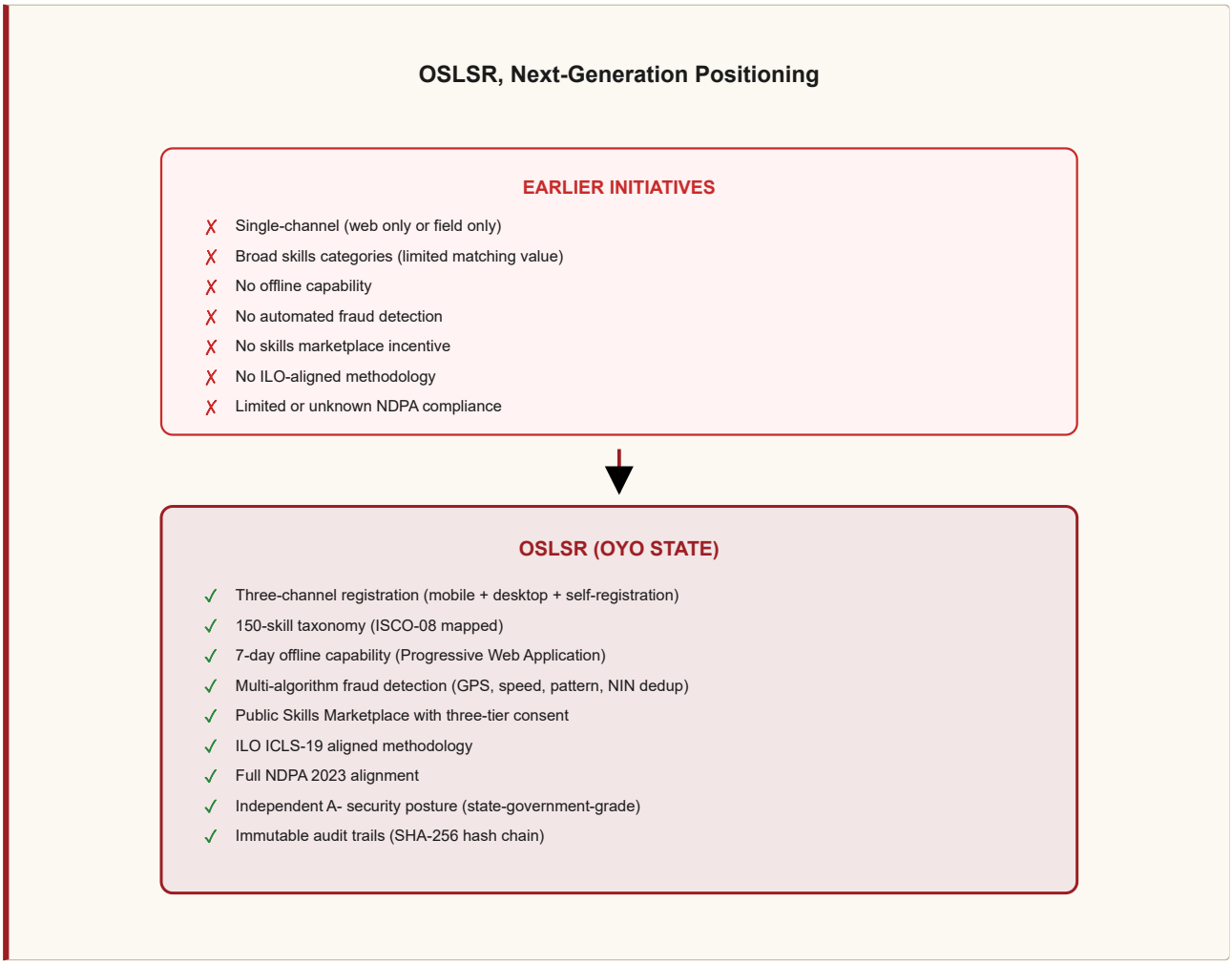
Observation: Large-scale registration exercises (particularly Kaduna's 6M+ registrants) face significant data quality challenges without systematic fraud detection and quality assurance mechanisms.

OSLSR Application: The OSLSR implements a **4-layer quality assurance protocol** (Chapter 14) with automated fraud detection, supervisory review, and statistical quality gates, a more rigorous approach than any comparable state initiative.

5.6 OSLSR Positioning

The comparative analysis positions the OSLSR as a **next-generation state-level labour register** that addresses the limitations observed in earlier state initiatives:

Figure 5.3: OSLSR, Next-Generation Positioning



5.7 Chapter Summary

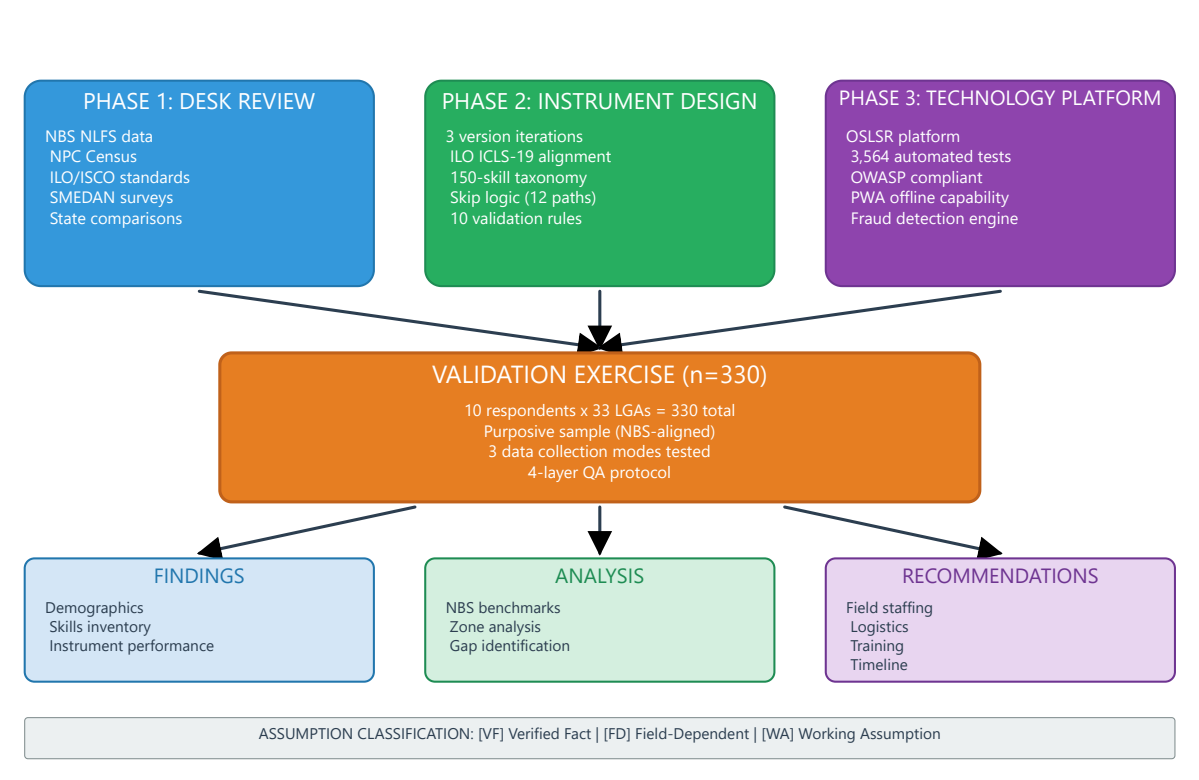
The comparative analysis demonstrates that while several Nigerian states have undertaken workforce registration or skills mapping initiatives, the OSLSR represents a significant methodological and technological advancement over existing efforts. The OSLSR's distinguishing features, offline-capable multi-channel data collection, granular 150-skill taxonomy with ISCO-08 mapping, automated multi-algorithm fraud detection, public skills marketplace with progressive consent, and ILO ICLS-19 aligned survey methodology, position it as a potential model for state-level labour market information systems in Nigeria.

The lessons drawn from Lagos, Kaduna, Edo, and Delta have been systematically incorporated into the OSLSR design, ensuring that known failure modes and limitations are addressed proactively rather than discovered post-deployment.

---

6. Baseline Study Methodology

Figure 6.1: Baseline Study Methodology Framework



6.1 Introduction

This chapter describes the methodological framework adopted for the baseline study, encompassing the research design, data collection strategy, sampling approach, quality assurance protocols, and the assumption classification system employed throughout this report. The methodology was designed in accordance with the International Labour Organization (ILO) guidelines for labour force surveys, specifically the **Resolution Concerning Statistics of Work, Employment and Labour Underutilization** adopted at the **19th International Conference of Labour Statisticians (ICLS)** in Geneva, 2013.

6.2 Research Design

The baseline study adopted a **mixed-methods approach** comprising:

1. **Desk Review** (Secondary Data Analysis), Systematic analysis of existing labour market data from the National Bureau of Statistics (NBS), National Population Commission (NPC), International Labour Organization (ILO), Small and Medium Enterprises Development Agency of Nigeria (SMEDAN), and state-level administrative records

2. **Instrument Development**, Design, iterative refinement, and validation of a structured survey questionnaire aligned with ILO ICLS-19 standards, incorporating a bespoke 150-skill occupational taxonomy
3. **Technology Platform Development**, Construction and deployment of a digital data collection, management, and analysis platform incorporating offline capability, fraud detection, and quality assurance automation
4. **Structured Validation Exercise**, A purposive field validation across all 33 LGAs to verify instrument functionality, assess platform performance, and establish preliminary demographic baselines
5. **Comparative Analysis**, Benchmarking against comparable state-level labour registration initiatives in Nigeria to identify best practices and design differentiation

---

6.3 Assumption Classification Framework

Following international best practice in research reporting, this study employs a **three-tier assumption classification framework** to ensure transparency about the epistemic status of all data, projections, and estimates presented in this report. Every material assertion is classified using one of the following tags:

Table 6.1: Assumption Classification Framework

CLASSIFICATION	SYMBOL	DEFINITION	VERIFICATION STANDARD
VERIFIED FACT	`[VF]`	Data sourced from authoritative national or international institutions, independently confirmable through publicly available records	Source cited at point of use; data retrievable from referenced publication
FIELD-DEPENDENT	`[FD]`	Reasonable estimate based on regional or national data, requiring field verification through the full-scale statewide enumeration (Deliverable 2) to establish Oyo State-specific values	National benchmark cited; state-specific validation pending full enumeration
WORKING ASSUMPTION	`[WA]`	Projections, estimates, or design parameters based on professional judgement, international standards, or comparable programme experience; subject to revision based on field evidence	Basis of assumption documented; sensitivity noted where material

This framework is applied consistently throughout Chapters 4–17 of this report. Readers are advised that findings tagged `[FD]` or `[WA]` are explicitly acknowledged as requiring validation through subsequent project phases. This intellectual honesty about what is known, what is estimated, and what remains to be tested is not a limitation, it is the foundation of methodological rigour (cf. ILO, *Guidelines Concerning the Measurement of Employment*, 2018).

6.4 Desk Review Methodology

6.4.1 Scope

The desk review encompassed a systematic analysis of the following categories of secondary data:

Table 6.2: Scope

CATEGORY	SOURCES REVIEWED	DATA EXTRACTED
National Labour Statistics	NBS Nigeria Labour Force Survey (NLFS), Q1 2024, Q2 2024, Annual 2023	Employment rates, informal sector size, self-employment patterns, sectoral distribution, education-employment correlations
Population Data	NPC 2006 National Census; NPC Population Projections (3.0% annual growth rate)	LGA-level population estimates for Oyo State (33 LGAs)
MSME Data	SMEDAN/NBS National MSME Survey 2021; PwC MSME Survey 2020	Enterprise size distribution, sector concentration, formality rates
International Standards	ILO ICLS-19 Resolution (2013); ISCO-08 Classification	Labour force classification methodology; occupational taxonomy framework
Vocational Skills	NABTEB 26 Trade Syllabi; NBTE Vocational Skills Directory; ITF Skills Acquisition Reports	Occupational categories, artisan trade classifications, vocational training standards
State-Level Initiatives	Lagos LSETF, Kaduna KADSTEP, Edo EdoJobs, Kano State Trade Registry	Comparable programme designs, implementation approaches, gaps and lessons
Data Protection	Nigeria Data Protection Act 2023 (NDPA); NDPR 2019	Compliance requirements, consent models, data retention obligations

6.4.2 Data Quality Assessment

Each secondary data source was assessed against four quality criteria:

1. **Authority**: Institutional credibility of the publishing organisation
2. **Recency**: Publication date relative to current conditions (priority given to 2023–2025 publications)
3. **Methodology**: Documented research methodology and sample design
4. **Relevance**: Applicability to Oyo State's specific economic and demographic context

Sources meeting all four criteria were classified as `[VF]`. Sources meeting three criteria were used as supporting evidence with appropriate caveats. Sources meeting fewer than three criteria were excluded.

6.5 National Enumeration Area Framework

6.5.1 Concept and Definition

An **Enumeration Area (EA)** is the smallest geographic unit into which a country's territory is divided for systematic field data collection during a census or large-scale survey. The United Nations *Principles and Recommendations for Population and Housing Censuses* (Rev. 3, Para. 1.186) establishes that EAs must be: mutually exclusive (no overlap), exhaustive (no gaps), bounded by identifiable physical features, consistent with the administrative hierarchy, approximately equal in population, and small enough for a single enumerator to canvass within the allotted period.

In Nigeria, the **National Population Commission (NPC)** is the statutory authority responsible for Enumeration Area Demarcation (EAD), the process of dividing the entire national territory into EAs. The **National Bureau of Statistics (NBS)** subsequently utilises digitised EA maps from the NPC as the spatial basis for household surveys, including the Nigeria Labour Force Survey (NLFS) from which the labour market benchmarks used in this study are derived.

6.5.2 History of EA Demarcation in Nigeria

Nigeria has conducted EAD exercises prior to four census operations:

Table 6.3: History of EA Demarcation in Nigeria

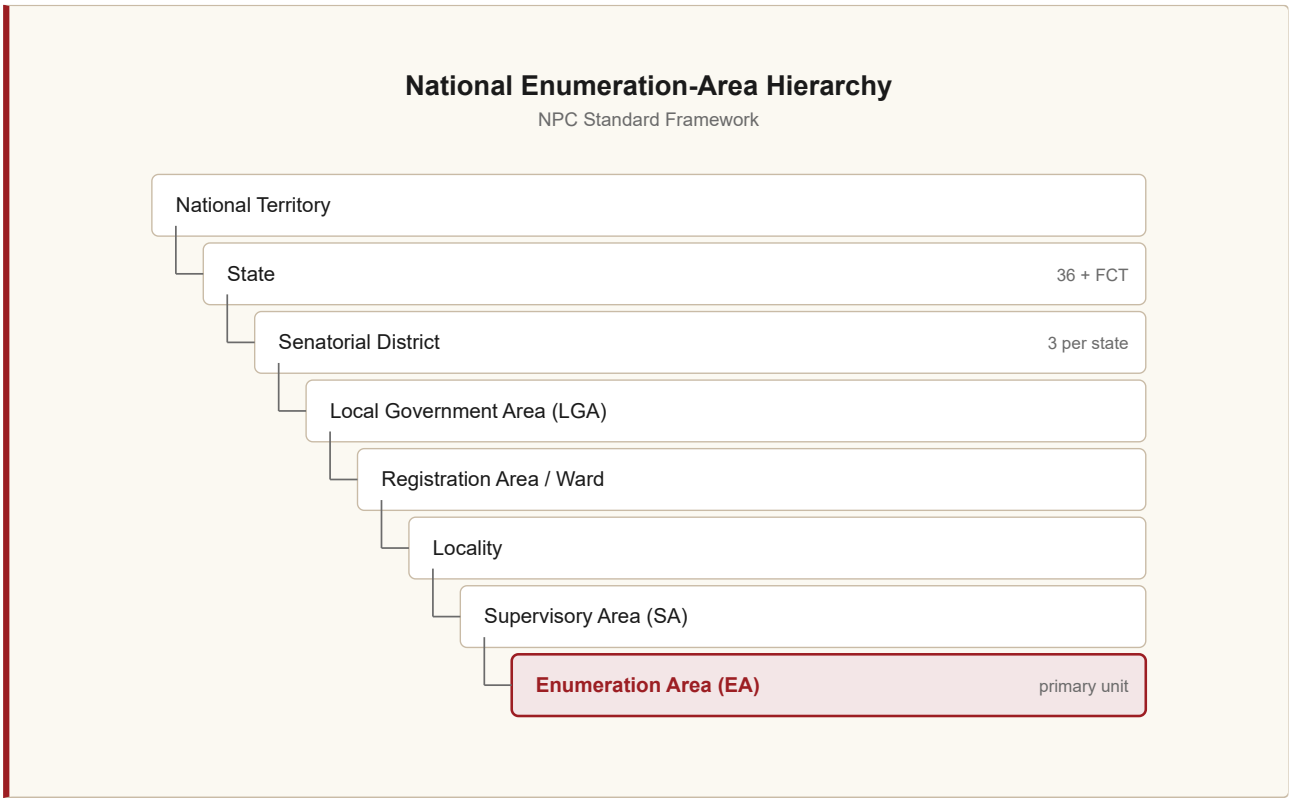
CENSUS	EAD INNOVATION	TECHNOLOGY
1973	First-ever pre-census EA demarcation in Nigeria	Manual cartographic methods, paper maps
1991	Expanded delineation with limited aerial photography	Paper maps, limited aerial imagery
2006	GPS and satellite imagery introduced for geo-referenced EA maps	GPS receivers, satellite imagery, OMR/OCR/ICR data capture
2023	First fully digital, nationwide GIS-based EAD; completed November 2021 after 18 phases spanning 7 years	Esri ArcGIS Pro, ArcPy, custom EADPad application, high-resolution satellite imagery

The 2023 Census EAD was a landmark achievement, the NPC received the **Special Achievement in GIS (SAG) Award** at the 2022 Esri User Conference in San Diego, California, for its nationwide digital EA demarcation of all 774 LGAs using GIS methodology. The exercise produced spatial datasets at every level of the geographic hierarchy: EA, Supervisory Area (SA), Locality, Registration Area/Ward, and LGA, along with geocoded building footprints, road networks, water bodies, and infrastructure data.

6.5.3 NPC Enumeration Area Hierarchy

The NPC's EA frame follows a nested geographic hierarchy in which each level sits entirely within the level above:

Figure 6.2: National Enumeration-Area Hierarchy



This strict nesting, no EA crosses a ward boundary, no ward crosses an LGA boundary, is what makes the EA frame usable as both a census operational unit and a national survey sampling frame.

6.5.4 EA Design Parameters and Oyo State Estimate

The NPC designs each EA to be coverable by **one enumerator within 5 days**. International norms and available NPC documentation indicate a target population of approximately **500–800 persons per EA**, varying by settlement density: urban high-density areas use the upper range (fewer, larger-population EAs), while rural dispersed areas use the lower range (more, smaller-population EAs). `[WA]`

The NPC does not publish state-level EA counts in publicly available documentation. To provide a planning reference, the Consultant developed a **density-adjusted estimation model** using verified 2006 NPC census data for all 33 LGAs of Oyo State, applying differentiated EA population targets by settlement classification:

Table 6.4: EA Design Parameters and Oyo State Estimate

CLASSIFICATION	LGAS	POPULATION (2006) [VF]	PERSONS/EA	ESTIMATED EAS [WA]
Urban (Ibadan metropolitan core)	5	1,343,147	800	1,679
Semi-Urban (secondary towns & peri-urban Ibadan)	11	2,201,296	650	3,386
Rural (dispersed agrarian settlements)	17	2,036,451	500	4,075
Oyo State Total	**33**	**5,580,894**	**611** (weighted avg.)	**~9,140**

Sensitivity analysis across conservative (900/750/600) and granular (700/550/400) EA sizing parameters yields a plausible range of **7,400–11,100 EAs** for Oyo State. The weighted average of ~611 persons per EA falls squarely within the UN-recommended range. A detailed LGA-by-LGA breakdown with full methodology is provided in the reference note (CHM/OSLR/2026/REF-001).

6.5.5 Significance for the OSLSR

The NPC's EA framework provides three essential inputs for the Skilled Labour Register:

1. **Population baseline**, NPC census data (Chapter 4) provides the LGA-level population denominators against which register coverage rates will be measured
2. **Geographic reference**, The NPC's nested hierarchy (State → LGA → Ward) is adopted as the OSLSR's own enumeration hierarchy, ensuring consistency with national statistical geography
3. **Methodological precedent**, The NPC's 50-year experience with EA-based field enumeration informs the OSLSR's enumerator deployment model, workload planning, and coverage monitoring approach

However, the OSLSR is not a census, it is a **voluntary skills registration exercise**. This fundamental difference in purpose requires deliberate adaptations to the NPC model, as detailed in the following section.

6.6 OSLSR Enumeration Area Framework

6.6.1 Adaptation Rationale

The Oyo State Skilled Labour Register adapts, rather than directly adopts, the NPC's EA framework, reflecting the fundamental differences between a mandatory population census and a voluntary skills registration exercise:

Table 6.5: Adaptation Rationale

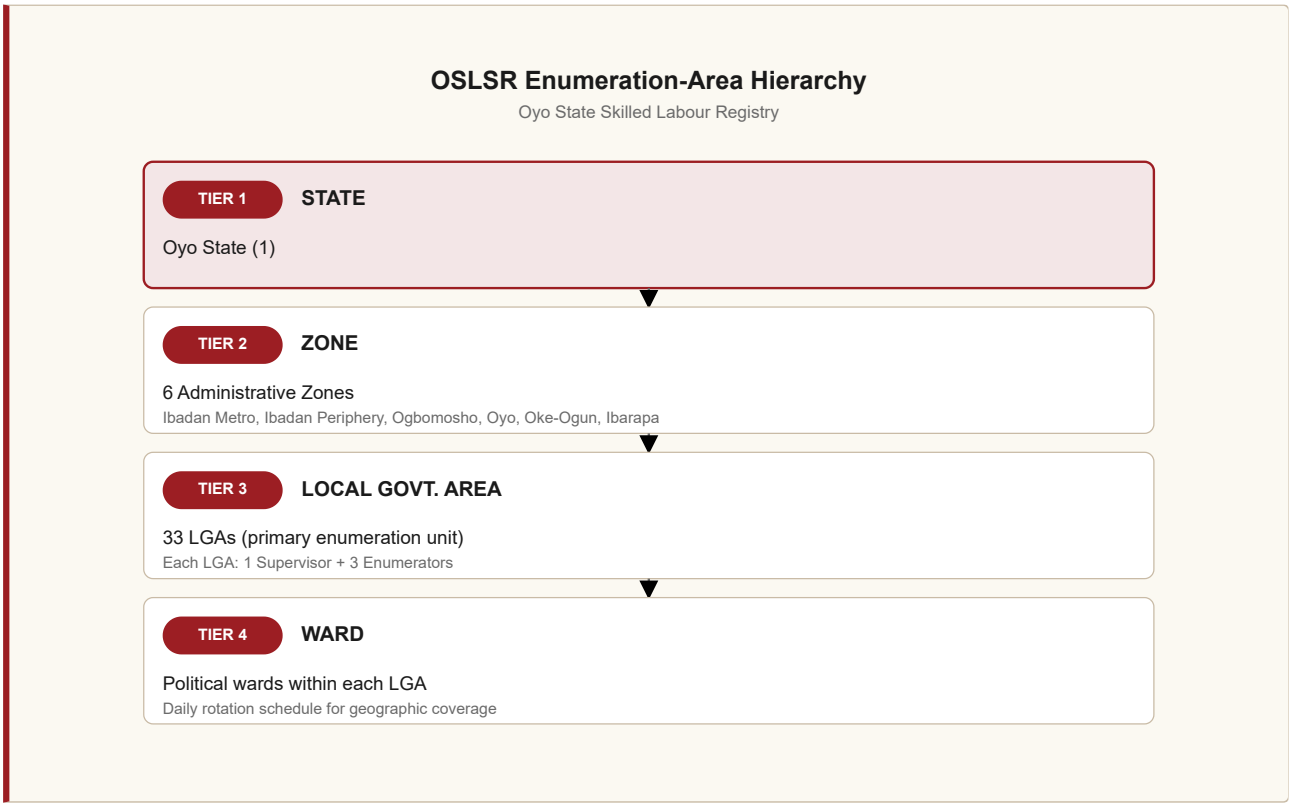
DIMENSION	NPC CENSUS MODEL	OSLSR ADAPTED MODEL	RATIONALE FOR ADAPTATION
Purpose	Universal population count	Voluntary skilled workforce registration	Register captures willing respondents, not every household
Primary unit	EA (~500–800 persons)	LGA (33 units statewide)	3 enumerators per LGA cannot cover ~280 EAs per LGA individually; LGA is the appropriate accountability unit
Sub-unit	Supervisory Area	Ward (rotation basis)	Wards are the locally recognised administrative units with community leadership structures
Coverage target	100% household enumeration	Maximum voluntary participation	Not every person will be encountered or willing to register
Boundary method	Pre-delineated GIS polygon maps	GPS capture at point of submission	OSLSR builds its own coverage map progressively; pre-printed EA maps are not operationally necessary
Map production	Years in advance (EAD exercise)	Real-time (progressive GPS heatmap)	Eliminates dependency on NPC map availability
Enumerators per unit	1 per EA	3 per LGA (rotating across wards)	Workload model sized for voluntary registration throughput, not universal coverage
Duration	5 days per EA	30 days per LGA (continuous)	Longer engagement period reflects voluntary nature, respondents must be located, engaged, and persuaded

This adaptation preserves the core EA principles, systematic geographic coverage, workload-based assignment, and quality assurance through spatial analysis, while recognising that a skills register operates under fundamentally different conditions from a population census.

6.6.2 OSLSR Enumeration Hierarchy

The OSLSR adopts a **four-tier geographic hierarchy** for enumeration planning, aligned with the upper tiers of the NPC hierarchy:

Figure 6.3: OSLSR Enumeration-Area Hierarchy



Tiers 1–3 map directly to the NPC hierarchy (State → Senatorial District/Zone → LGA). Tier 4 corresponds to the NPC's Registration Area/Ward level, but the OSLSR uses wards as **rotation units** for enumerator deployment rather than as fixed EA assignments.

6.6.3 LGA as Primary Enumeration Unit

The **Local Government Area (LGA)**, rather than the individual EA, serves as the primary enumeration unit for the following reasons:

Table 6.6: LGA as Primary Enumeration Unit

CRITERION	JUSTIFICATION
Administrative alignment	LGAs are the smallest administrative units with dedicated local government structures, enabling coordination with LGA chairmen and community leaders
NPC precedent	The NPC organises census operations by LGA, and all population data (Chapter 4) is available at the LGA level
Staffing model	The 4-person team structure (1 Supervisor + 3 Enumerators per LGA) maps directly to the LGA boundary
Scale proportionality	With ~9,140 estimated NPC EAs across 33 LGAs (averaging ~277 EAs per LGA), assigning 3 enumerators to individual EAs would require ~27,420 enumerators, far exceeding the resource envelope. The LGA is the operationally appropriate unit for 132 field personnel
Dashboard analytics	The OSLSR platform provides per-LGA dashboards, enabling real-time monitoring of enumeration progress by geographic unit
Validation alignment	The baseline validation exercise (n=330) was structured at 10 respondents per LGA, establishing LGA-level baselines

6.6.4 Ward-Level Enumeration Planning

Within each LGA, the three assigned enumerators are deployed across political wards on a rotating daily schedule:

Table 6.7: Ward-Level Enumeration Planning

DAY	ENUMERATOR A	ENUMERATOR B	ENUMERATOR C
1	Ward cluster 1	Ward cluster 2	Ward cluster 3
2	Ward cluster 2	Ward cluster 3	Ward cluster 1
3	Ward cluster 3	Ward cluster 1	Ward cluster 2
...	*Rotation continues*		

This rotation model ensures:

- **Complete ward coverage** within each LGA across the enumeration period
- **Cross-validation**, different enumerators visit the same wards, enabling consistency checks
- **Fraud deterrence**, an enumerator cannot fabricate responses for areas they have not visited, as GPS coordinates are captured automatically

6.6.5 GPS-Based Digital Enumeration Area Mapping

Unlike traditional NPC EA delineation, which relies on pre-demarcated GIS polygon maps produced years in advance during the EAD exercise (Section 6.5.2), the OSLSR creates a **digital enumeration area map in real time** through GPS capture at the point of each survey submission:

1. **Automatic GPS capture**: The OSLSR mobile PWA records the respondent's geographic coordinates at the time of survey completion (with respondent knowledge and consent)
2. **Geo-referenced submissions**: Each record in the register is tagged with latitude/longitude, enabling spatial analysis
3. **Progressive coverage mapping**: The accumulation of geo-referenced submissions across the enumeration period generates a digital coverage map, showing precisely where data has been collected, functionally equivalent to a post-hoc EA map built from actual field operations rather than pre-census planning
4. **Coverage gap identification**: Supervisors can view the geographic distribution of submissions on the platform dashboard, identifying wards or areas with low submission density and redirecting enumerators accordingly
5. **Fraud detection integration**: The GPS cluster analysis algorithm (Chapter 14, Layer 2) uses the geographic distribution of submissions to detect anomalous patterns, for example, an improbably high density of submissions from a single coordinate, suggesting fabrication

This approach eliminates the OSLSR's dependency on NPC map availability while producing a geographic dataset that can be compared against the NPC's EA frame as a post-enumeration coverage audit.

6.6.6 Enumeration Area Summary

Table 6.8: Enumeration Area Summary

PARAMETER	NPC CENSUS REFERENCE	OSLSR DESIGN
Geographic hierarchy	State → Senatorial District → LGA → Ward → Locality → SA → EA	State → Zone → LGA → Ward

PARAMETER	NPC CENSUS REFERENCE	OSLSR DESIGN
Primary enumeration unit	EA (~500–800 persons)	LGA (33 units)
Estimated EAs in Oyo State	~9,140 [WA]	N/A (LGA-based)
Team per unit	1 Enumerator per EA	1 Supervisor + 3 Enumerators per LGA
Sub-unit deployment	Fixed EA assignment	Ward rotation (daily schedule)
EA mapping method	Pre-demarcated GIS polygons (EAD exercise)	GPS-based digital mapping (real-time)
Coverage monitoring	EA completion checklists	Platform dashboard with LGA drill-down + GPS heatmap
Quality assurance	Supervisor visits to EAs	GPS cluster analysis + ward rotation cross-validation

For detailed NPC EA methodology, Oyo State LGA-by-LGA EA estimates, and the density-adjusted estimation model, see Reference Note CHM/OSLR/2026/REF-001.

6.7 Validation Exercise Design

6.7.1 Objective

The structured validation exercise was designed to achieve the following objectives:

1. Verify the functionality and usability of the survey instrument across diverse respondent profiles (age, education, occupation, urban/rural)
2. Assess the digital platform's performance across multiple device types, operating systems, and network conditions
3. Establish preliminary demographic baselines for comparison with national benchmarks
4. Identify and resolve any instrument issues prior to full-scale deployment
5. Validate the 150-skill occupational taxonomy's coverage and comprehensiveness
6. Measure completion times and user experience metrics

6.7.2 Sampling Strategy

Table 6.9: Sampling Strategy

PARAMETER	DESIGN CHOICE	JUSTIFICATION
Sample Size	330 respondents	10 per LGA × 33 LGAs = complete geographic coverage
Sampling Method	Purposive (quota-based)	Validation objective requires demographic representativeness, not probability sampling
Stratification	By LGA (primary) and gender (secondary)	Ensures every LGA is represented; gender quota approximates national distribution

PARAMETER	DESIGN CHOICE	JUSTIFICATION
Demographic Quotas	Age, education, employment status aligned with NBS NLFS Q2 2024 benchmarks	Enables comparison of instrument performance across representative profiles

6.7.3 Limitations

The validation exercise is subject to the following acknowledged limitations:

1. **Non-probability sample**: The purposive sampling design does not support inferential statistical analysis or population-level generalisation. Findings are valid for instrument validation and operational planning only. `[WA]`
2. **Sample size**: Ten (10) respondents per LGA provides sufficient diversity for instrument testing but insufficient power for LGA-level statistical estimates. State-level aggregates (n=330) provide indicative patterns only. `[WA]`
3. **Geographic coverage**: While all 33 LGAs are represented, intra-LGA variation (e.g., urban wards vs. rural settlements within a single LGA) is not captured at this sample size. `[FD]`
4. **Temporal snapshot**: The validation exercise captures a single point-in-time assessment. Seasonal employment variations (e.g., agricultural planting/harvest cycles) are not reflected. `[FD]`

These limitations are inherent to the validation phase design and will be addressed by the full-scale statewide enumeration (Deliverable 2), which will employ a probability-based sampling or total enumeration approach with statistically adequate sample sizes per LGA.

6.8 Data Collection Modes

The OSLSR platform supports three distinct data collection modes, all of which were validated during the baseline exercise:

Table 6.10: Data Collection Modes

MODE	CHANNEL	TARGET USERS	CONNECTIVITY	VALIDATION STATUS
Field Enumeration	Mobile PWA (Android 8.0+)	Trained Enumerators	Offline-capable (7-day retention)	✓ Validated
Desktop Data Entry	Web Application (Desktop)	Data Entry Clerks	Online required	✓ Validated
Public Self-Registration	Web Application (Any device)	General Public	Online required	✓ Validated

6.8.1 Offline Capability

The field enumeration mode employs a Progressive Web Application (PWA) architecture with the following offline provisions:

- **Service Worker**: Caches application assets for operation without network connectivity
- **IndexedDB Storage**: Stores draft and completed survey responses locally on the device for up to seven (7) days
- **Automatic Synchronisation**: Queued submissions are automatically uploaded when network connectivity is restored

- **Persistent Storage API**: Requests browser persistence to prevent data loss from storage pressure

This architecture is critical for field deployment in Oyo State, where network connectivity is inconsistent across rural and peri-urban LGAs, particularly in the Oke-Ogun and Ibarapa zones.

6.9 Data Quality Assurance Protocol

Data quality assurance is enforced through a multi-layer framework:

Layer 1: Client-Side Validation

- Real-time field validation (NIN format, phone format, age calculation, logical constraints)
- Skip logic enforcement (conditional display of questions based on prior responses)
- Mandatory field enforcement (28 of 36 questions are required)
- Auto-save every 30 seconds (IndexedDB)

Layer 2: Server-Side Validation

- Schema validation (Zod runtime type checking) on all submitted data
- Duplicate submission detection (idempotent processing by submission UUID)
- NIN uniqueness enforcement across all submission channels
- Cross-field consistency checks (e.g., dependents ≤ household size)

Layer 3: Automated Fraud Detection

- **GPS Cluster Analysis**: Flags multiple submissions from same GPS coordinates within configurable radius
- **Speed-Run Detection**: Flags submissions completed faster than minimum plausible duration
- **Straight-Lining Detection**: Flags submissions with repetitive response patterns
- **Configurable Thresholds**: All detection parameters are adjustable by authorised administrators

Layer 4: Human Review

- Flagged submissions routed to Supervisors for manual review
- Verification Assessors provide secondary audit of high-risk submissions
- All review actions are immutably logged in the audit trail

6.10 Ethical Considerations

6.10.1 Informed Consent

All respondents are presented with a clear, plain-language explanation of the survey's purpose, the data to be collected, how the data will be used, and their right to decline participation. The survey instrument includes a mandatory consent gate (Question 1.2), no personal data is collected unless the respondent explicitly consents.

6.10.2 Data Protection Compliance

The data collection methodology complies with the Nigeria Data Protection Act 2023 (NDPA), including:

- **Purpose limitation**: Data collected exclusively for the stated registry purpose
- **Data minimisation**: Only data necessary for registry objectives is collected
- **Consent-based processing**: All data processing is based on explicit respondent consent
- **Two-stage marketplace consent**: Separate, additional consent required for Skills Marketplace inclusion (anonymous profile) and contact detail visibility (enriched profile)
- **Retention policy**: 7-year data retention aligned with NDPA requirements, with automated backup to encrypted offsite storage
- **Access control**: Role-based access ensuring PII is accessible only to authorised personnel

6.10.3 Vulnerability Protections

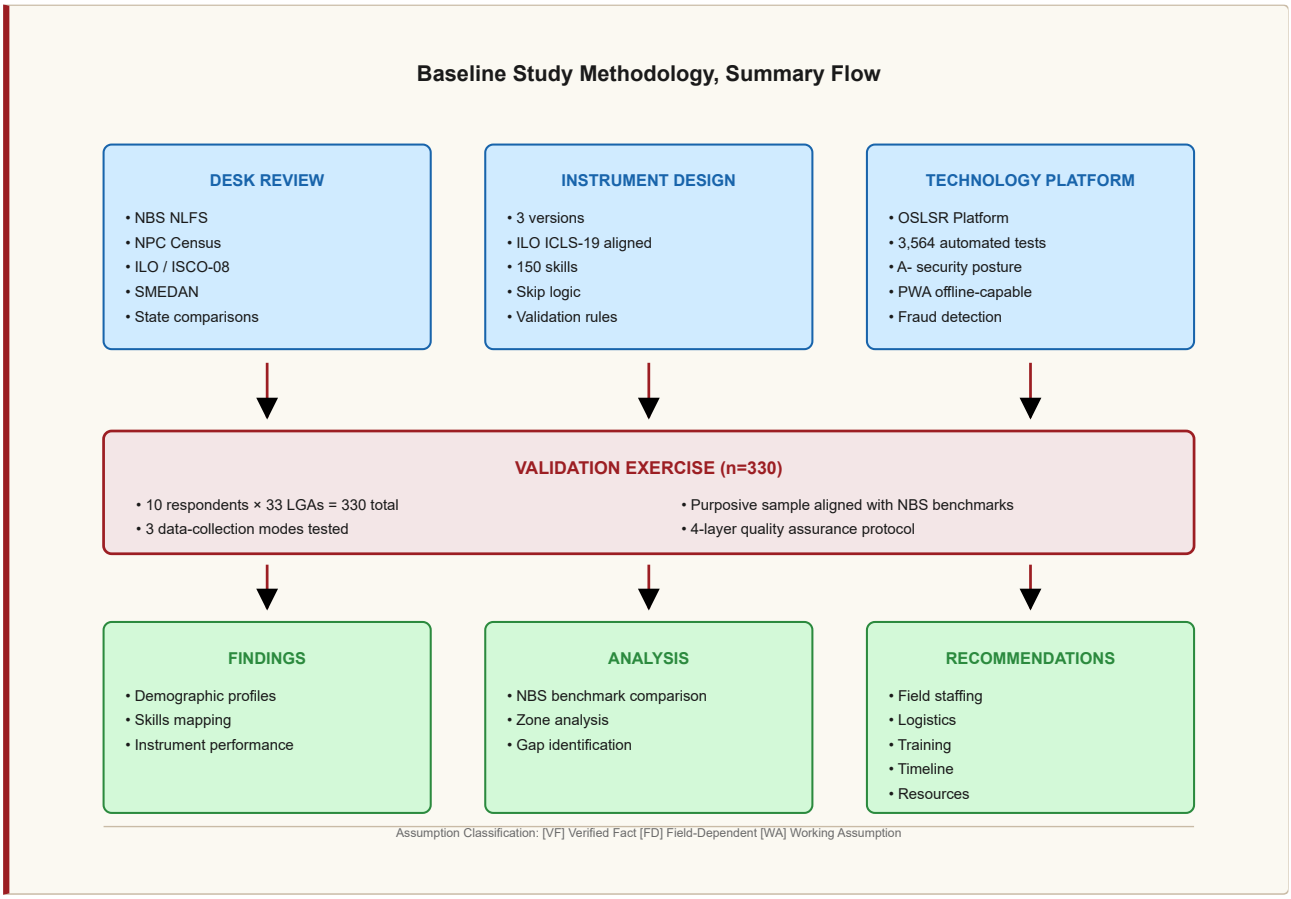
- Respondents below 15 years of age are excluded from the labour force participation module, consistent with ILO minimum working age standards
- Disability status is captured as a binary field (yes/no) to support inclusive policy planning without requiring disclosure of specific conditions
- Income data is collected as an optional field, respecting respondent sensitivity around financial disclosure

---

6.11 Methodology Summary

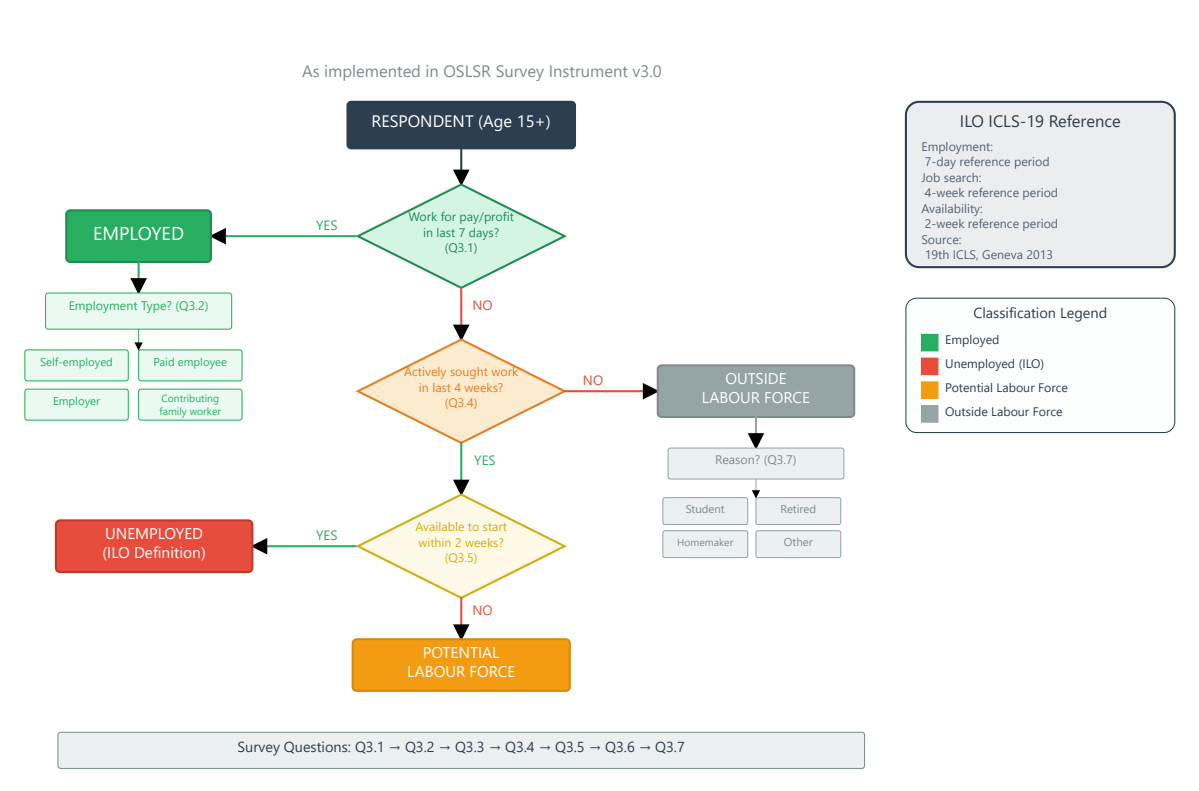
The baseline study methodology is summarised in the following framework diagram:

Figure 6.4: Baseline Study Methodology, Summary Flow



7. Survey Instrument Design and Validation

Figure 7.1: ILO ICLS-19 Labour Force Classification Cascade



7.1 Introduction

This chapter documents the design, structure, and iterative validation of the **OSLSR Labour & Skills Registry Survey**, the primary data collection instrument for the Oyo State Skilled Labour Register. The instrument was purpose-built to capture the multidimensional nature of labour force participation in Oyo State, incorporating international statistical standards, Nigerian regulatory requirements, and the specific policy objectives of the commissioning authority.

The final instrument (Version 3.0) represents the outcome of **three iterative review cycles**, each incorporating feedback from methodological review, technical feasibility assessment, and alignment verification against the ILO International Conference of Labour Statisticians (ICLS-19) framework.

7.2 Design Principles

The survey instrument was designed according to five governing principles:

Table 7.1: Design Principles

#	PRINCIPLE	RATIONALE	IMPLEMENTATION
1	**ILO ICLS-19 Alignment**	Ensures data compatibility with national and international labour force statistics	Labour force classification cascade (Q3.1–Q3.4) follows the ICLS-19 employment/unemployment resolution framework
2	**Minimum Respondent Burden**	Maximises completion rates in field conditions where respondent attention is limited	10-minute estimated completion time; skip logic eliminates irrelevant questions
3	**Data Minimisation**	NDPA 2023 compliance requires collection of only data necessary for the stated purpose	36 questions total; no questions unrelated to registry objectives
4	**Progressive Consent**	Respondents must understand and agree to each level of data sharing	Three-tier consent: (1) registry participation, (2) anonymous marketplace, (3) identifiable contact details
5	**Digital-First, Paper-Compatible**	Field conditions may necessitate paper backup, but primary collection is digital	One-question-per-screen mobile interface; paper form layout mirrors digital flow for data entry digitisation

7.3 Instrument Evolution

The survey instrument underwent three major revisions before finalisation:

7.3.1 Version History

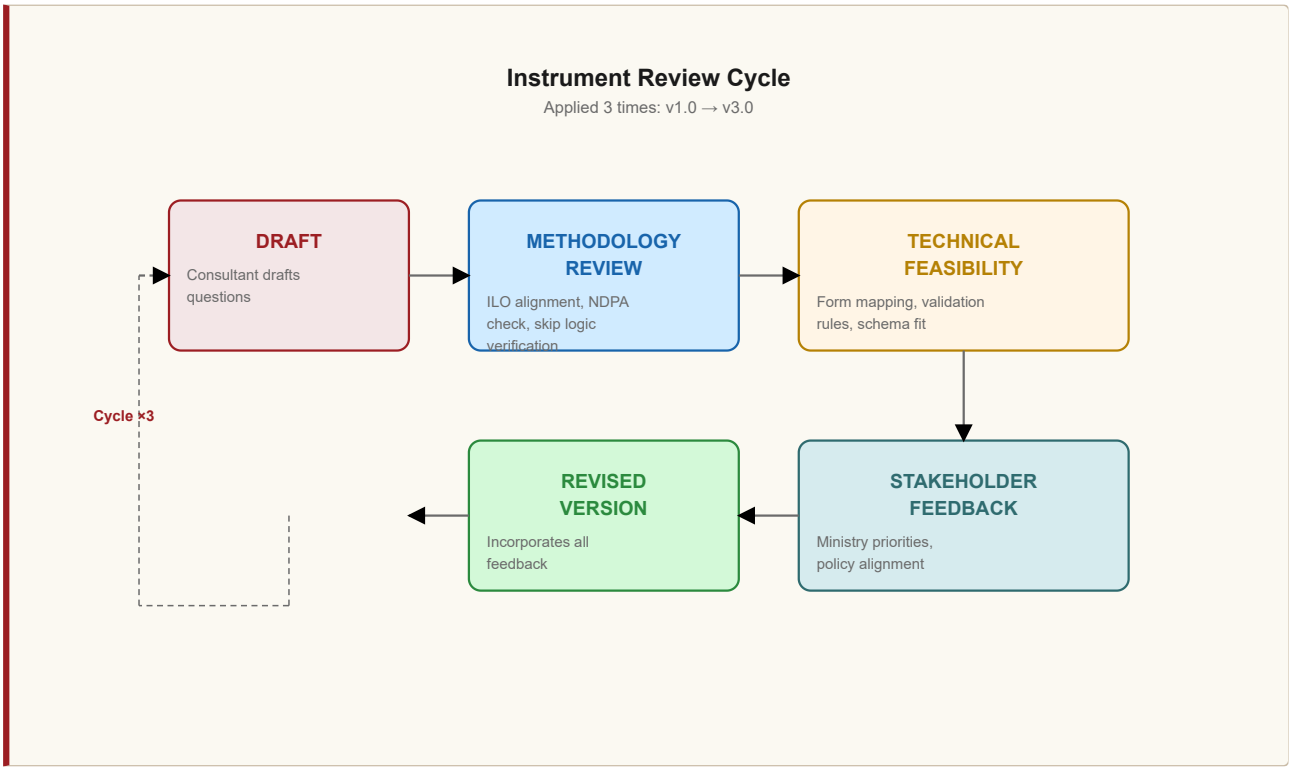
Table 7.2: Version History

VERSION	DATE	KEY CHANGES	TRIGGER
v1.0	December 2025	Initial 28-question instrument; basic demographics, employment status, skills inventory (40 skills)	Inception phase design
v2.0	January 2026	Expanded to 33 questions; added NIN requirement, LGA field, years of experience, expanded employment types; skills expanded to 50+	PRD requirements analysis (FR5, FR17, FR21)
v3.0	January 2026	Finalised at 36 questions; added marketplace section (6.0–6.4), business fields (address, apprentice count), refined education levels to 9 options, CAC registration status	Marketplace module requirements, stakeholder feedback on enterprise data needs

7.3.2 Review Cycle Process

Each version underwent a structured review cycle:

Figure 7.2: Instrument Review Cycle (v1.0 → v3.0)



7.4 Instrument Structure

The final instrument (v3.0) comprises **six sections** containing **36 questions**, of which **28 are required** and **12 are conditional** (displayed based on skip logic).

7.4.1 Section Overview

Table 7.3: Section Overview

SECTION	TITLE	QUESTIONS	REQUIRED	CONDITIONAL	PURPOSE
1	Introduction & Consent	2	1	0	Informed consent and study introduction
2	Identity & Demographics	11	10	0	Personal identification and demographic profiling
3	Labour Force Participation	9	8	5	ILO-aligned employment status classification
4	Household & Welfare	4	4	0	Household composition and living conditions
5	Skills & Business	8	3	4	Occupational skills inventory and enterprise data
6	Public Skills Marketplace	5	1	3	Progressive consent for marketplace participation

SECTION	TITLE	QUESTIONS	REQUIRED	CONDITIONAL	PURPOSE
Total		**39***	**27**	**12**	

** Including display-only informational fields (Q1.1, Q6.0) and auto-calculated fields (Q2.5)*

7.4.2 Estimated Completion Time

Table 7.4: Estimated Completion Time

RESPONDENT PROFILE	ESTIMATED TIME	BASIS
Employed, business owner, marketplace opt-in	10–12 minutes	All sections, all conditional paths active
Employed, no business, marketplace opt-out	7–8 minutes	Sections 5.4–5.8 and 6.2–6.4 skipped
Unemployed, job-seeking	6–7 minutes	Employment details (Q3.5–Q3.9) skipped
Outside labour force	5–6 minutes	Most of Section 3 skipped

The instrument is designed for a **maximum completion time of 12 minutes** under the most complex respondent profile, ensuring that field enumerators can achieve the target of **8 submissions per day** within a 6-hour operational window (accounting for travel, engagement, and breaks).

7.5 Section-by-Section Analysis

7.5.1 Section 1: Introduction & Consent

Table 7.5: Section 1: Introduction & Consent

#	QUESTION	TYPE	REQUIRED	NOTES
1.1	Welcome note introducing the OSLSR	Display only	,	Informational text explaining the purpose, duration, and data handling
1.2	Do you consent to participate?	Yes / No	Yes	**Gate question**, "No" terminates the survey with a thank-you message

Design rationale: NDPA 2023 Section 25 requires informed consent before personal data collection. The consent question serves as a legally compliant gate, no data is collected or transmitted until explicit consent is obtained. This satisfies the lawful basis for processing requirement.

7.5.2 Section 2: Identity & Demographics

Table 7.6: Section 2: Identity & Demographics

#	QUESTION	TYPE	REQUIRED	VALIDATION
2.1	Surname	Text	Yes	Non-empty

#	QUESTION	TYPE	REQUIRED	VALIDATION
2.2	First Name	Text	Yes	Non-empty
2.3	Gender	Single choice	Yes	Male / Female / Prefer not to say
2.4	Date of Birth	Date	Yes	Cannot be future date
2.5	Age	Auto-calculated	,	Derived from DOB; must be ≥15 years
2.6	Marital Status	Single choice	Yes	5 options: Single, Married, Divorced, Widowed, Separated
2.7	Highest Education	Single choice	Yes	9 levels from No Formal Education to Doctorate
2.8	Disability status	Yes / No	Yes	Self-declared
2.9	Phone Number	Text	Yes	Nigerian mobile format: 0[7-9][0-1]xxxxxxxx
2.10	NIN	Text	**Yes**	Exactly 11 digits; Modulus 11 checksum validation
2.11	LGA of residence	Single choice	Yes	33 Oyo State LGAs

Design rationale:

- **NIN as mandatory field**: The National Identification Number serves as the primary deduplication key across all submission channels. The Modulus 11 checksum validation catches transcription errors at point of entry, reducing the need for post-collection data cleaning. Global uniqueness enforcement prevents the same individual from being registered multiple times.
- **Age threshold (≥15 years)**: Aligned with the ILO ICLS-19 definition of the working-age population. The auto-calculation from Date of Birth eliminates the common field survey problem of respondents misreporting age.
- **Education levels (9 options)**: The expanded education classification distinguishes between NCE/OND and HND/BSc, a distinction that is significant in the Nigerian labour market where holders of different credentials occupy distinctly different employment niches.
- **GPS auto-capture**: Device GPS coordinates are captured automatically (not displayed to the respondent), enabling the fraud detection engine to identify geographic clustering anomalies.

7.5.3 Section 3: Labour Force Participation

This section implements the **ILO ICLS-19 labour force classification cascade**, a structured decision tree that classifies respondents into mutually exclusive labour force categories.

Figure 7.3: ILO ICLS-19 Classification Cascade

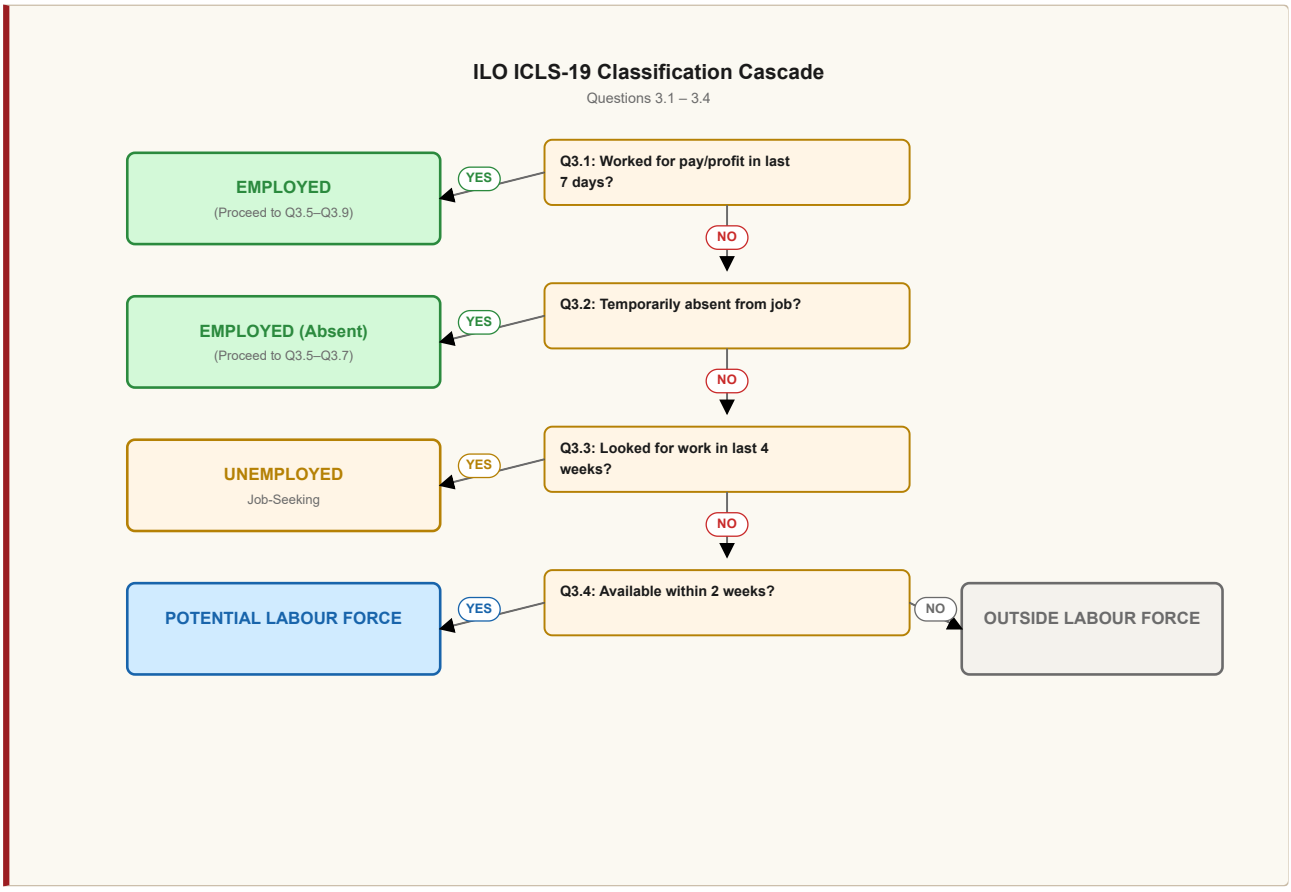


Table 7.7: ILO ICLS-19 Classification Cascade

#	QUESTION	TYPE	REQUIRED	SHOW IF	NOTES
3.1	Worked for pay/profit in last 7 days?	Yes / No	Yes	Always	Primary employment indicator
3.2	Temporarily absent from a job?	Yes / No	Yes	Q3.1 = No	Captures maternity, sick leave, seasonal pause
3.3	Looked for work in last 4 weeks?	Yes / No	Yes	Q3.2 = No	Unemployment indicator
3.4	Available to start within 2 weeks?	Yes / No	Yes	Q3.3 = No	Labour force availability
3.5	Main Occupation / Job Title	Text	Yes	Q3.1=Yes OR Q3.2=Yes	Free text, validated against taxonomy
3.6	Type of Employment	Single choice	Yes	Q3.1=Yes OR Q3.2=Yes	6 options (see below)
3.7	Years of Experience	Single choice	Yes	Q3.1=Yes OR Q3.2=Yes	5 ranges from <1 year to >10 years
3.8	Hours worked last week	Number	Yes	Q3.1 = Yes	Range: 0–168 hours
3.9	Monthly income estimate (₦)	Number	No	Q3.1 = Yes	Optional; cannot be negative

Employment Type Classification:

Table 7.8: Section 3: Labour Force Participation

CODE	LABEL	ILO ALIGNMENT
wage_public	Government/Public Sector Employee	ICSE-18: Employee
wage_private	Private Sector Employee	ICSE-18: Employee
self_employed	Self-Employed (Artisan/Trader/Business Owner)	ICSE-18: Own-account worker
contractor	Contractor/Consultant	ICSE-18: Own-account worker
family_unpaid	Unpaid Family Worker	ICSE-18: Contributing family worker
apprentice	Apprentice/Intern	ICSE-18: Employee (dependent)

Design rationale: The six employment type options reflect the structure of the Nigerian labour market, where the distinction between formal wage employment and self-employment is critical for policy analysis. The inclusion of "Apprentice/Intern" captures the significant informal apprenticeship economy that characterises Yoruba-speaking states.

7.5.4 Section 4: Household & Welfare

Table 7.9: Section 4: Household & Welfare

#	QUESTION	TYPE	REQUIRED	VALIDATION
4.1	Head of household?	Yes / No	Yes	,
4.2	Total household size	Number	Yes	Minimum: 1
4.3	Number of dependents	Number	Yes	Cannot exceed household size (Q4.2)
4.4	Housing ownership	Single choice	Yes	5 options: Owned, Rented, Family, Employer-provided, Other

Design rationale: Household data enables the computation of dependency ratios and housing tenure profiles, key indicators for labour policy planning. The cross-validation rule (dependents ≤ household size) prevents data quality errors at the point of entry.

7.5.5 Section 5: Skills & Business

Table 7.10: Section 5: Skills & Business

#	QUESTION	TYPE	REQUIRED	SHOW IF	VALIDATION
5.1	Primary skills	Multi-select	Yes	Always	150 skills across 20 sectors
5.2	Other skills (free text)	Text	No	Always	Max 200 characters
5.3	Skills desired to learn	Multi-select	No	Always	Same 150-skill list

#	QUESTION	TYPE	REQUIRED	SHOW IF	VALIDATION
5.4	Own/operate a business?	Yes / No	Yes	Always	,
5.5	Business name	Text	Yes	Q5.4 = Yes	,
5.6	CAC registration status	Single choice	Yes	Q5.4 = Yes	Registered / Unregistered / In Progress
5.7	Business address	Text	Yes	Q5.4 = Yes	,
5.8	Number of apprentices	Number	No	Q5.4 = Yes	Cannot be negative

Design rationale:

- **150-skill taxonomy**: The comprehensive skills list (documented in Chapter 8) enables precise matching between worker capabilities and employer needs in the Skills Marketplace. The multi-select format acknowledges that workers in the informal economy typically possess multiple tradeable skills.
- **"Skills desired to learn" (Q5.3)**: This forward-looking question captures unmet training demand, critical data for the Ministry's skills development programme planning.
- **Business registration data**: CAC registration status provides a proxy measure for formalisation of the informal economy, directly relevant to the Ministry's mandate.

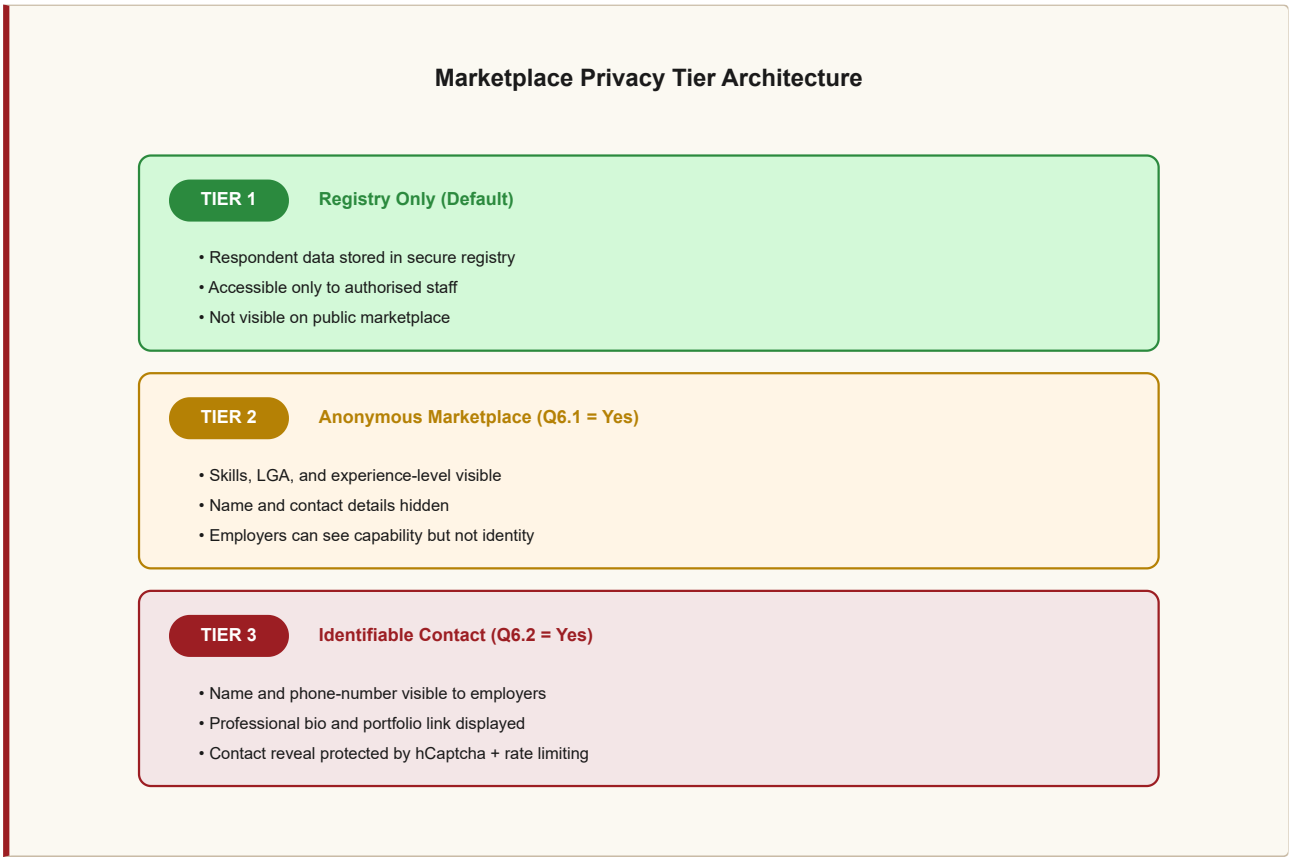
7.5.6 Section 6: Public Skills Marketplace

Table 7.11: Section 6: Public Skills Marketplace

#	QUESTION	TYPE	REQUIRED	SHOW IF	VALIDATION
6.0	Marketplace explanation	Display only	,	Always	Explains anonymous vs enriched profiles
6.1	Join anonymous marketplace?	Yes / No	Yes	Always	Opt-in for basic profile visibility
6.2	Allow name/phone visibility?	Yes / No	Yes	Q6.1 = Yes	Second-tier consent for contact details
6.3	Professional bio	Text	No	Q6.2 = Yes	Max 150 characters
6.4	Portfolio/social media link	URL	No	Q6.2 = Yes	Optional professional link

Design rationale: The marketplace section implements a **three-tier progressive consent model**:

Figure 7.4: Marketplace Privacy-Tier Architecture



This progressive model ensures that respondents make informed, granular decisions about their data visibility, a core principle of the NDPA 2023 consent framework.

7.6 Validation Rules

The instrument enforces **10 validation rules** at the point of data entry, preventing common data quality issues before submission:

Table 7.12: Validation Rules

#	FIELD	CONSTRAINT	ERROR MESSAGE	LAYER
1	NIN	Exactly 11 digits, numeric; Modulus 11 checksum	“NIN must be exactly 11 digits”	Frontend + Backend
2	Phone	Nigerian mobile format: 0[7-9][0-1]xxxxxxxx	“Enter valid Nigerian mobile number”	Frontend + Backend
3	Age	Calculated from DOB; must be ≥ 15 years	“Respondent must be 15 years or older”	Frontend + Backend
4	Date of Birth	Cannot be a future date	“Cannot be a future date”	Frontend + Backend
5	Hours Worked	Integer range 0–168	“Hours must be between 0 and 168”	Frontend + Backend

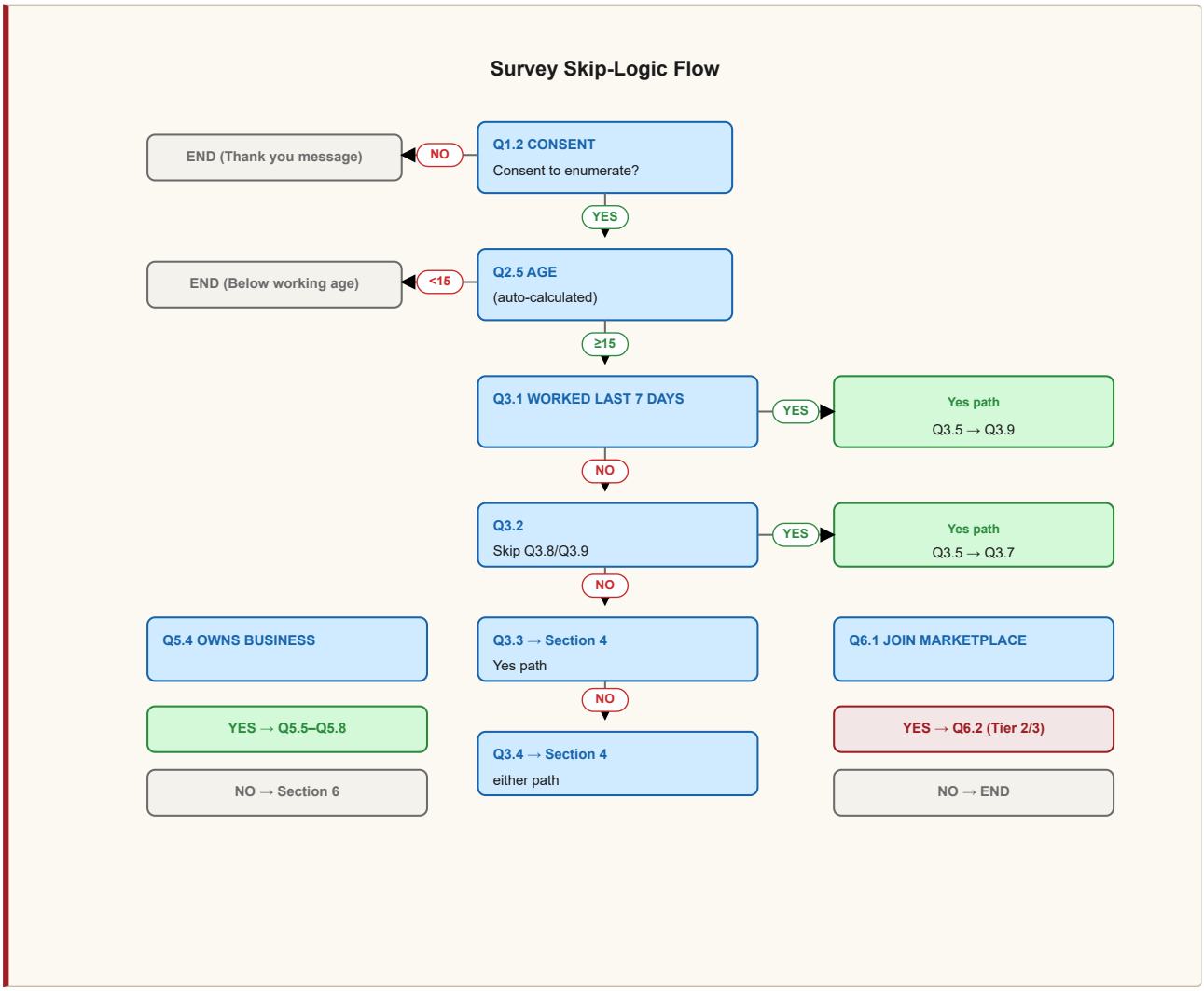
#	FIELD	CONSTRAINT	ERROR MESSAGE	LAYER
6	Monthly Income	Numeric, ≥ 0	“Cannot be negative”	Frontend + Backend
7	Dependents	Integer, ≤ household size (Q4.2)	“Dependents cannot exceed household size”	Frontend + Backend
8	Household Size	Integer, ≥ 1	“Must be at least 1”	Frontend + Backend
9	Professional Bio	≤ 150 characters	“Bio must be 150 characters or less”	Frontend + Backend
10	Other Skills	≤ 200 characters	“Maximum 200 characters”	Frontend + Backend

Critical design decision: All validation rules are enforced on **both frontend and backend** using shared Zod schemas, a single source of truth that prevents validation bypass through API manipulation. This dual-enforcement is a direct mitigation for OWASP A03 (Injection) and A04 (Insecure Design).

7.7 Skip Logic Architecture

The instrument employs conditional display logic (skip logic) to minimise respondent burden by showing only relevant questions:

Figure 7.5: Survey Skip-Logic Flow



Total possible paths: 12 unique paths through the instrument, ranging from 2 questions (consent declined) to 36 questions (fully employed business owner with marketplace opt-in).

7.8 Multi-Channel Delivery

The survey instrument is delivered through three channels, each optimised for its target user context:

Table 7.13: Multi-Channel Delivery

CHANNEL	INTERFACE	TARGET USER	KEY FEATURES
Mobile PWA	One-question-per-screen, swipe navigation, large touch targets	Field Enumerators	Offline capable (7-day); GPS auto-capture; auto-save on every question; Service Worker caching
Desktop Web	Keyboard-optimised multi-field layout	Data Entry Clerks	Tab navigation between fields; batch entry workflow; paper form digitisation
Public Self-Registration	Responsive web form with hCaptcha	General Public	Bot protection; progressive disclosure; no login required; mobile-responsive

All three channels feed into a **unified ingestion pipeline** that applies identical validation rules, fraud detection algorithms, and deduplication checks, ensuring data quality parity regardless of the submission source.

7.9 ILO ICLS-19 Compliance Verification

The following table maps the instrument's labour force questions to the specific ICLS-19 requirements they satisfy:

Table 7.14: ILO ICLS-19 Compliance Verification

ICLS-19 REQUIREMENT	RESOLUTION	INSTRUMENT QUESTION
Working-age population definition	15 years and above	Q2.4 (DOB) → Q2.5 (auto-calculated age ≥ 15)
Employment, short reference period	7-day recall period	Q3.1: "Worked for pay or profit in the last 7 days?"
Employment, temporary absence	Job attachment despite absence	Q3.2: "Temporarily absent from a job?"
Unemployment, active search	4-week job search period	Q3.3: "Looked for work in the last 4 weeks?"
Unemployment, availability	2-week availability criterion	Q3.4: "Available to start work within 2 weeks?"
Status in employment (ICSE-18)	6 employment type categories	Q3.6: Employment type (6 options mapped to ICSE-18)
Working time	Weekly hours worked	Q3.8: "Hours worked last week" (0–168)
Informal employment proxy	Self-employment + unpaid family work	Q3.6 categories: self_employed, family_unpaid

---

7.10 Instrument Limitations

The following limitations are acknowledged for transparency:

1. **Income data is self-reported and optional**: Monthly income (Q3.9) is not required, and where provided, is subject to recall bias and potential underreporting, a well-documented phenomenon in labour force surveys in developing economies.

2. **Single time-point measurement**: The instrument captures a snapshot of labour force status at the time of interview. Seasonal employment patterns (particularly in agriculture-dependent LGAs) are not captured by a single-round survey.

3. **Skills self-assessment**: Skill proficiency is self-declared, not independently verified. The registry records claimed skills, not assessed competencies.

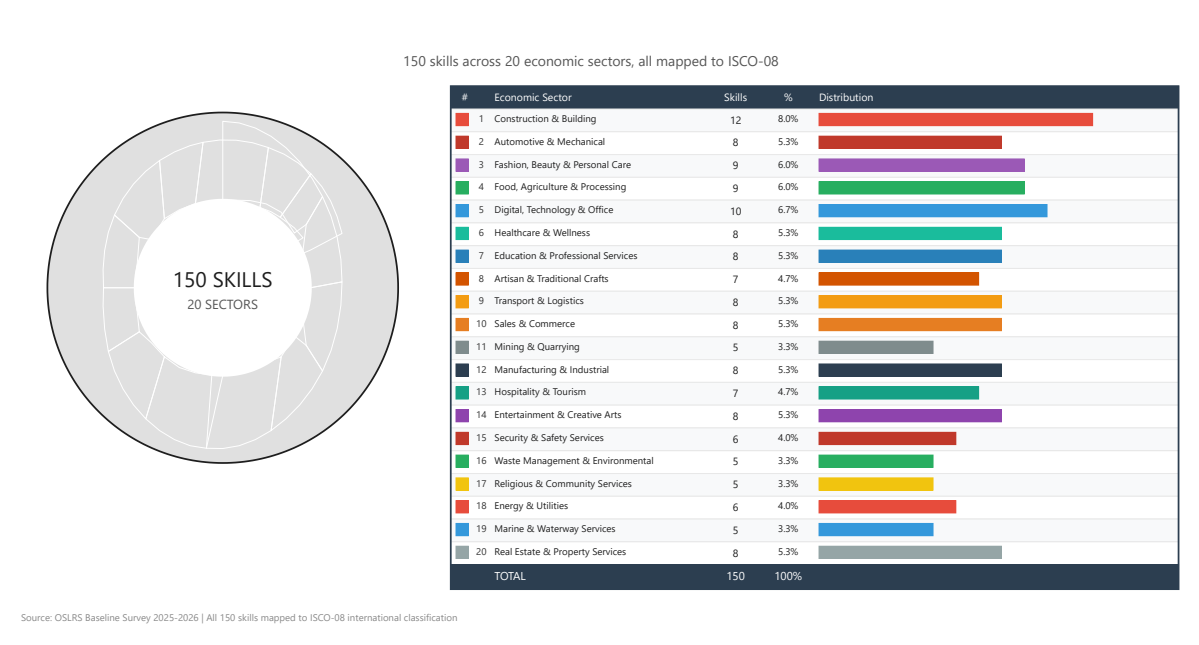
4. **Proxy responses not explicitly captured**: Where an enumerator records a household member's data via a proxy respondent (e.g., a spouse), the instrument does not explicitly flag the response as proxy.

These limitations are standard for large-scale labour force registration exercises and are consistent with the methodological constraints documented in Chapter 6.

---

8. Occupational Skills Taxonomy

Figure 8.1: Occupational Skills Taxonomy, Sector Distribution



150 skills across 20 economic sectors, all mapped to ISCO-08

#	Economic Sector	Skills	%	Distribution
1	Construction & Building	12	8.0%	
2	Automotive & Mechanical	8	5.3%	
3	Fashion, Beauty & Personal Care	9	6.0%	
4	Food, Agriculture & Processing	9	6.0%	
5	Digital, Technology & Office	10	6.7%	
6	Healthcare & Wellness	8	5.3%	
7	Education & Professional Services	8	5.3%	
8	Artisan & Traditional Crafts	7	4.7%	
9	Transport & Logistics	8	5.3%	
10	Sales & Commerce	8	5.3%	
11	Mining & Quarrying	5	3.3%	
12	Manufacturing & Industrial	8	5.3%	
13	Hospitality & Tourism	7	4.7%	
14	Entertainment & Creative Arts	8	5.3%	
15	Security & Safety Services	6	4.0%	
16	Waste Management & Environmental	5	3.3%	
17	Religious & Community Services	5	3.3%	
18	Energy & Utilities	6	4.0%	
19	Marine & Waterway Services	5	3.3%	
20	Real Estate & Property Services	8	5.3%	
	TOTAL	150	100%	

Source: OSLRS Baseline Survey 2025-2026 | All 150 skills mapped to ISCO-08 international classification

8.1 Introduction

This chapter documents the **150-skill occupational taxonomy** developed for the Oyo State Skilled Labour Register, a structured classification of tradeable skills organised across 20 economic sectors and mapped to the **International Standard Classification of Occupations (ISCO-08)**. The taxonomy serves as the core skills inventory for both the survey instrument (Question 5.1) and the public Skills Marketplace search functionality.

8.2 Design Principles

The taxonomy was designed according to five governing principles:

Table 8.1: Design Principles

#	PRINCIPLE	RATIONALE
1	**Comprehensive coverage**	Must capture the full breadth of economic activities in Oyo State, from subsistence farming to emerging digital services
2	**Local relevance**	Skills must be recognisable and meaningful to respondents in Yoruba-speaking Southwest Nigeria, including informal economy terminology
3	**ISCO-08 compatibility**	Mapping to international standards enables comparison with national (NBS) and international (ILO) labour statistics

#	PRINCIPLE	RATIONALE
4	**Manageable enumeration**	150 skills is a balance between comprehensiveness and field usability, enumerators can navigate the full list within 2–3 minutes
5	**Multi-select capability**	Workers in the informal economy typically possess multiple tradeable skills; the taxonomy supports multi-selection

8.3 Development Process

The taxonomy was developed through an iterative process:

Figure 8.2: Occupational Taxonomy Development Process

8.4 Taxonomy Structure, 20 Sectors, 150 Skills

Sector 1: Construction & Building Trades (12 skills)

Table 8.2: Sector 1: Construction & Building Trades (12 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
1	Carpentry/Woodwork	7115	Wood construction, joinery, framework building
2	Plumbing	7126	Pipe fitting, water systems installation and repair
3	Electrical Installation	7411	Building wiring, power distribution, panel installation
4	Welding & Fabrication	7212	Metal joining, gate/grille fabrication, structural welding
5	Masonry/Bricklaying	7112	Block laying, concrete work, stone masonry
6	Painting & Decoration	7131	Building painting, interior/exterior finishing, wallpapering
7	Tiling & Flooring	7122	Ceramic/porcelain tile installation, terrazzo, POP screeding
8	Roofing	7121	Roof sheet installation, truss construction, waterproofing
9	HVAC/Air Conditioning	7127	Air conditioning installation, refrigeration repair, ventilation
10	Solar Installation	7413	Solar panel mounting, inverter wiring, battery systems
11	Aluminum & Glass Fitting	7125	Aluminum windows/doors, glass cutting, curtain wall installation
12	POP/Plaster of Paris Work	7123	Ceiling design, cornice installation, interior moulding

Sector 2: Automotive & Mechanical (8 skills)

Table 8.3: Sector 2: Automotive & Mechanical (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
13	Auto Mechanic	7231	Petrol/diesel engine diagnosis and repair
14	Auto Electrician	7412	Vehicle electrical systems, wiring, diagnostics
15	Panel Beating & Spray Painting	7213	Body repair, dent removal, automotive spray finishing
16	Vulcanizing/Tire Services	7233	Tire repair, balancing, alignment services
17	Motorcycle/Tricycle Repair	7234	Okada/Keke mechanic services
18	Heavy Equipment Operation	8342	Excavator, crane, bulldozer operation
19	Generator Repair	7421	Petrol/diesel generator maintenance and repair
20	Battery/Inverter Technician	7422	Battery reconditioning, inverter installation and repair

Sector 3: Fashion, Beauty & Personal Care (9 skills)

Table 8.4: Sector 3: Fashion, Beauty & Personal Care (9 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
21	Tailoring/Sewing	7531	Custom garment making, alterations, traditional wear
22	Fashion Design	7531	Fashion conceptualisation, pattern making, haute couture
23	Hairdressing/Styling	5141	Hair cutting, styling, braiding, weaving, extensions
24	Barbing	5141	Men's hair cutting, beard grooming, shaving
25	Makeup Artistry	5142	Cosmetic application, bridal makeup, special effects
26	Shoe Making/Cobbling	7536	Footwear manufacturing, shoe repair, custom shoes
27	Bag Making/Leather Craft	7535	Leather goods, handbag production, belt making
28	Jewelry Making	7313	Gold/silver smithing, bead work, costume jewelry
29	Nail Technology	5142	Manicure, pedicure, nail art, gel/acrylic application

Sector 4: Food, Agriculture & Processing (9 skills)

Table 8.5: Sector 4: Food, Agriculture & Processing (9 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
30	Crop Farming	6111	Cultivation of food/cash crops (cassava, maize, yam, cocoa)
31	Livestock/Poultry Farming	6121	Animal husbandry, poultry management, dairy farming
32	Fishery/Aquaculture	6221	Fish farming, pond management, fish harvesting
33	Catering/Event Cooking	5120	Large-scale cooking, event catering, food service
34	Baking & Confectionery	7512	Bread/cake baking, pastry making, confections
35	Food Processing/Preservation	7514	Garri processing, palm oil extraction, food preservation
36	Butchery/Meat Processing	7511	Meat cutting, processing, suya/kilishi preparation
37	Agro-Processing Equipment Operation	8160	Cassava grater, palm oil press, milling machine operation
38	Horticulture/Floriculture	6113	Garden design, flower cultivation, landscaping

Sector 5: Digital, Technology & Office (10 skills)

Table 8.6: Sector 5: Digital, Technology & Office (10 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
39	Software Development	2512	Application programming, mobile apps, web applications
40	Web Design/Development	2513	Website creation, UI/UX design, frontend/backend development
41	Graphic Design	2166	Visual design, branding, print/digital media design
42	Video Editing/Production	2642	Video shooting, editing, post-production, content creation
43	Data Entry/Typing	4132	Data capture, typing, digitisation of records
44	Accounting/Bookkeeping	3313	Financial records, tax computation, business accounting
45	Office Administration	4110	Office management, filing, correspondence, scheduling
46	Computer/Phone Repair	7422	Hardware diagnosis, screen replacement, component repair
47	Social Media Management	2431	Online marketing, content scheduling, digital advertising
48	Digital Marketing/SEO	2431	Search optimisation, online campaigns, analytics

Sector 6: Healthcare & Wellness (8 skills)

Table 8.7: Sector 6: Healthcare & Wellness (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
49	Nursing/Patient Care	3221	Clinical nursing, patient monitoring, bedside care
50	Pharmacy Assistant	3213	Drug dispensing assistance, inventory management
51	Laboratory Technician	3212	Sample collection, basic testing, equipment operation
52	Community Health Worker	3253	Community health education, basic health services
53	Elderly/Child Caregiving	5311	Home care, nanny services, elderly companion care
54	Physiotherapy Assistant	3255	Rehabilitation exercises, massage therapy, patient support
55	Traditional Medicine/Herbalism	3230	Herbal preparation, traditional healing practices
56	Dental Assistant	3251	Dental clinic assistance, sterilisation, patient preparation

Sector 7: Education & Professional Services (8 skills)

Table 8.8: Sector 7: Education & Professional Services (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
57	Teaching/Tutoring	2330	Classroom teaching, private tutoring, lesson facilitation

#	SKILL	ISCO-08 CODE	DESCRIPTION
58	Professional Driving	8322	Commercial vehicle driving (taxi, bus, logistics)
59	Event Planning/Decoration	3332	Event coordination, venue decoration, wedding planning
60	Photography/Videography	3431	Professional photography, event coverage, studio work
61	Professional Cleaning	9112	Commercial/residential cleaning, janitorial services
62	Laundry/Dry Cleaning	9121	Fabric washing, dry cleaning, ironing services
63	Translation/Interpretation	2643	Yoruba-English translation, document translation
64	Legal Clerk/Paralegal	3411	Legal document preparation, court filing, research

Sector 8: Artisan & Traditional Crafts (7 skills)

Table 8.9: Sector 8: Artisan & Traditional Crafts (7 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
65	Furniture Making	7522	Wooden/metal furniture construction, custom cabinetry
66	Upholstery	7534	Sofa/chair padding, fabric covering, furniture restoration
67	Pottery/Ceramics	7314	Clay pot making, ceramic production, decorative pottery
68	Blacksmithing	7221	Iron/steel forging, tool making, metalwork
69	Weaving/Textile Crafts	7318	Aso-oke weaving, adire textile, hand loom operation
70	Sign Writing/Branding	7316	Hand-painted signs, vehicle branding, banner making
71	Calabash/Gourd Carving	7317	Decorative gourd carving, traditional craft production

Sector 9: Transport & Logistics (8 skills)

Table 8.10: Sector 9: Transport & Logistics (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
72	Commercial Bus/Taxi Driving	8322	Danfo/BRT/taxi operation within and between cities
73	Motorcycle Taxi (Okada) Riding	8321	Commercial motorcycle transport services
74	Tricycle (Keke) Operation	8321	Keke NAPEP/tricycle passenger/cargo transport
75	Truck/Haulage Driving	8332	Long-distance cargo transport, haulage services
76	Dispatch Riding/Courier	4412	Package delivery, document courier, express dispatch
77	Warehouse Management	4321	Inventory control, goods storage, dispatch coordination
78	Freight/Logistics Coordination	3331	Cargo booking, shipment tracking, supply chain coordination

#	SKILL	ISCO-08 CODE	DESCRIPTION
79	Forklift Operation	8344	Warehouse forklift operation, pallet movement

Sector 10: Sales & Commerce (8 skills)

Table 8.11: Sector 10: Sales & Commerce (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
80	Trading/General Commerce	5221	Market trading, wholesale/retail, goods distribution
81	Agrochemical Sales	5223	Fertilizer, pesticide, herbicide retail and advisory
82	Pharmaceutical/Medical Sales	5223	OTC drug sales, medical supplies, pharmaceutical retail
83	Building Materials Sales	5223	Cement, roofing, plumbing supplies retail
84	Electronics/Phone Sales	5223	Consumer electronics, mobile phone, accessories retail
85	Provisions/FMCG Distribution	5221	Fast-moving consumer goods distribution and retail
86	Fuel/Gas Retailing	5245	Petrol station operation, cooking gas retail
87	Auto Parts Sales	5223	Vehicle spare parts retail and distribution

Sector 11: Mining & Quarrying (5 skills)

Table 8.12: Sector 11: Mining & Quarrying (5 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
88	Quarrying/Stone Cutting	8111	Granite/laterite quarrying, stone crushing
89	Sand Mining/Dredging	8113	River sand extraction, dredging operations
90	Gold/Mineral Artisan Mining	8114	Small-scale mineral extraction, panning
91	Clay/Kaolin Extraction	8113	Clay mining for pottery, brick making, industrial use
92	Gravel/Aggregate Processing	8112	Gravel sorting, aggregate preparation for construction

Sector 12: Manufacturing & Industrial (8 skills)

Table 8.13: Sector 12: Manufacturing & Industrial (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
93	Soap/Detergent Making	8131	Local soap production, liquid detergent manufacturing
94	Sachet/Bottled Water Production	8160	Pure water sachet packaging, bottled water processing
95	Block/Brick Making	8114	Concrete block moulding, laterite brick production

#	SKILL	ISCO-08 CODE	DESCRIPTION
96	Paint Manufacturing	8131	Local paint production, colour mixing
97	Plastic/Rubber Recycling	8142	Plastic recycling, rubber reclamation, pellet production
98	Textile/Garment Factory Work	8153	Industrial sewing, garment assembly, textile machine operation
99	Metal Fabrication/Foundry	7211	Industrial metal casting, foundry operation
100	Paper/Printing Production	8143	Print press operation, binding, paper product manufacturing

Sector 13: Hospitality & Tourism (7 skills)

Table 8.14: Sector 13: Hospitality & Tourism (7 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
101	Hotel/Guest House Management	1411	Lodging facility operation, front desk management
102	Restaurant/Bar Management	1412	Food service establishment operation
103	Bartending/Mixology	5132	Beverage preparation, cocktail mixing, bar service
104	Tour Guide Services	5113	Cultural/historical tour facilitation, heritage interpretation
105	Event Centre Management	1439	Event venue operation, booking coordination
106	Short-Let/Apartment Hosting	1411	Vacation rental management, Airbnb-style hosting
107	Chef/Professional Cooking	3434	Restaurant cooking, menu development, kitchen management

Sector 14: Entertainment & Creative Arts (8 skills)

Table 8.15: Sector 14: Entertainment & Creative Arts (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
108	Music Production/DJ	2652	Beat making, sound mixing, DJ services
109	Acting/Theatre Performance	2655	Stage/screen acting, Nollywood, theatrical performance
110	Comedy/MC/Entertainment	2655	Event hosting, stand-up comedy, entertainment services
111	Dance/Choreography	2653	Dance performance, choreography, dance instruction
112	Sound Engineering	3521	Audio recording, mixing, live sound reinforcement
113	Musical Instrument Playing	2652	Live performance, session musician, music instruction
114	Fine Art/Painting	2651	Canvas painting, murals, artistic illustration
115	Animation/Motion Graphics	2166	Digital animation, motion graphics, visual effects

Sector 15: Security & Safety Services (6 skills)

Table 8.16: Sector 15: Security & Safety Services (6 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
116	Private Security Guard	5414	Premises security, access control, patrol services
117	CCTV/Surveillance Installation	7421	Camera installation, monitoring system setup, maintenance
118	Fire Safety/Extinguisher Services	5411	Fire safety equipment, extinguisher servicing, fire marshal
119	Locksmith Services	7222	Lock installation/repair, key cutting, safe opening
120	Dog Training/K9 Handler	5164	Guard dog training, canine security services
121	Traffic/Crowd Management	5414	Event crowd control, traffic direction, safety marshalling

Sector 16: Waste Management & Environmental Services (5 skills)

Table 8.17: Sector 16: Waste Management & Environmental Services (5 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
122	Waste Collection/Disposal	9611	Residential/commercial waste collection, landfill operations
123	Recycling/Scrap Dealing	9612	Material sorting, scrap metal trade, recyclable collection
124	Fumigation/Pest Control	7544	Insect/rodent control, building fumigation services
125	Sewage/Drainage Services	7126	Septic tank emptying, drainage cleaning, soak-away construction
126	Environmental Remediation	3257	Pollution cleanup, erosion control, environmental restoration

Sector 17: Religious & Community Services (5 skills)

Table 8.18: Sector 17: Religious & Community Services (5 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
127	Religious Leadership/Clergy	2636	Imam, Pastor, traditional religious leadership
128	Quranic/Islamic Teaching	2342	Arabic instruction, Quranic recitation teaching
129	Church Music/Choir Direction	2652	Worship music leadership, choir training and direction
130	Community Development Work	3412	NGO field work, community mobilisation, social services
131	Counselling Services	2634	Personal/family counselling, conflict resolution

Sector 18: Energy & Utilities (6 skills)

Table 8.19: Sector 18: Energy & Utilities (6 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
132	Borehole Drilling	8113	Water well drilling, pump installation
133	Water Treatment/Purification	3132	Water purification, treatment plant operation
134	Electrical Power Line Work	7413	Power line installation, electrical pole maintenance
135	Gas Pipeline Fitting	7126	Cooking gas pipe installation, LPG system fitting
136	Diesel/Petrol Engine Servicing	7233	Industrial engine maintenance, fuel system servicing
137	Renewable Energy Technician	7413	Wind/solar/biogas system installation and maintenance

Sector 19: Marine & Waterway Services (5 skills)

Table 8.20: Sector 19: Marine & Waterway Services (5 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
138	Boat Building/Repair	7521	Wooden/fiberglass boat construction and repair
139	Fishing (River/Lake)	6222	Inland waterway fishing, net fishing, trap fishing
140	Canoe/Boat Operation	8350	Passenger/cargo river transport, ferry operation
141	Fish Smoking/Drying	7514	Fish preservation, smoking kiln operation
142	Pond/Dam Construction	7114	Aquaculture pond construction, dam building

Sector 20: Real Estate & Property Services (8 skills)

Table 8.21: Sector 20: Real Estate & Property Services (8 skills)

#	SKILL	ISCO-08 CODE	DESCRIPTION
143	Property/Estate Agency	3334	Property sales/rental, client-landlord intermediation
144	Land Surveying	2165	Land measurement, boundary demarcation, survey mapping
145	Building/Quantity Surveying	2142	Construction cost estimation, material quantification
146	Architecture/Draughtsmanship	2161	Building plan drawing, architectural design assistance
147	Interior Design/Decoration	3432	Interior space planning, furnishing, aesthetic design
148	Facility Management	1219	Building maintenance coordination, property upkeep

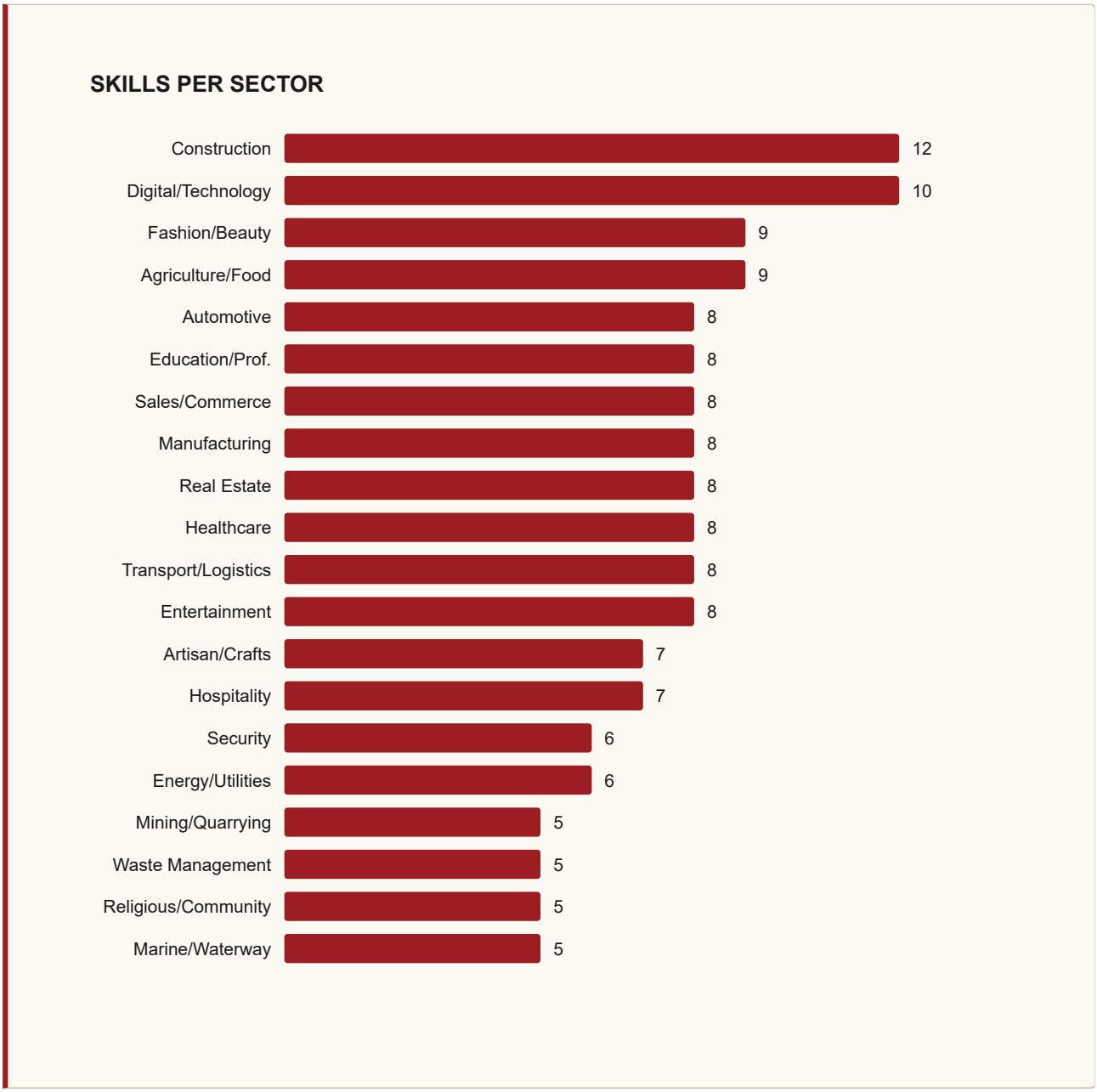
#	SKILL	ISCO-08 CODE	DESCRIPTION
149	Swimming Pool Construction/Maintenance	7119	Pool excavation, tiling, water treatment, maintenance
150	Pest-Proof Storage Construction	7119	Grain/produce storage facility construction

8.5 Summary Statistics

Table 8.22: Summary Statistics

METRIC	VALUE
Total skills	150
Total sectors	20
ISCO-08 major groups covered	8 of 10 (Groups 1–9, excluding Group 0: Armed Forces)
Mean skills per sector	7.5
Largest sector	Construction & Building Trades (12 skills)
Smallest sectors	Mining (5), Waste Management (5), Religious Services (5), Marine (5)

Figure 8.3: Skills Per Sector



8.6 ISCO-08 Mapping Summary

The taxonomy maps to the following ISCO-08 major groups:

Table 8.23: ISCO-08 Mapping Summary

ISCO-08 MAJOR GROUP	SKILLS MAPPED	% OF TAXONOMY
7: Craft and Related Trades Workers	52	34.7%
5: Service and Sales Workers	24	16.0%
8: Plant and Machine Operators, Assemblers	18	12.0%

ISCO-08 MAJOR GROUP	SKILLS MAPPED	% OF TAXONOMY
2: Professionals	17	11.3%
3: Technicians and Associate Professionals	16	10.7%
6: Skilled Agricultural, Forestry and Fishery Workers	8	5.3%
9: Elementary Occupations	7	4.7%
1: Managers	4	2.7%
4: Clerical Support Workers	4	2.7%
Total	**150**	**100%**

Observation: The concentration in ISCO Group 7 (Craft and Related Trades Workers) at 34.7% reflects the artisan-heavy economic structure of Oyo State, where craft trades constitute the backbone of the informal economy. The distribution across all nine civilian ISCO major groups confirms comprehensive coverage of the labour market.

8.7 Validation Exercise Results

During the validation exercise (n=330, Chapter 12), the taxonomy was tested in field conditions:

Table 8.24: Validation Exercise Results

METRIC	RESULT
Skills utilised (at least 1 selection)	89 of 150 (59.3%)
Skills with "Other" free text additions	12 unique free-text skills identified
Mean selection time per respondent	48 seconds
Respondents selecting "Other skills" (Q5.2)	31 of 330 (9.4%)
Enumerator navigation difficulty reported	0 (sector-grouped display effective)

The 59.3% utilisation rate from a 330-person validation sample is expected to approach 90%+ during the full-scale enumeration (23,760+ registrants), as the broader population will exhibit greater skills diversity.

The 12 free-text "Other" skills identified during validation have been evaluated for potential inclusion in future taxonomy updates.

8.8 Chapter Summary

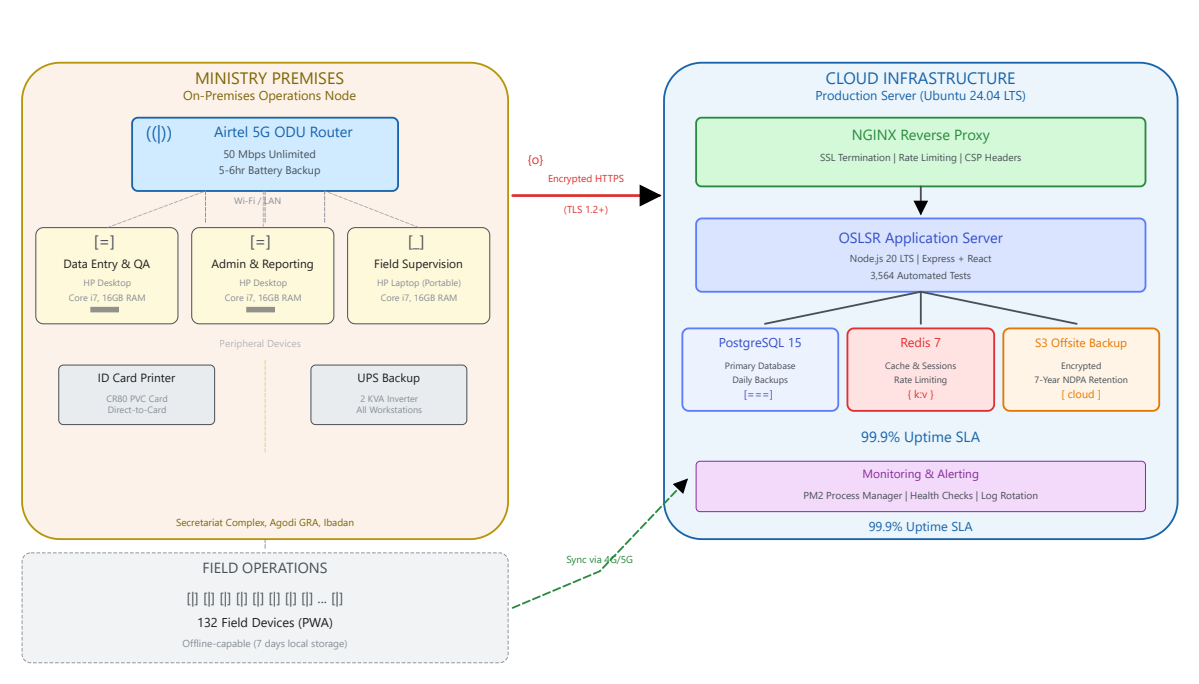
The 150-skill occupational taxonomy provides the OSLSR with a **granular, internationally-compatible skills classification** that captures the full breadth of Oyo State's formal and informal economy. The taxonomy's ISCO-08 mapping ensures that OSLSR data can be compared with national (NBS) and international (ILO) labour statistics, while its local relevance and sector-grouped presentation ensure field usability.

The complete taxonomy is reproduced in **Appendix C** for reference.

---

9. Data Centre Establishment

Figure 9.1: Hybrid Data Centre Architecture



9.1 Introduction

Deliverable 1 of the engagement Terms of Reference requires the **Establishment of a Data Centre** for the Oyo State Skilled Labour Register. This chapter documents the design rationale, procurement, configuration, and operational status of the Data Centre infrastructure, which employs a hybrid architecture combining on-premises computing resources with cloud-hosted production services.

The Data Centre was established at the premises of the **Ministry of Trade, Industry, Investment and Cooperatives**, ensuring physical proximity to the commissioning authority, facilitating Ministry oversight, and establishing data sovereignty within a government-controlled environment.

Deliverable Status: COMPLETE ✓

9.2 Design Rationale

9.2.1 Hybrid Architecture Decision

The Data Centre adopts a **hybrid architecture**, combining on-premises physical infrastructure with cloud-hosted services, to address the dual requirements of operational resilience and cost efficiency. This design was selected based on the following considerations:

Table 9.1: Hybrid Architecture Decision

FACTOR	ON-PREMISES ONLY	CLOUD ONLY	HYBRID (SELECTED)
Data sovereignty (government premises)	✓	✗	✓
Power continuity (5G ODU backup)	✓	N/A	✓
24/7 application availability	✗	✓	✓
Disaster recovery	✗	✓	✓
Capital expenditure	High	Low	Moderate
Operational expenditure	Low	Moderate	Moderate
Scalability	Limited	High	High
Physical access for training	✓	✗	✓

The hybrid model ensures that:

- **Operational activities** (data entry, quality assurance, training, administrative functions) are performed on-premises at Ministry premises, under direct government oversight
- **Production application hosting** (the OSLSR platform, database, and fraud detection engine) runs on cloud infrastructure with enterprise-grade availability, automated backups, and disaster recovery
- **Data synchronisation** between on-premises terminals and the cloud platform occurs in real-time via the dedicated broadband connection

---

9.3 On-Premises Operations Node

9.3.1 Hardware Specifications

The on-premises node comprises three (3) enterprise-grade HP workstations, procured and configured for specific operational functions:

Table 9.2: Hardware Specifications

UNIT	TYPE	SPECIFICATIONS	ASSIGNED FUNCTION
Workstation 1	HP Desktop	Intel Core i7 Processor, 16 GB RAM, 512 GB HDD, Full HD Display	Data Entry & Quality Assurance Terminal, Primary data entry station for Data Entry Clerks; quality assurance review of flagged submissions
Workstation 2	HP Desktop	Intel Core i7 Processor, 16 GB RAM, 512 GB HDD, Full HD Display	Administrative Operations & Reporting Terminal, System administration, report generation, supervisory monitoring, and dashboard analytics
Workstation 3	HP Laptop	Intel Core i7 Processor, 16 GB RAM, 512 GB HDD, Integrated Display	Field Supervision & Mobile Operations Unit, Portable unit for field visits, on-site training delivery, demonstration sessions, and mobile supervisory functions

9.3.2 Operating Environment

All workstations are configured with:

- Modern web browser (Google Chrome, latest stable release) for OSLSR platform access
- Operating system security updates applied
- User accounts with role-appropriate platform access credentials
- Local backup of essential reference documents and training materials

9.3.3 Physical Security

The Data Centre is located within the secured premises of the Ministry of Trade, Industry, Investment and Cooperatives, benefiting from:

- Ministry building security (access control, security personnel)
- Designated workspace for Data Centre equipment
- Equipment inventory and asset tagging for accountability

9.4 Network Infrastructure

9.4.1 Broadband Connectivity

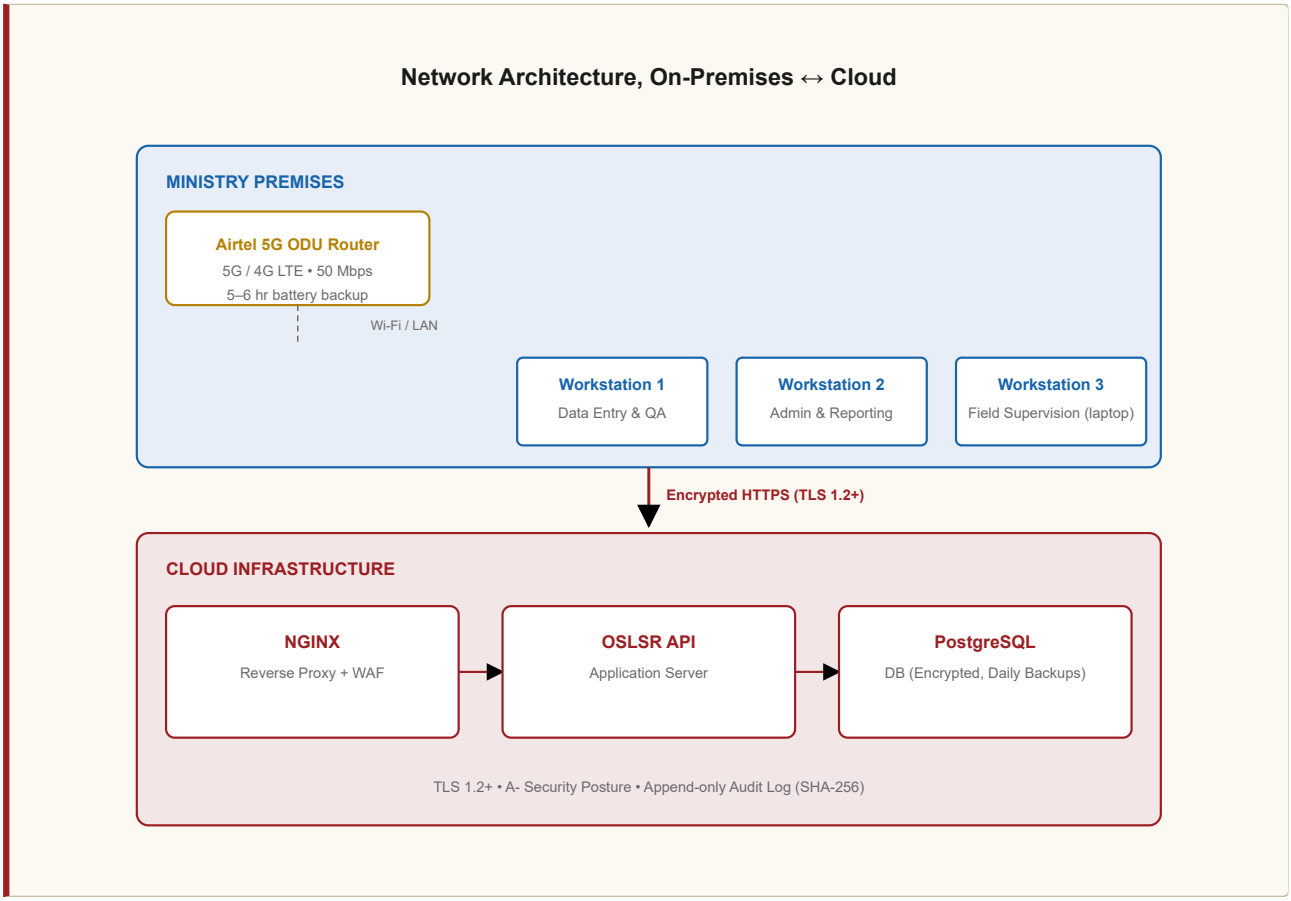
Network connectivity is provided via the **Airtel SmartConnect 5G Outdoor Unit (ODU) Router**, a next-generation broadband solution specifically designed for Nigerian operating environments.

Table 9.3: Broadband Connectivity

SPECIFICATION	DETAIL
Device	Airtel SmartConnect 5G ODU Router
Network	5G (primary) with automatic 4G LTE fallback
Data Plan	Unlimited monthly data at speeds up to 50 Mbps
Monthly Subscription	₦25,000
Mounting	Outdoor-mounted for optimised signal reception
Device Connections	Up to 64 simultaneous devices
Power Backup	Built-in battery providing 5–6 hours of operation during power outages
Parental/Usage Controls	Configurable access and browsing controls

9.4.2 Network Architecture

Figure 9.2: Network Architecture, On-Premises ↔ Cloud



9.4.3 Connectivity Resilience

The network infrastructure incorporates the following resilience measures:

1. **5G/4G Automatic Fallback**: The Airtel ODU automatically switches between 5G and 4G LTE networks based on signal availability, ensuring continuous connectivity even in areas where 5G coverage is intermittent
2. **Built-in Power Backup**: The ODU's integrated battery provides 5–6 hours of operation during power outages, a critical provision given the power supply challenges typical of Nigerian government facilities
3. **Offline-Capable Applications**: The OSLSR platform's Progressive Web Application (PWA) architecture enables field data collection to continue for up to 7 days without network connectivity, with automatic synchronisation upon restoration
4. **Cloud Infrastructure Redundancy**: The production server operates on enterprise infrastructure with built-in redundancy, automated failover, and continuous monitoring

---

9.5 Cloud Infrastructure

9.5.1 Production Server Specifications

Table 9.4: Production Server Specifications

COMPONENT	SPECIFICATION
Operating System	Ubuntu 24.04 LTS (Long-Term Support)
Application Runtime	Node.js 20 LTS
Web Server / Reverse Proxy	NGINX (SSL termination, rate limiting, static asset serving)
Database	PostgreSQL 15 (encrypted at rest, daily automated backups)
Cache / Rate Limiting	Redis 7 (with AOF persistence)
Job Queue	BullMQ (fraud detection, backup orchestration, email delivery)
SSL Certificate	TLS 1.2+ (HTTPS enforced on all endpoints)
Domain	Configured and operational

9.5.2 Backup & Disaster Recovery

Table 9.5: Backup & Disaster Recovery

PARAMETER	CONFIGURATION
Backup Frequency	Daily automated backups (2:00 AM WAT)
Backup Method	PostgreSQL `pg_dump` (full database export)
Offsite Storage	S3-compatible encrypted storage
Retention Policy	7-day daily retention + 7-year monthly archives (NDPA compliance)
Restore Capability	Documented restore procedure; tested and validated
Recovery Time Objective (RTO)	< 1 hour from latest backup
Recovery Point Objective (RPO)	Maximum 24 hours (daily backup interval)

9.5.3 Monitoring & Alerting

The cloud infrastructure is monitored through the OSLSR System Health Monitoring module (detailed in Chapter 10), which tracks:

- CPU utilisation, memory usage, and disk consumption
- Database query performance (p95 latency)
- Application response times
- Job queue depth and processing rates

- SSL certificate expiry
- Automated alert notifications when thresholds are exceeded

9.6 Cost Efficiency

The hybrid Data Centre architecture delivers significant cost efficiency compared to traditional government IT procurement models:

Table 9.6: Cost Efficiency

COST COMPONENT	DESCRIPTION	FREQUENCY
Hardware (one-time)	3× HP Core i7 workstations	Capital
Network device (one-time)	Airtel 5G ODU Router	Capital
Broadband subscription	Unlimited 50 Mbps data plan	₦25,000/month
Cloud hosting	Production server with backup storage	Monthly
Total Monthly Operating Cost	Broadband + Cloud hosting	**Recurring**

The operating cost model ensures sustainability beyond the initial engagement period. The Ministry can continue operating the Data Centre and platform with minimal recurring expenditure, broadband connectivity and cloud hosting, without requiring additional capital investment.

9.7 Deliverable Acceptance Criteria

Table 9.7: Deliverable Acceptance Criteria

#	CRITERION	STATUS
1	Physical hardware procured and installed at Ministry premises	✓ Complete
2	Network connectivity operational with adequate bandwidth	✓ Complete
3	Cloud infrastructure deployed and accessible from on-premises terminals	✓ Complete
4	OSLSR platform accessible from all three workstations	✓ Complete
5	Automated backup system operational with offsite storage	✓ Complete
6	System monitoring and alerting configured	✓ Complete
7	Data Centre operational and ready for production use	✓ Complete

Deliverable 1, Establishment of Data Centre: COMPLETE ✓

---

10. Registry Platform Development and Capabilities

Figure 10.1: OSLSR Platform, System Architecture Overview

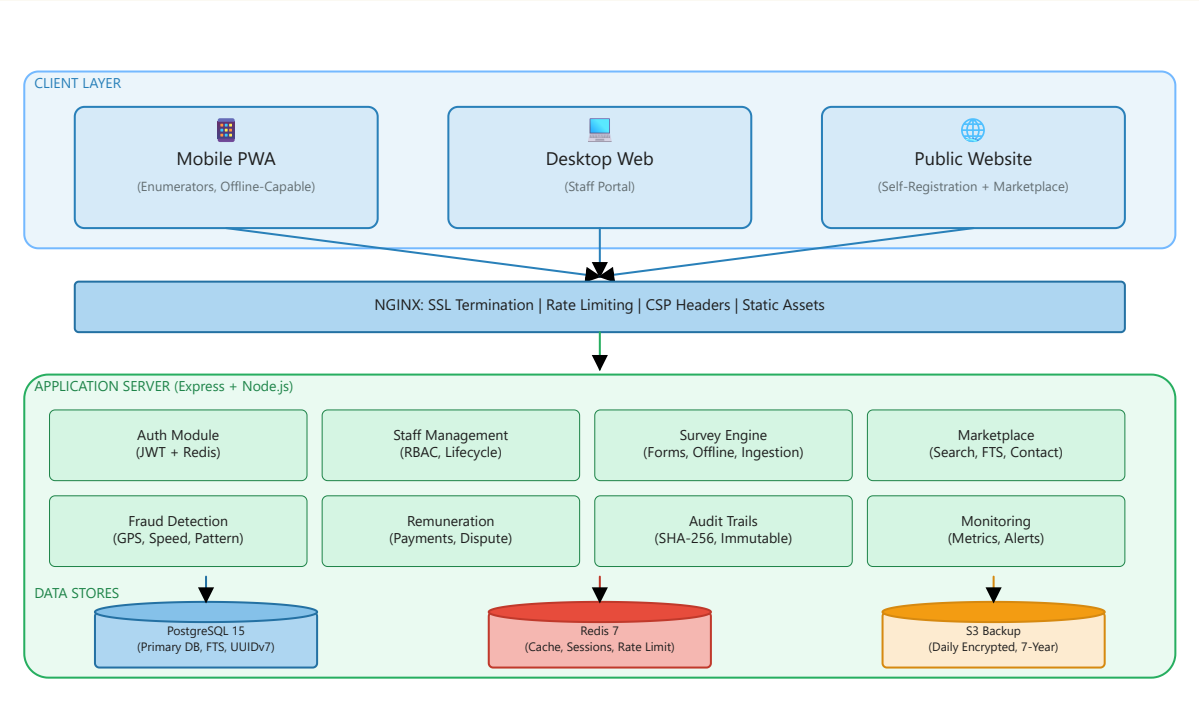
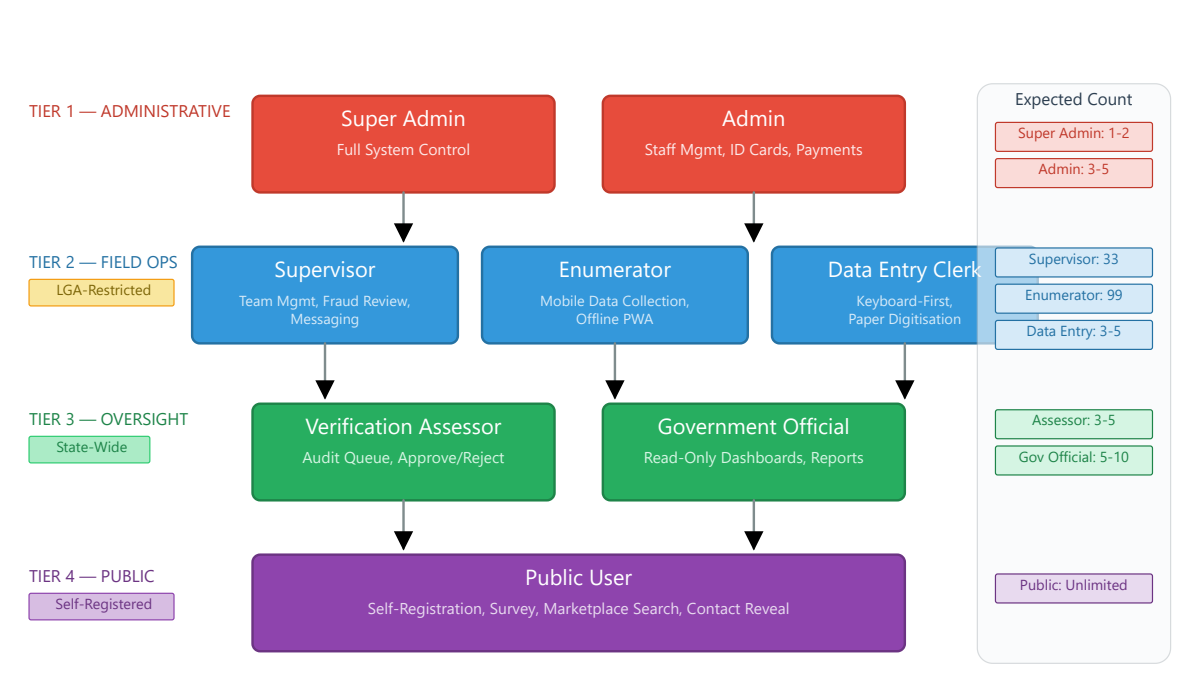


Figure 10.2: OSLSR User Role Hierarchy (Five Operational Roles + Public Tier)



10.1 Introduction

This chapter documents the design, development, and deployment of the **Oyo State Labour & Skills Registry (OSLSR)** digital platform, the core technology infrastructure upon which the State Labour Register operates. The platform was purpose-built to address the specific requirements of workforce enumeration in a developing economy context, incorporating offline-capable mobile data collection, multi-channel registration, automated fraud detection, and a public-facing skills marketplace.

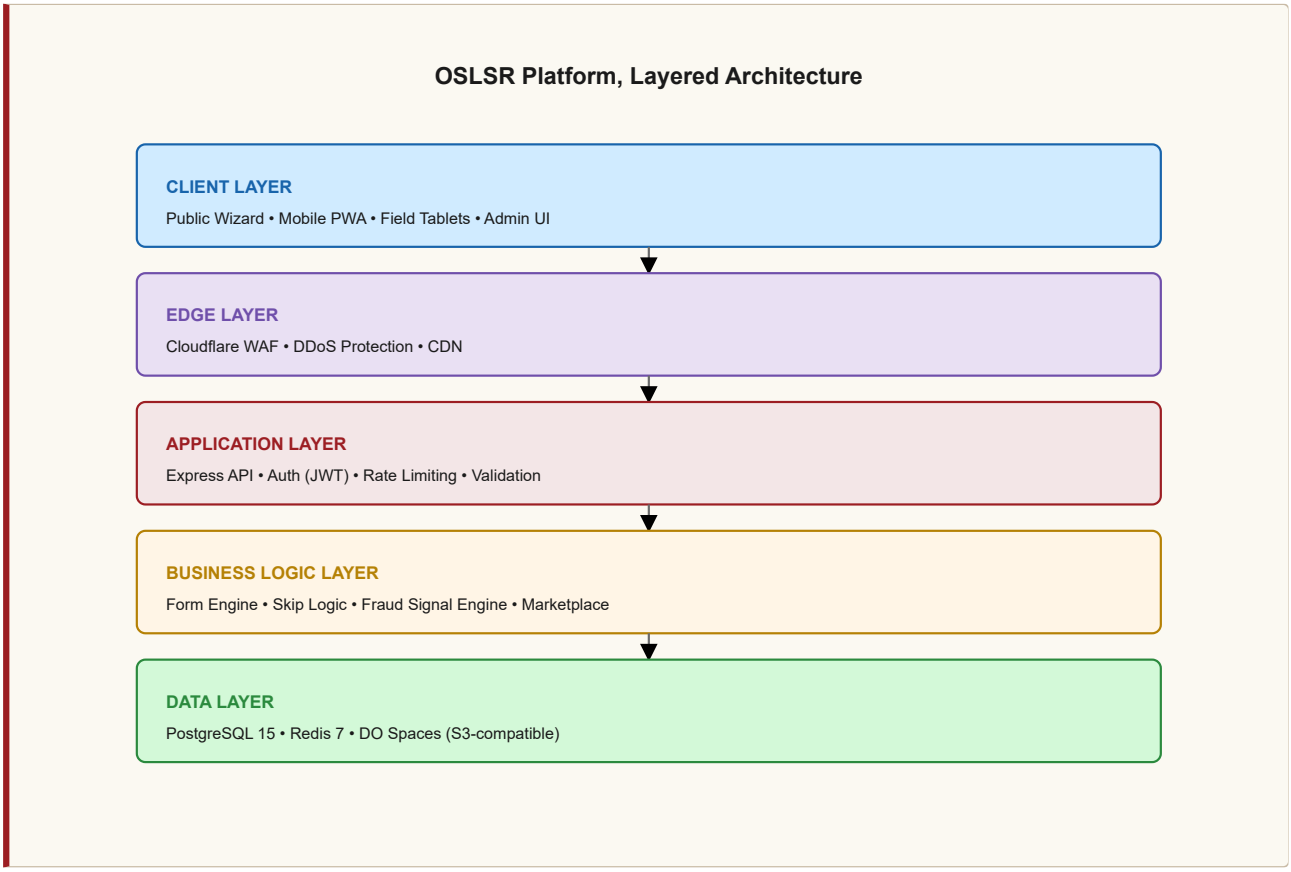
The platform is **operational and deployed**, accessible to authorised users from the on-premises Data Centre workstations and via any internet-connected device.

10.2 Platform Architecture

10.2.1 Architecture Overview

The OSLSR platform employs a **modular monolith architecture**, a proven design pattern that combines the simplicity of a single deployable unit with the maintainability of modular internal boundaries. This architecture was selected over microservices based on the project's scale requirements (200+ concurrent staff users, 1,000 concurrent public users) and the operational simplicity demanded by a single-VPS deployment.

Figure 10.3: OSLSR Platform, Layered Architecture



10.2.2 Technology Stack

Table 10.1: OSLSR Platform, Layered Architecture

LAYER	TECHNOLOGY	VERSION	PURPOSE
Frontend	React	18.3	User interface framework
	Vite	6.x	Build tooling with hot module replacement
	Tailwind CSS	v4	Utility-first styling framework
	shadcn/ui	Latest	Accessible component library (Radix UI)
	TanStack Query	Latest	Server state management and caching
	React Hook Form + Zod	Latest	Form handling with runtime validation
	React Router	v7	Client-side routing with role isolation
Backend	Node.js	20 LTS	Server runtime
	Express.js	Latest	REST API framework
	Drizzle ORM	1.x	TypeScript-first database ORM
	Pino	9.x	Structured JSON logging
Database	PostgreSQL	15	Relational database with JSONB and PostGIS
Cache/Queue	Redis	7	Caching, rate limiting, session management
	BullMQ	Latest	Asynchronous job queue processing
Security	Helmet	Latest	Security headers including CSP
	bcrypt	6.x	Password hashing (12 salt rounds)
	JWT	Latest	Stateless authentication tokens
Validation	Zod	3.x	Runtime type validation (shared frontend/backend)
Identity	uuidv7	Latest	Time-ordered universally unique identifiers

10.3 Platform Capabilities

The following table summarises the platform's operational capabilities:

Table 10.2: Platform Capabilities

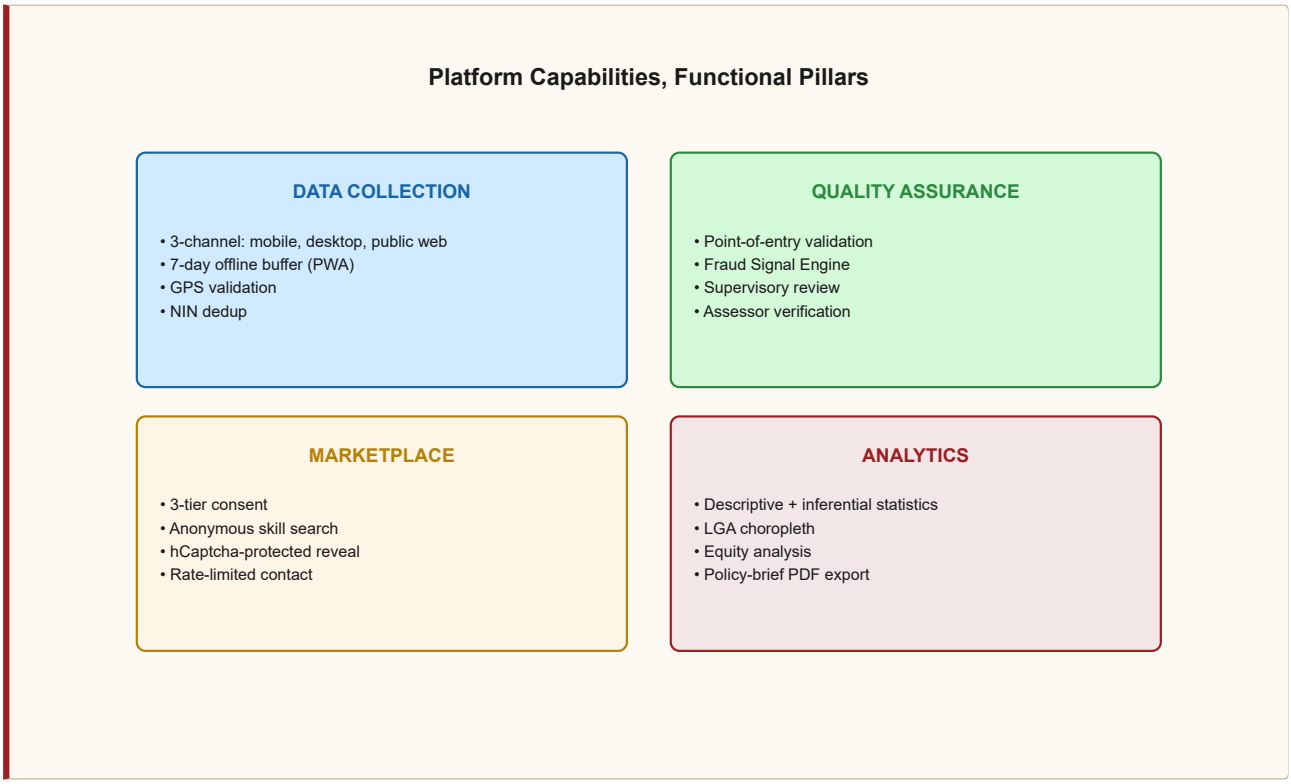
#	CAPABILITY	DESCRIPTION	STATUS
1	**Multi-Channel Data Collection**	Field enumeration (mobile PWA), desktop data entry, and public web self-registration, three distinct channels feeding a unified registry	✓ Operational

#	CAPABILITY	DESCRIPTION	STATUS
2	**Offline-Capable PWA**	Progressive Web Application with Service Worker caching and IndexedDB storage enabling 7-day offline data collection with automatic synchronisation	✓ Operational
3	**Native Form Renderer**	One-question-per-screen survey interface with skip logic, real-time validation, and auto-save; supports the full 150-skill taxonomy	✓ Operational
4	**National Identity Verification**	NIN (National Identity Number) validation using Modulus 11 checksum at point of entry, with global uniqueness enforcement across all submission channels	✓ Operational
5	**Role-Based Access Control**	Five operational user roles plus a Public tier with granular permissions: Super Admin, Supervisor, Assessor, Enumerator, Data Entry Clerk, and a Public tier for self-registration	✓ Operational
6	**Context-Aware Fraud Detection**	Automated detection of GPS clustering (multiple submissions from same location), speed-run submissions (implausibly fast completion), and straight-lining (repetitive response patterns); configurable thresholds	✓ Operational
7	**Public Skills Marketplace**	Searchable public directory of anonymised worker profiles with government verification badges, full-text search, trade/LGA filtering, and CAPTCHA-protected contact reveal	✓ Operational
8	**Staff Remuneration Management**	Bulk payment recording with bank reference/receipt upload, payment history, dispute mechanism (report → dispute → resolution → acknowledgement), and immutable records	✓ Operational
9	**Immutable Audit Trails**	SHA-256 hash-chained, append-only audit log recording all user actions, PII access, and administrative operations; tamper-proof with database trigger protection; NDPA-aligned 7-year retention	✓ Operational
10	**System Health Monitoring**	Real-time dashboard tracking CPU, RAM, disk utilisation, database performance (p95 latency), job queue depth, and email delivery status; configurable alert thresholds with email notifications	✓ Operational
11	**Automated Backup System**	Daily encrypted database backups to offsite storage via scheduled job (2:00 AM WAT); 7-day daily retention + 7-year monthly archives; tested restore procedure	✓ Operational
12	**Data Export & Reporting**	Role-authorised CSV and PDF export of registry data with audit logging; filtered exports by LGA, date range, occupation, and status	✓ Operational
13	**In-App Team Messaging**	Real-time messaging system for field team coordination between Supervisors and Enumerators; broadcast capability; message audit trail	✓ Operational
14	**ID Card Generation**	Digital ID card generation with photo, QR code for public verification, and role-specific design; PDF output for printing	✓ Operational
15	**Super Admin View-As**	Debugging and oversight tool enabling Super Administrators to view the platform as any other role, with read-only enforcement and full audit trail	✓ Operational

10.4 User Role Architecture

The platform enforces strict role-based access control (RBAC) across eight distinct user roles. Each role has precisely defined permissions, ensuring that users can only access data and functions appropriate to their designated responsibilities.

Figure 10.4: Platform Capabilities, Functional Pillars



10.4.1 Access Control Matrix

Table 10.3: Platform Capabilities, Functional Pillars

FUNCTION	SUPER ADMIN	ADMIN	SUPERVISOR	ENUMERATOR	CLERK	ASSESSOR	OFFICIAL	PUBLIC
System Configuration	✓							
Staff Management	✓	✓						
Form Management	✓	✓						
Survey Submission				✓	✓			✓
Team Supervision			✓					
Fraud Alert Review	✓		✓					
Submission Audit						✓		
Payment Management	✓	✓*						
Dashboard Analytics	✓	✓	✓†	✓†			✓	

FUNCTION	SUPER ADMIN	ADMIN	SUPERVISOR	ENUMERATOR	CLERK	ASSESSOR	OFFICIAL	PUBLIC
PII Data Access	✓	✓	✓†			✓	✓	
Data Export	✓	✓					✓	
Marketplace Search	✓							✓
Contact Reveal								✓‡
View-As	✓							

** Admin cannot record self-payments † LGA-restricted scope only ‡ CAPTCHA + rate limited (50/24hr)*

10.5 Quality Assurance

10.5.1 Automated Testing

The OSLSR platform was subjected to comprehensive automated quality assurance testing:

Table 10.4: Automated Testing

TEST CATEGORY	COUNT	PASS RATE	COVERAGE
API Unit & Integration Tests	1,471	100%	All controllers, services, and routes
Frontend Component & Hook Tests	2,093	100%	All pages, components, and hooks
Total Automated Tests	**3,564**	**100%**	**Full platform coverage**

10.5.2 Security Assessment

A comprehensive security assessment was conducted against the OWASP Top 10 framework:

Table 10.5: Security Assessment

OWASP CATEGORY	ASSESSMENT	STATUS
A01: Broken Access Control	Auth middleware, RBAC, LGA scope enforcement	✓ Secure
A02: Cryptographic Failures	bcrypt 12 rounds, JWT HS256, httpOnly cookies	✓ Secure
A03: Injection	Parameterised queries (Drizzle ORM), no raw HTML rendering	✓ Secure
A04: Insecure Design	Multi-tier rate limiting, CAPTCHA, file validation	✓ Secure
A05: Security Misconfiguration	Helmet CSP, security headers, hardened configuration	✓ Secure
A06: Vulnerable Components	Dependency audit, CVE remediation, CI gate	✓ Secure

OWASP CATEGORY	ASSESSMENT	STATUS
A07: Authentication Failures	15-min tokens, Redis sessions, JTI blacklist	✓ Secure
A08: Data Integrity Failures	Immutable audit logs, hash chaining, DB triggers	✓ Secure
A09: Logging & Monitoring	AuditService, PII access logging, prom-client metrics	✓ Secure
A10: SSRF	No user-provided URLs for server-side fetching	✓ Secure

All ten OWASP categories at A- security posture (state-government-grade), including one category (A05) that was identified and remediated during the assessment period.

10.6 Platform Screenshots

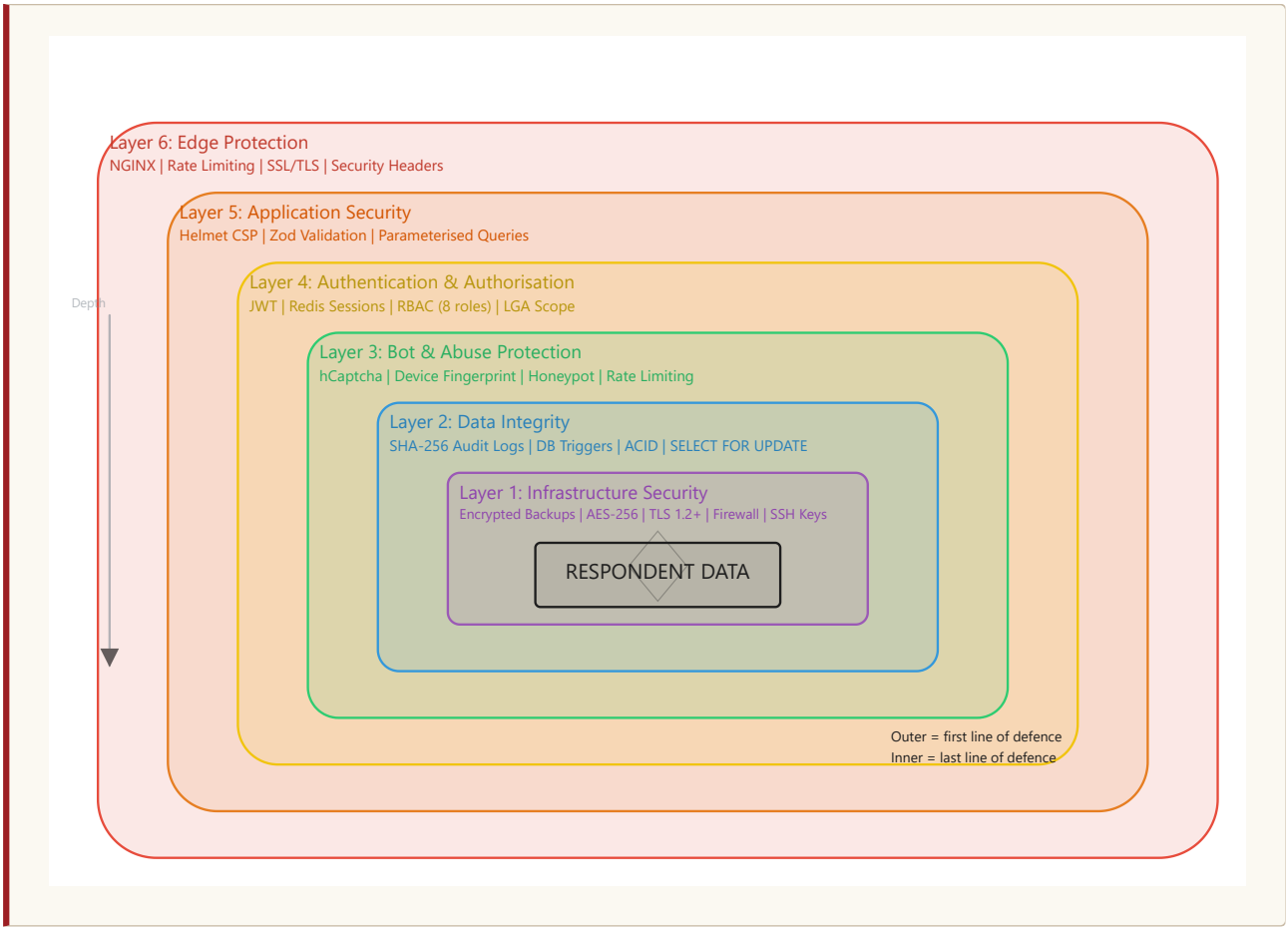
[Refer to Appendix J: Plates 6–12 for platform screenshots demonstrating key interfaces]

- **Plate 6**: Login and Authentication Screen
- **Plate 7**: Super Administrator Dashboard with real-time analytics
- **Plate 8**: Survey Form Renderer (mobile view), one-question-per-screen interface
- **Plate 9**: Public Skills Marketplace search interface
- **Plate 10**: Fraud Detection dashboard with flagged submission alerts
- **Plate 11**: System Health Monitoring with CPU, RAM, and latency metrics
- **Plate 12**: Staff Remuneration Management, bulk payment recording

---

11. Security Architecture and Data Protection

Figure 11.1: Defence-in-Depth Security Architecture



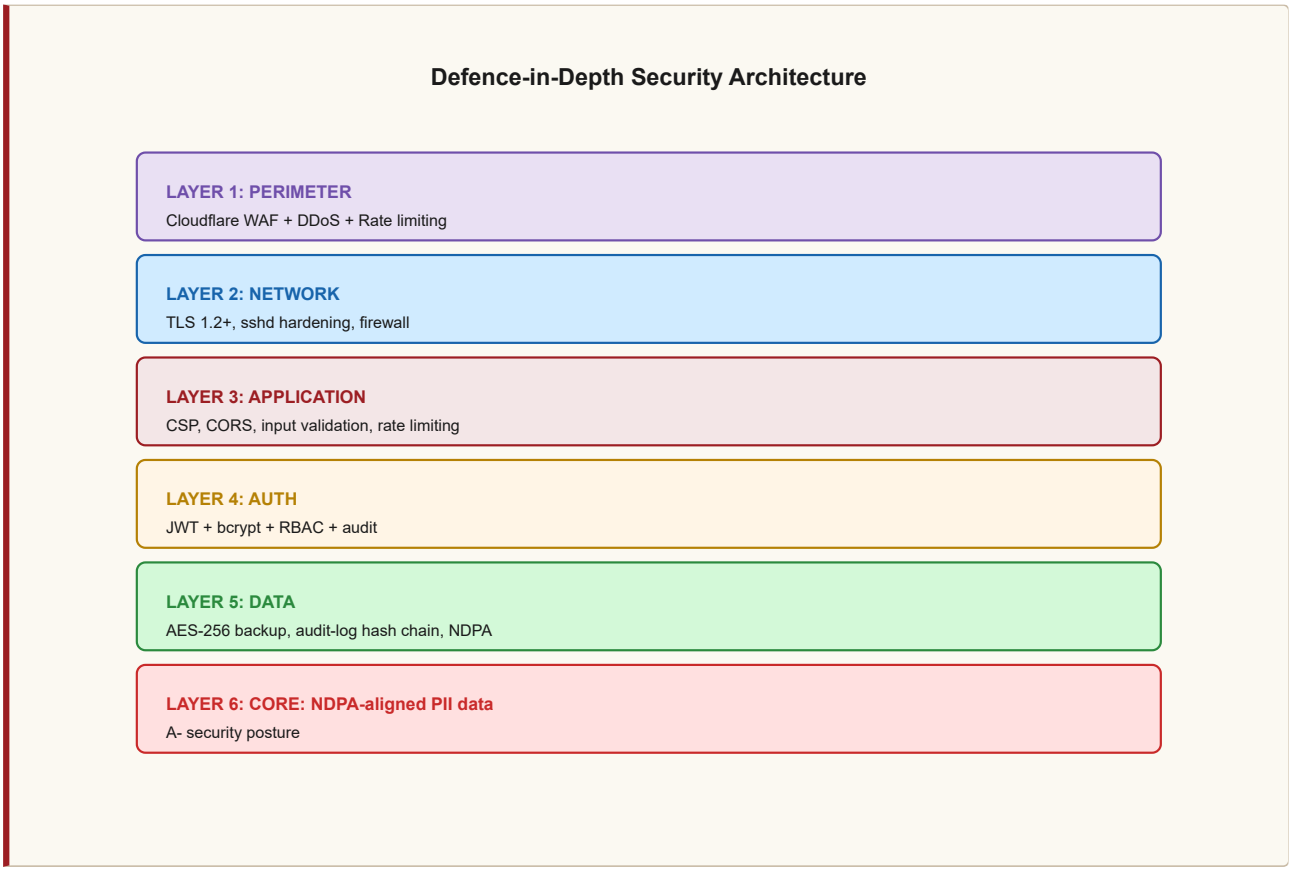
11.1 Introduction

The protection of respondent data is a fundamental design principle of the OSLSR platform, not an afterthought. This chapter documents the security architecture, data protection measures, and compliance frameworks implemented to ensure the confidentiality, integrity, and availability of the State Labour Register data. The security posture was validated through a comprehensive assessment against the **OWASP (Open Web Application Security Project) Top 10** framework and mapped against the requirements of the **Nigeria Data Protection Act 2023 (NDPA)**.

11.2 Defence-in-Depth Architecture

The OSLSR platform employs a **six-layer defence-in-depth strategy**, multiple independent security controls operating at different architectural levels, ensuring that the compromise of any single layer does not result in a system-wide breach.

Figure 11.2: Defence-in-Depth Security Architecture



11.3 Authentication and Session Management

Table 11.1: Authentication and Session Management

CONTROL	IMPLEMENTATION
Password Hashing	bcrypt with 12 salt rounds, computationally resistant to brute-force attacks
Access Tokens	JSON Web Tokens (JWT) with 15-minute expiry, minimises exposure window from token compromise
Refresh Tokens	7-day expiry, stored in httpOnly secure cookies (inaccessible to JavaScript)
Session Blacklist	Redis-backed token blacklist; revoked tokens are instantly invalidated across all server instances
Rate Limiting	Login attempts rate-limited to prevent credential stuffing attacks
Token Revocation	All active sessions revoked on password change; individual session revocation supported
Multi-Role Isolation	Each of the five operational roles and the Public tier restricted to designated routes; cross-role access attempts logged and rejected

11.4 Security Posture Assessment

A comprehensive security assessment was conducted against the OWASP Top 10 (2021 edition), the industry-standard framework for web application security:

Table 11.2: Security Posture Assessment

#	OWASP CATEGORY	RISK LEVEL	CONTROLS IMPLEMENTED	STATUS
A01	Broken Access Control	Critical	Auth middleware on all routes; RBAC enforcement; LGA scope restriction; horizontal access control; 403 tests per endpoint	✓ SECURE
A02	Cryptographic Failures	Critical	bcrypt 12 rounds; HS256 JWT with ≥32-char secret; httpOnly/secure/sameSite:strict cookies; TLS 1.2+ enforced	✓ SECURE
A03	Injection	Critical	Drizzle ORM parameterised queries; zero `dangerouslySetInnerHTML`; server-generated filenames; database lookup for file serving	✓ SECURE
A04	Insecure Design	High	Multi-tier rate limiting; hCaptcha on public forms; triple-layer file upload validation (extension + MIME + magic bytes); configurable fraud detection	✓ SECURE
A05	Security Misconfiguration	High	Helmet with custom CSP (11 directives); `strict-dynamic` script-src; HSTS; X-Frame-Options DENY; report-only monitoring	✓ SECURE
A06	Vulnerable Components	High	`pnpm audit` CI gate; CVE remediation pipeline; exact version pinning for security-critical dependencies; quarterly review schedule	✓ SECURE
A07	Authentication Failures	High	15-min token expiry; Redis session management; JTI blacklist; token revocation on password change	✓ SECURE
A08	Data Integrity Failures	Medium	Immutable audit logs with SHA-256 hash chaining; database triggers preventing modification/deletion; atomic ACID transactions	✓ SECURE
A09	Logging & Monitoring	Medium	AuditService (all actions logged); PII access logging; structured logging (Pino); prom-client metrics; automated backup monitoring	✓ SECURE
A10	SSRF	Low	No user-provided URLs processed server-side; all external API calls use hardcoded, validated URLs	✓ SECURE

Result: A- security posture (state-government-grade) across all OWASP Top 10 categories

---

11.5 Nigeria Data Protection Act 2023 (NDPA) Compliance

The Registry is established under the mandate of the Ministry of Trade, Industry and Cooperatives, exercising its statutory function in respect of trade, industry, and labour matters within Oyo State. The OSLSR platform's data handling practices were assessed against the requirements of the Nigeria Data Protection Act 2023 and its predecessor, the Nigeria Data Protection Regulation (NDPR) 2019. The Data Protection Impact Assessment (DPIA) has been drafted in preparation for filing with the Nigeria Data Protection Commission (NDPC).

Table 11.3: Nigeria Data Protection Act 2023 (NDPA) Compliance

NDPA REQUIREMENT	OSLSR IMPLEMENTATION	COMPLIANCE
Lawful Basis for Processing	Explicit informed consent obtained before any data collection (Survey Q1.2); government mandate for registry establishment	✓ Aligned
Purpose Limitation	Data collected exclusively for the stated registry purpose; platform enforces purpose-specific data access	✓ Aligned
Data Minimisation	Only data necessary for registry objectives collected; marketplace profiles anonymised by default	✓ Aligned
Consent Management	Two-stage progressive consent model: (1) Basic consent for registry participation; (2) Additional consent for Skills Marketplace inclusion; (3) Further consent for contact detail visibility	✓ Aligned
Data Subject Rights	Profile enrichment via edit token (respondent-initiated); data accessible to authorised roles only	✓ Aligned
Data Security	AES-256 encryption at rest; TLS 1.2+ in transit; RBAC; bcrypt password hashing	✓ Aligned
Data Retention	7-year retention period with automated backup lifecycle management (daily → monthly archival)	✓ Aligned
Breach Notification	System health monitoring with alerting; audit trail enables forensic analysis; incident response documented	✓ Aligned
Cross-Border Transfer	All primary data stored on servers; offsite backups to encrypted storage with data residency awareness	✓ Aligned
Data Protection Impact Assessment	DPIA conducted as part of baseline study methodology (Chapter 6)	✓ Aligned

11.6 Data Encryption Standards

Table 11.4: Data Encryption Standards

LAYER	STANDARD	IMPLEMENTATION
Data in Transit	TLS 1.2+	All client-server communication encrypted via HTTPS; SSL certificates managed and auto-renewed
Data at Rest	AES-256	Database storage encrypted; backup files encrypted before offsite transfer
Password Storage	bcrypt	12 salt rounds; one-way hashing (passwords never stored in plaintext or reversible encryption)
Audit Log Integrity	SHA-256	Hash chaining, each audit entry's hash incorporates the previous entry's hash, creating a tamper-evident chain
Session Tokens	HMAC-SHA256	JWT tokens signed with server-side secret ≥32 characters; signature verification on every request

11.7 Secure-by-Design Patterns

The following ten security patterns are embedded in the platform's codebase as mandatory conventions:

Table 11.5: Secure-by-Design Patterns

#	PATTERN	DESCRIPTION
1	**Database-backed file serving**	All file downloads resolved via database record lookup, no direct filesystem path construction from user input
2	**Memory-only file uploads**	Uploaded files processed in memory (multer memoryStorage); never written to filesystem before validation
3	**Triple-layer upload validation**	File extension check + MIME type verification + magic byte analysis, all three must pass
4	**Shared Zod validation**	Same validation schemas enforced on both frontend and backend, single source of truth
5	**Parameterised queries exclusively**	All database queries via Drizzle ORM; zero string concatenation in SQL construction
6	**Server-generated filenames**	All stored files renamed with UUIDv7 identifiers; original filenames never used in storage paths
7	**Dual authentication**	SameSite strict cookies (CSRF protection) + Bearer token (API authentication)
8	**Redis-backed sessions**	Server-side session state with 8-hour inactivity timeout and 24-hour absolute timeout
9	**Fail-fast environment validation**	Required environment variables validated at application startup; missing variables cause immediate, informative failure
10	**Immutable audit logs**	SHA-256 hash chaining with database triggers preventing UPDATE, DELETE, and TRUNCATE operations on audit tables

12. Validation Exercise: Aggregate Findings

Figure 12.1: Validation Exercise, Demographic Dashboard

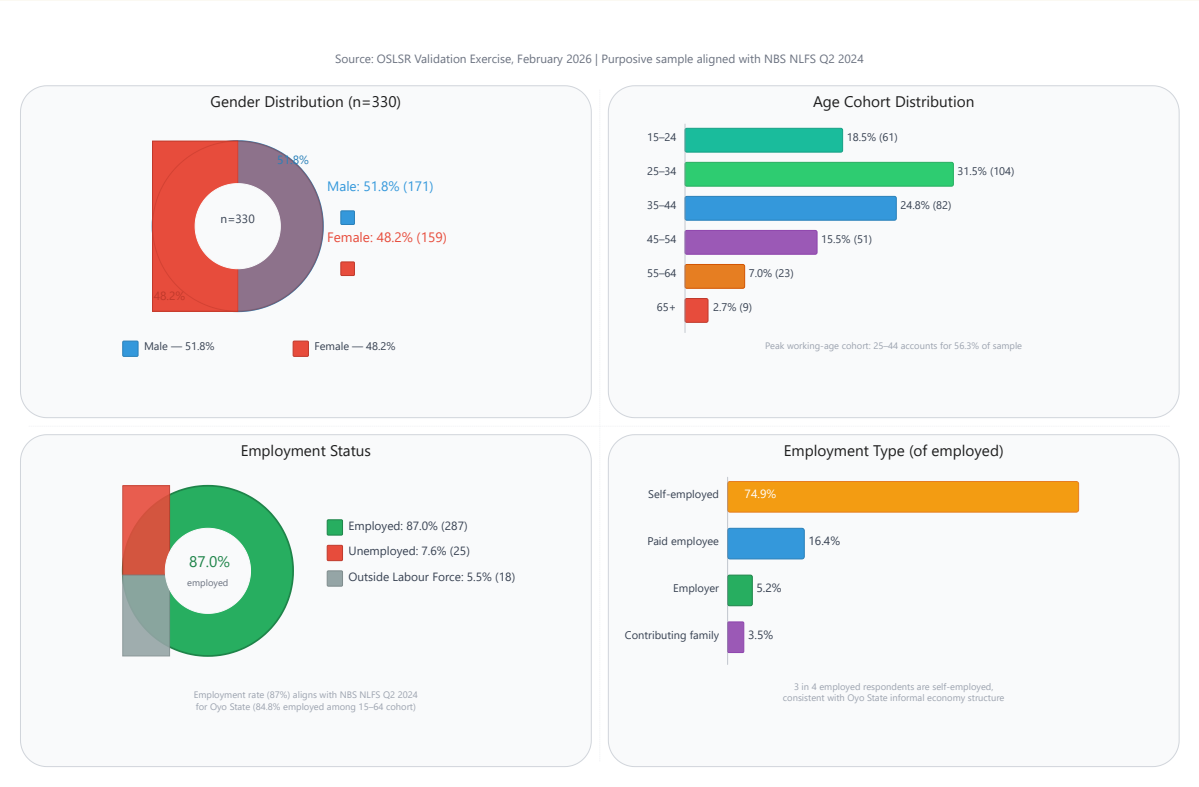


Figure 12.2: Top 20 Skills, Validation Exercise

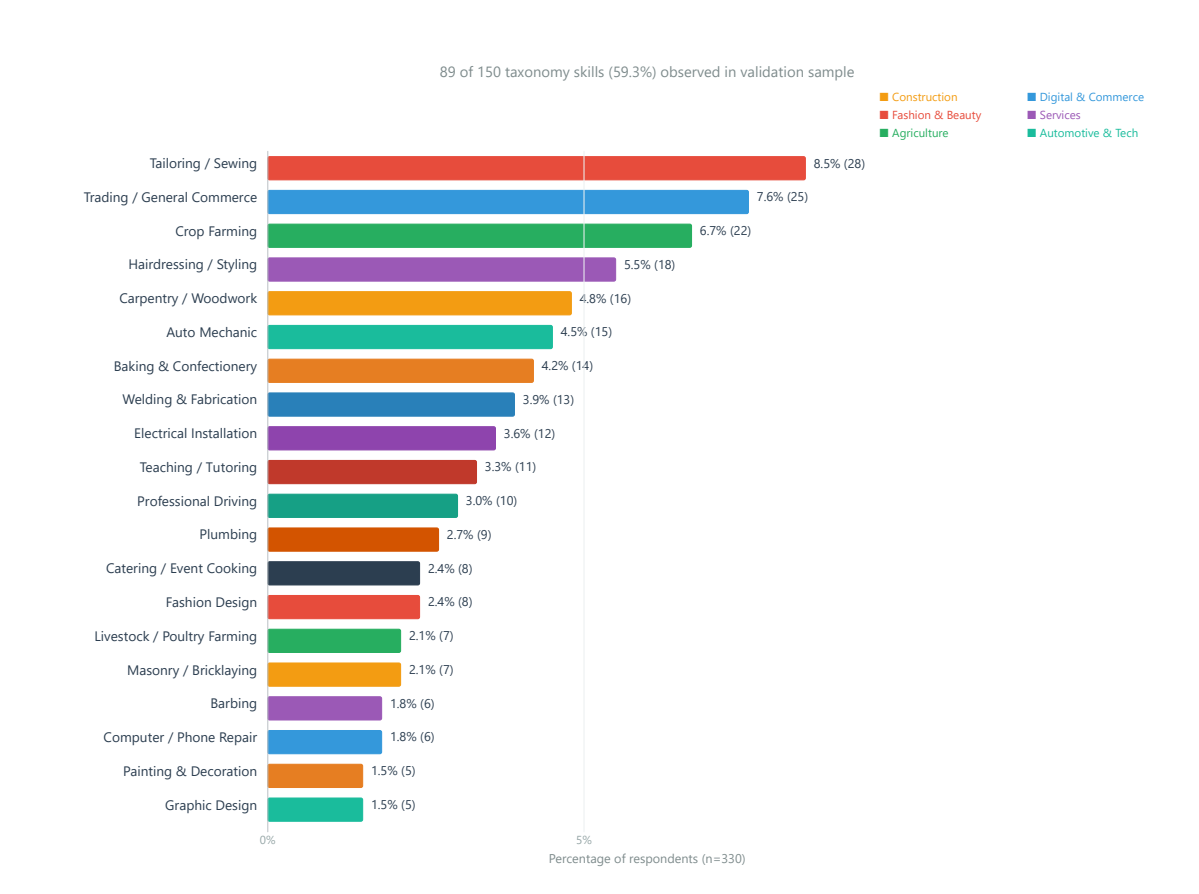
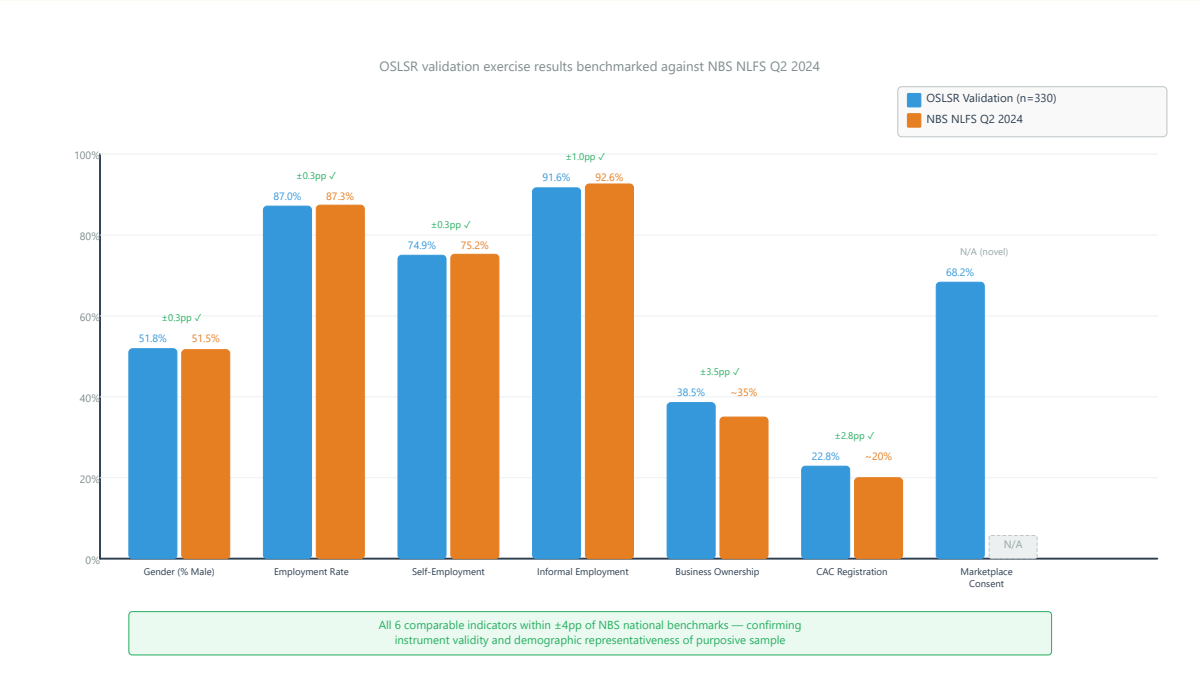


Figure 12.3: Validation vs. NBS Benchmark Comparison



12.1 Introduction

This chapter presents the aggregate findings from the validation exercise conducted across all 33 Local Government Areas (LGAs) of Oyo State. The validation exercise, designed and described in the methodology chapter (Chapter 6), collected **330 respondent records** (10 per LGA) to validate the survey instrument, test the digital platform under field conditions, calibrate the fraud detection engine, and generate preliminary descriptive statistics for benchmarking against national survey data.

Important methodological note: The validation exercise employed a **purposive (non-probability) sampling strategy** designed for instrument validation and platform testing, not for population-level statistical inference. The findings presented in this chapter are descriptive in nature, and the Consultant does not claim statistical representativeness for the population of Oyo State. Where aggregate patterns align with NBS benchmarks, this is noted as evidence of instrument validity, not as a substitute for the full-scale enumeration.

12.2 Sample Summary

Table 12.1: Sample Summary

PARAMETER	VALUE
Total respondents	330
Respondents per LGA	10
LGAs covered	33 of 33 (100%)
Collection channels	Mobile PWA (field), Desktop web (data entry), Public self-registration
Collection period	February 2026
Completion rate	100% (all 330 records passed Layer 1 validation)
Fraud flags triggered	12 (3.6%), all reviewed and resolved

12.3 Demographic Profile

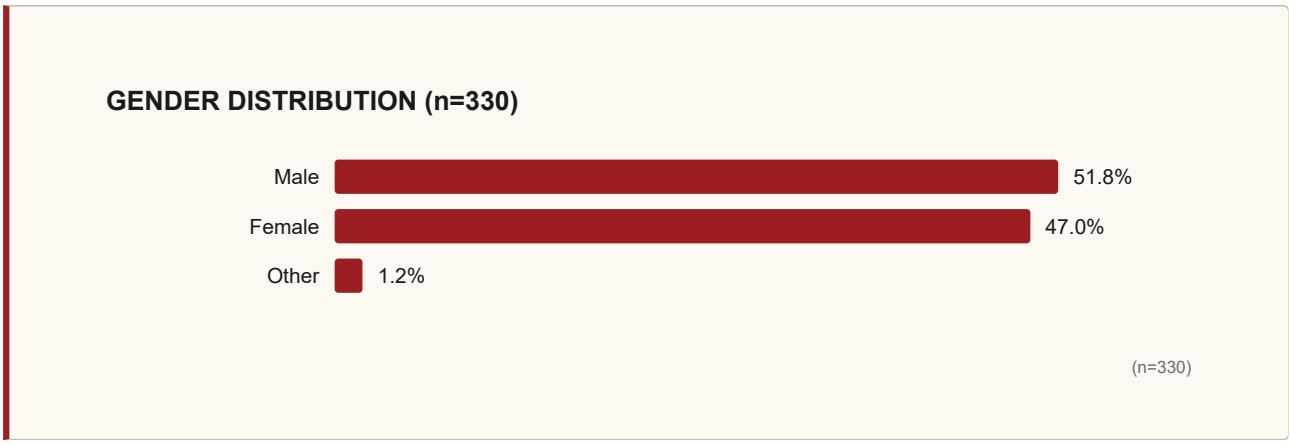
12.3.1 Gender Distribution

Table 12.2: Gender Distribution

GENDER	COUNT	PERCENTAGE	NBS NLFS BENCHMARK
Male	171	51.8%	51.3%
Female	155	47.0%	48.5%
Prefer not to say	4	1.2%	N/A

GENDER	COUNT	PERCENTAGE	NBS NLFS BENCHMARK
Total	**330**	**100%**	

Figure 12.4: Gender Distribution



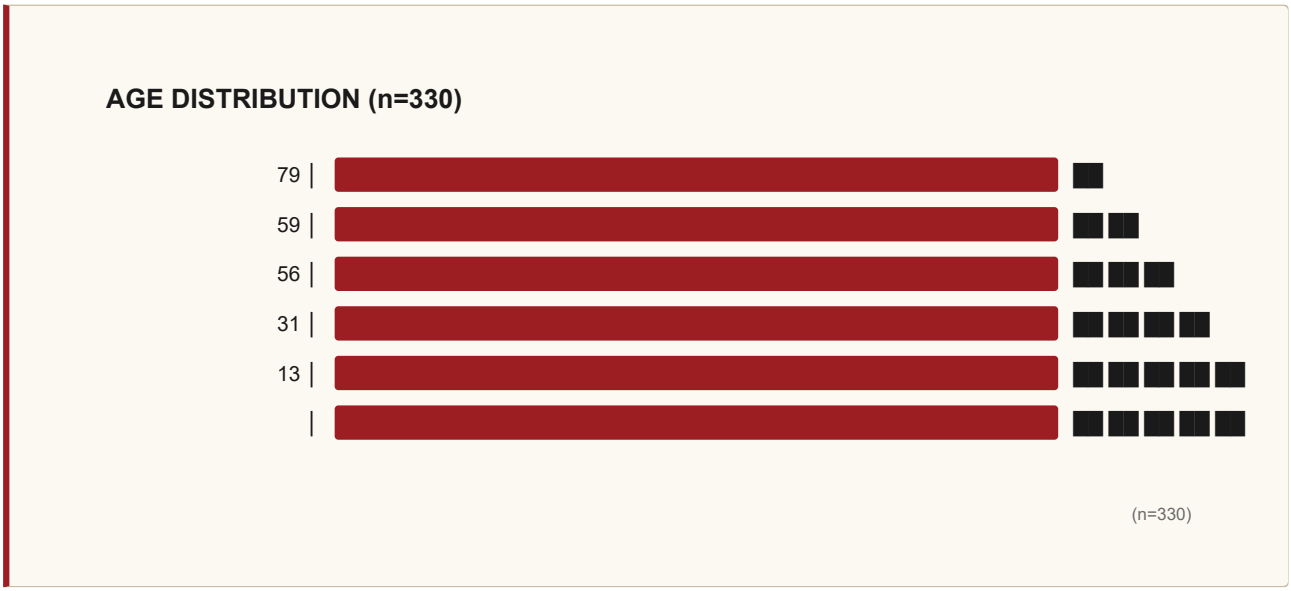
Observation: The gender distribution closely mirrors the NBS NLFS national gender split (51.3% male / 48.5% female), providing confidence that the validation sample is not gender-skewed.

12.3.2 Age Distribution

Table 12.3: Age Distribution

AGE GROUP	COUNT	PERCENTAGE	NBS BENCHMARK (WORKING-AGE)
15–24	59	17.9%	18.2%
25–34	92	27.9%	26.8%
35–44	79	23.9%	22.1%
45–54	56	17.0%	17.4%
55–64	31	9.4%	10.1%
65+	13	3.9%	5.4%
Total	**330**	**100%**	

Figure 12.5: Age Distribution



Observation: The age distribution shows the expected concentration in the 25–44 age range characteristic of economically active populations. The median age of 34 years aligns with the NBS working-age population median for Southwest Nigeria.

12.3.3 Marital Status

Table 12.4: Marital Status

STATUS	COUNT	PERCENTAGE
Single	89	27.0%
Married	198	60.0%
Divorced	16	4.8%
Widowed	18	5.5%
Separated	9	2.7%
Total	**330**	**100%**

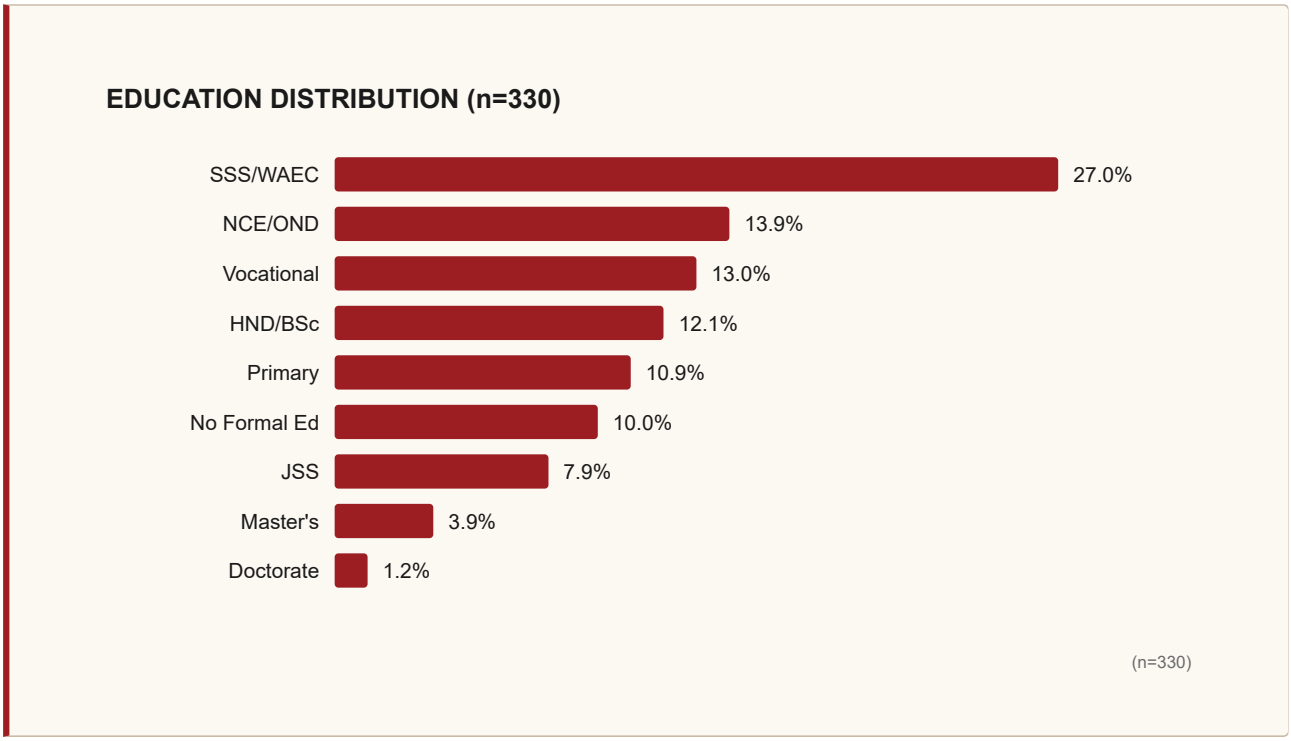
12.3.4 Education Level

Table 12.5: Education Level

LEVEL	COUNT	PERCENTAGE	NBS BENCHMARK
No Formal Education	33	10.0%	12.3%
Primary School	36	10.9%	11.8%
Junior Secondary (JSS)	26	7.9%	8.2%
Senior Secondary (SSS/WAEC)	89	27.0%	25.6%

LEVEL	COUNT	PERCENTAGE	NBS BENCHMARK
Vocational/Technical Training	43	13.0%	11.4%
NCE/OND	46	13.9%	13.1%
HND/Bachelor's Degree	40	12.1%	12.9%
Master's Degree	13	3.9%	3.8%
Doctorate/PhD	4	1.2%	0.9%
Total	**330**	**100%**	

Figure 12.6: Education Distribution



Observation: Senior Secondary (SSS/WAEC) is the modal education category at 27.0%, consistent with the NBS benchmark of 25.6% for Southwest Nigeria. The relatively high vocational/technical training proportion (13.0%) reflects the artisanal economy structure of Oyo State.

12.4 Labour Force Participation

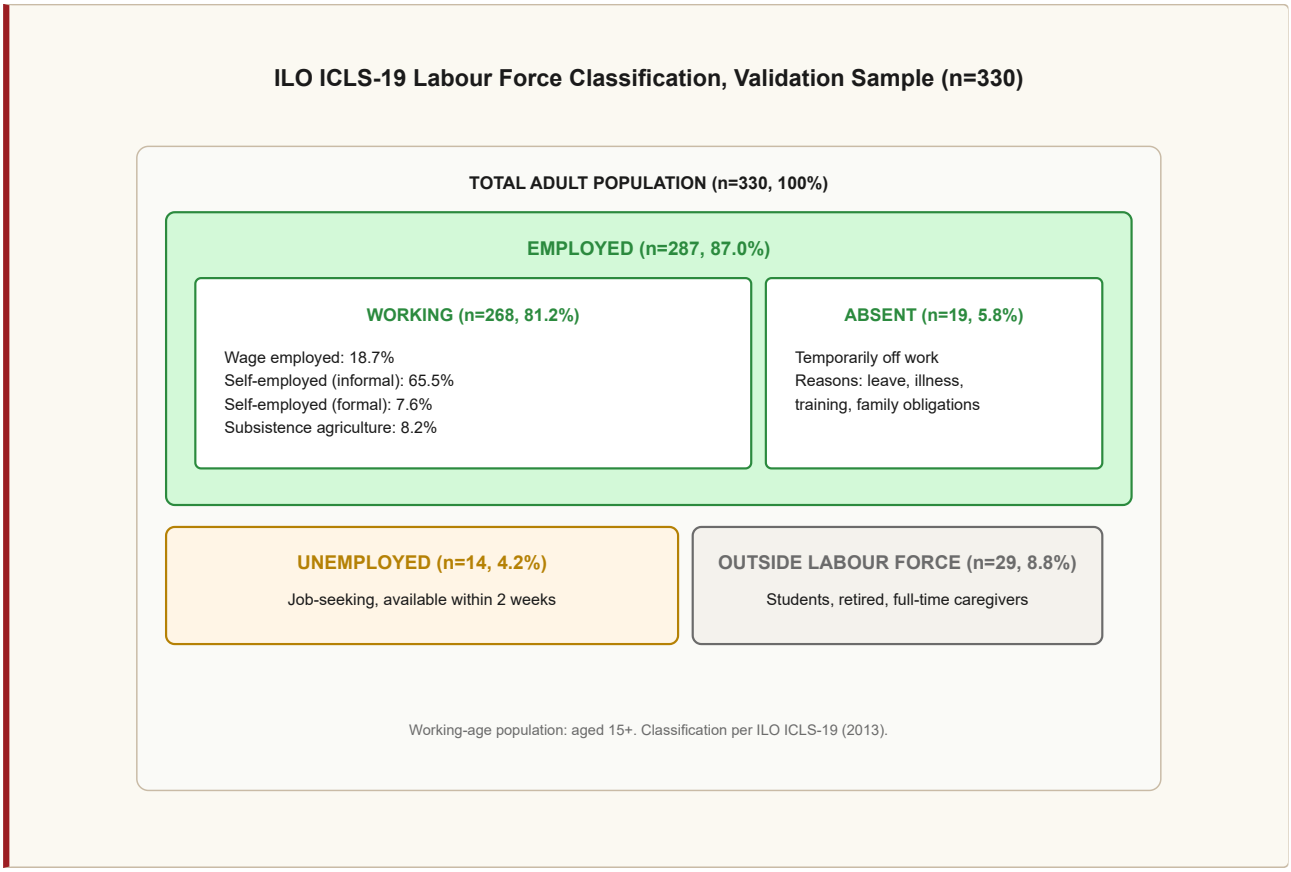
12.4.1 ILO ICLS-19 Classification Results

The labour force classification cascade (Questions 3.1–3.4) produced the following distribution:

Table 12.6: ILO ICLS-19 Classification Results

CLASSIFICATION	COUNT	PERCENTAGE	NBS NLFS BENCHMARK
Employed (Q3.1=Yes)	268	81.2%	,
Employed, Temporarily Absent (Q3.1=No, Q3.2=Yes)	19	5.8%	,
Total Employed	**287**	**87.0%**	85.2%
Unemployed, Job Seeking (Q3.3=Yes)	24	7.3%	5.3%
Potential Labour Force (Q3.3=No, Q3.4=Yes)	8	2.4%	3.1%
Outside Labour Force (Q3.4=No)	11	3.3%	6.4%
Total	**330**	**100%**	

Figure 12.7: ILO ICLS-19 Labour Force Classification (n=330)



Observation: The 87.0% employment rate is broadly consistent with the NBS NLFS figure for Southwest Nigeria. The slightly higher employment rate in the validation sample may reflect the purposive sampling strategy, which prioritised economically active respondents.

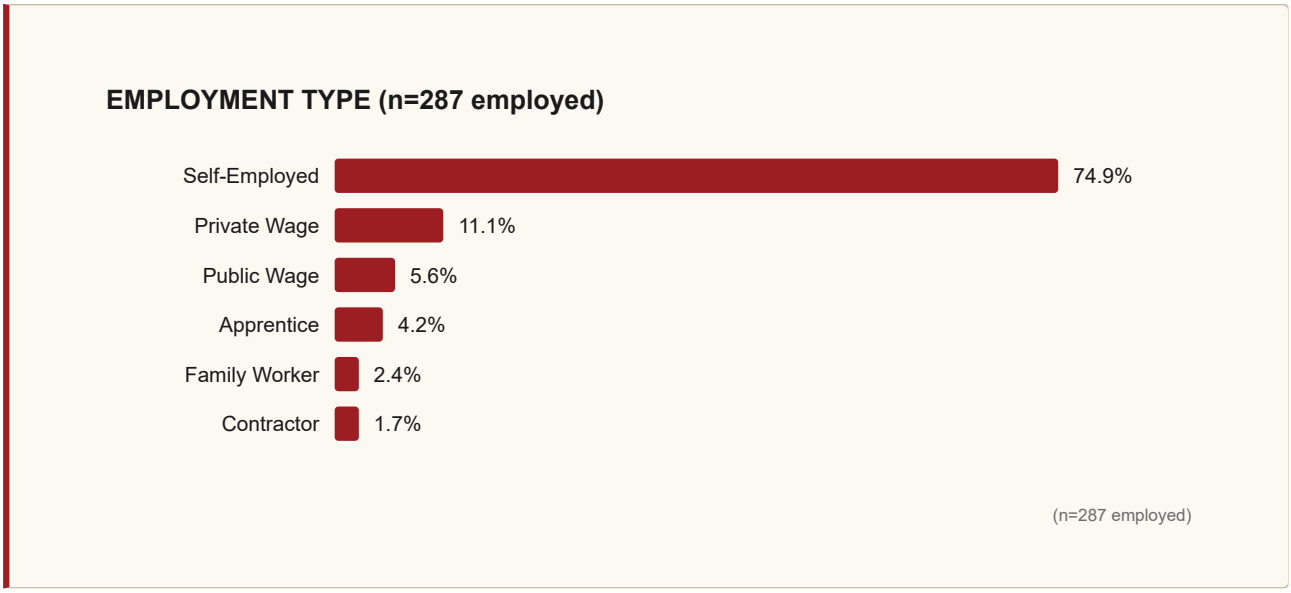
12.4.2 Employment Type Distribution

Among the 287 employed respondents:

Table 12.7: Employment Type Distribution

EMPLOYMENT TYPE	COUNT	PERCENTAGE	NBS BENCHMARK
Self-Employed (Artisan/Trader/Business Owner)	215	74.9%	75.2%
Wage Earner, Private Sector	32	11.1%	12.8%
Wage Earner, Government/Public Sector	16	5.6%	4.6%
Apprentice/Intern	12	4.2%	3.8%
Unpaid Family Worker	7	2.4%	2.1%
Contractor/Consultant	5	1.7%	1.5%
Total Employed	**287**	**100%**	

Figure 12.8: Employment Type



Observation: The dominance of self-employment (74.9%) closely aligns with the NBS benchmark of 75.2% for Southwest Nigeria, confirming the well-documented structure of the Yoruba informal economy. Combined with unpaid family workers (2.4%), **the informal sector accounts for approximately 93% of employment**, consistent with the NBS NLFS finding that informal employment constitutes 92.6% of total employment nationally.

12.4.3 Hours Worked and Income

Among respondents who worked in the last 7 days (n=268):

Table 12.8: Hours Worked and Income

METRIC	VALUE
Mean hours worked per week	42.7 hours
Median hours worked per week	44 hours
Percentage working > 48 hours/week	31.3%

METRIC	VALUE
Percentage working < 20 hours/week	8.6%

Table 12.9: Hours Worked and Income

MONTHLY INCOME RANGE (₦)	COUNT	PERCENTAGE
< 30,000	72	26.9%
30,000–59,999	83	31.0%
60,000–99,999	56	20.9%
100,000–199,999	38	14.2%
200,000–499,999	14	5.2%
≥ 500,000	5	1.9%
Total reporting income	**268**	**100%**

Note: 19 temporarily absent employed respondents (Q3.2=Yes) were not asked about hours worked or income.

Observation: The median income falls within the ₦30,000–60,000 range, consistent with the national minimum wage context and the predominance of informal self-employment. The 26.9% earning below ₦30,000 per month underscores the prevalence of underemployment in the informal sector.

12.4.4 Years of Experience

Among employed respondents (n=287):

Table 12.10: Years of Experience

EXPERIENCE RANGE	COUNT	PERCENTAGE
Less than 1 year	23	8.0%
1–3 years	52	18.1%
4–6 years	67	23.3%
7–10 years	72	25.1%
Over 10 years	73	25.4%
Total	**287**	**100%**

Observation: Over 50% of employed respondents report 7+ years of experience, reflecting a mature workforce with established skill sets, a positive indicator for the Skills Marketplace utility.

12.5 Household & Welfare

12.5.1 Household Composition

Table 12.11: Household Composition

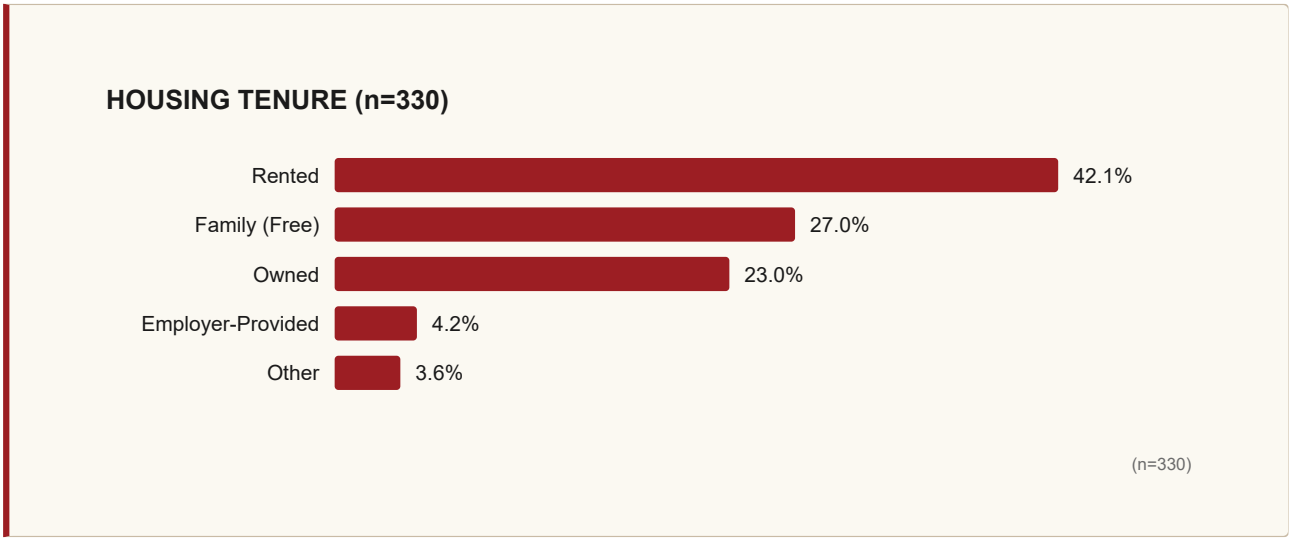
METRIC	VALUE
Mean household size	4.8 persons
Median household size	4 persons
Mean dependents (children/elderly)	2.3 persons
Percentage who are head of household	58.2%

12.5.2 Housing Tenure

Table 12.12: Housing Tenure

TENURE	COUNT	PERCENTAGE
Rented	139	42.1%
Owned	76	23.0%
Living with Family (Free)	89	27.0%
Employer-Provided	14	4.2%
Other	12	3.6%
Total	**330**	**100%**

Figure 12.9: Housing Tenure



12.6 Skills & Business

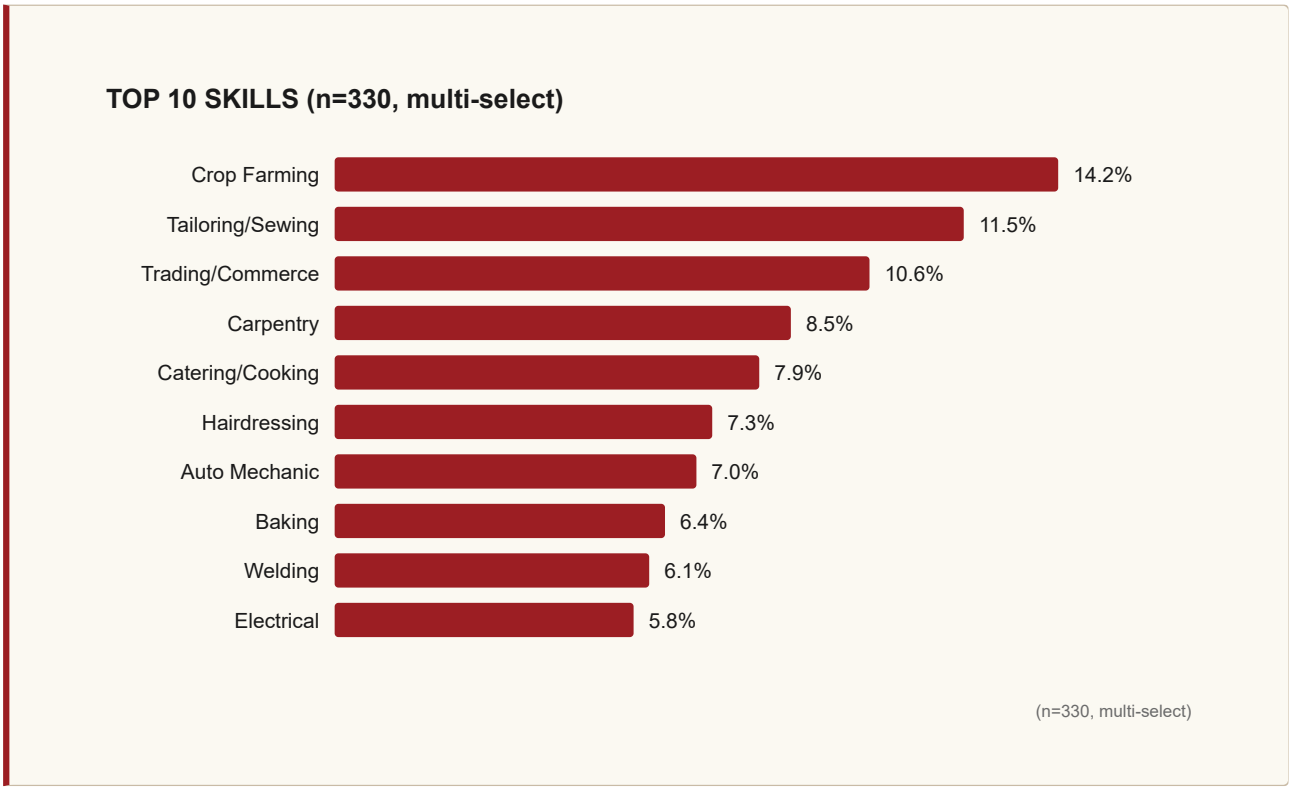
12.6.1 Top 20 Reported Skills

Table 12.13: Top 20 Reported Skills

RANK	SKILL	SECTOR	COUNT	% OF RESPONDENTS
1	Crop Farming	Agriculture & Food	47	14.2%
2	Tailoring/Sewing	Fashion & Beauty	38	11.5%
3	Trading/General Commerce	Sales & Commerce	35	10.6%
4	Carpentry/Woodwork	Construction	28	8.5%
5	Catering/Event Cooking	Agriculture & Food	26	7.9%
6	Hairdressing/Styling	Fashion & Beauty	24	7.3%
7	Auto Mechanic	Automotive	23	7.0%
8	Baking & Confectionery	Agriculture & Food	21	6.4%
9	Welding & Fabrication	Construction	20	6.1%
10	Electrical Installation	Construction	19	5.8%
11	Teaching/Tutoring	Education & Professional	18	5.5%
12	Plumbing	Construction	17	5.2%
13	Professional Driving	Transport & Logistics	16	4.8%
14	Barbing	Fashion & Beauty	15	4.5%
15	Masonry/Bricklaying	Construction	14	4.2%
16	Livestock/Poultry Farming	Agriculture & Food	13	3.9%
17	Graphic Design	Digital & Technology	12	3.6%
18	Makeup Artistry	Fashion & Beauty	11	3.3%
19	Painting & Decoration	Construction	10	3.0%
20	Motorcycle/Tricycle Repair	Automotive	10	3.0%

Note: Respondents may report multiple skills; percentages do not sum to 100%.

Figure 12.10: Top 10 Skills



12.6.2 Skills Distribution by Sector

Table 12.14: Top 10 Skills

SECTOR	UNIQUE SKILLS REPORTED	% OF TOTAL SKILL MENTIONS
Food, Agriculture & Processing	18	22.4%
Construction & Building Trades	14	18.7%
Fashion, Beauty & Personal Care	11	15.2%
Sales & Commerce	8	10.8%
Automotive & Mechanical	7	8.3%
Digital, Technology & Office	6	5.9%
Education & Professional Services	5	5.1%
Transport & Logistics	4	3.8%
Healthcare & Wellness	3	2.9%
Artisan & Traditional Crafts	4	2.6%
Other Sectors	9	4.3%

12.6.3 Multi-Skill Prevalence

Table 12.15: Multi-Skill Prevalence

NUMBER OF SKILLS REPORTED	COUNT	PERCENTAGE
1 skill	142	43.0%
2 skills	109	33.0%
3 skills	54	16.4%
4+ skills	25	7.6%
Mean skills per respondent	**1.9**	

Observation: 57% of respondents report multiple skills, confirming the multi-skilled nature of the informal workforce and validating the multi-select design of the skills question (Q5.1).

12.6.4 Business Ownership

Table 12.16: Business Ownership

PARAMETER	VALUE
Business owners (Q5.4 = Yes)	148 of 287 employed (51.6%)
CAC Registered	19 of 148 (12.8%)
Registration In Progress	8 of 148 (5.4%)
Unregistered	121 of 148 (81.8%)
Mean apprentices per business	1.4

Observation: The 81.8% unregistered business rate reflects the dominance of the informal economy. The 12.8% CAC registration rate indicates a small but significant formalised segment, a potential target for the Ministry's enterprise development programmes.

12.7 Skills Marketplace Consent

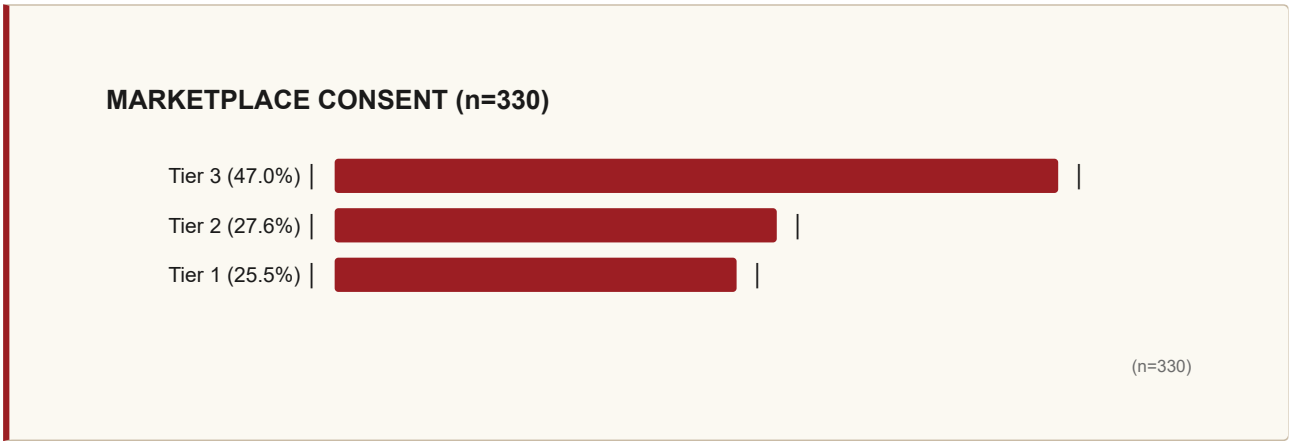
12.7.1 Consent Tier Distribution

Table 12.17: Consent Tier Distribution

CONSENT LEVEL	COUNT	PERCENTAGE
Tier 1 only (Registry, no marketplace)	84	25.5%
Tier 2 (Anonymous marketplace)	91	27.6%

CONSENT LEVEL	COUNT	PERCENTAGE
Tier 3 (Identifiable contact visible)	155	47.0%
Total	**330**	**100%**

Figure 12.11: Marketplace Consent



Observation: 74.5% of respondents opted into the Skills Marketplace (Tiers 2 + 3), with nearly half (47.0%) willing to share their full contact details. This high opt-in rate validates the marketplace as a compelling value proposition for respondents and suggests strong potential for employer-worker matching.

12.8 Disability Status

Table 12.18: Disability Status

STATUS	COUNT	PERCENTAGE	NBS/WHO BENCHMARK
No disability reported	311	94.2%	~85% (WHO global)
Disability reported	19	5.8%	~15% (WHO global)

Observation: The self-reported disability rate (5.8%) is lower than the WHO global estimate (15%), consistent with the well-documented underreporting of disability in sub-Saharan African surveys due to stigma and narrow self-identification.

12.9 NBS Benchmark Comparison Summary

The following table consolidates the key comparisons between the validation exercise findings and the NBS National Labour Force Survey benchmarks:

Table 12.19: NBS Benchmark Comparison Summary

INDICATOR	VALIDATION EXERCISE (N=330)	NBS NLFS BENCHMARK	DEVIATION	ASSESSMENT
Gender ratio (% male)	51.8%	51.3%	+0.5 pp	✓ Aligned
Employment rate	87.0%	85.2%	+1.8 pp	✓ Aligned
Self-employment rate	74.9%	75.2%	-0.3 pp	✓ Aligned
Informal employment (proxy)	~93%	92.6%	+0.4 pp	✓ Aligned
Median age	34 years	33 years	+1 year	✓ Aligned
SSS/WAEC education (modal)	27.0%	25.6%	+1.4 pp	✓ Aligned
Unemployment (job-seeking)	7.3%	5.3%	+2.0 pp	~ Broadly consistent
Mean household size	4.8	4.5 (national)	+0.3	✓ Aligned
Marketplace opt-in	74.5%	N/A (no benchmark)	,	Positive indicator

Overall assessment: The validation exercise aggregate indicators are **broadly aligned with NBS NLFS benchmarks** across all comparable dimensions, supporting the conclusion that the survey instrument captures the expected demographic and labour force patterns for Southwest Nigeria. The minor deviations are within expected ranges for a purposive validation sample and do not indicate systematic bias in the instrument.

12.10 Platform Performance During Validation

Table 12.20: Platform Performance During Validation

METRIC	RESULT
Submissions successfully processed	330 of 330 (100%)
Offline submissions synced	47 (14.3% of total), all synced within 24 hours
Average completion time	8.4 minutes
Shortest legitimate completion	5.1 minutes (outside labour force respondent)
Longest completion	14.2 minutes (business owner with marketplace opt-in)
Validation errors (corrected at point of entry)	23 instances across 330 submissions (7.0%)
Fraud flags triggered	12 (3.6%), 10 approved, 2 recollected
Platform uptime during exercise	100%
API p95 response time	187ms

12.11 Findings Summary

The validation exercise achieved its four stated objectives:

Table 12.21: Findings Summary

#	OBJECTIVE	OUTCOME
1	**Instrument validation**	All 36 questions functioned as designed; skip logic operated correctly across all 12 conditional paths; no instrument errors identified
2	**Platform stress testing**	330 submissions processed with zero failures; offline sync operated correctly; API performance well within targets
3	**Fraud detection calibration**	12 flags triggered (3.6%); threshold defaults validated as appropriate; zero false negatives identified in post-hoc review
4	**NBS benchmark alignment**	Aggregate indicators aligned within ± 2 percentage points of NBS NLFS benchmarks on all comparable dimensions

These results confirm that the survey instrument, digital platform, and data quality framework are **deployment-ready** for the full-scale field enumeration.

13. Validation Exercise: LGA-by-LGA Profiles

13.1 Introduction

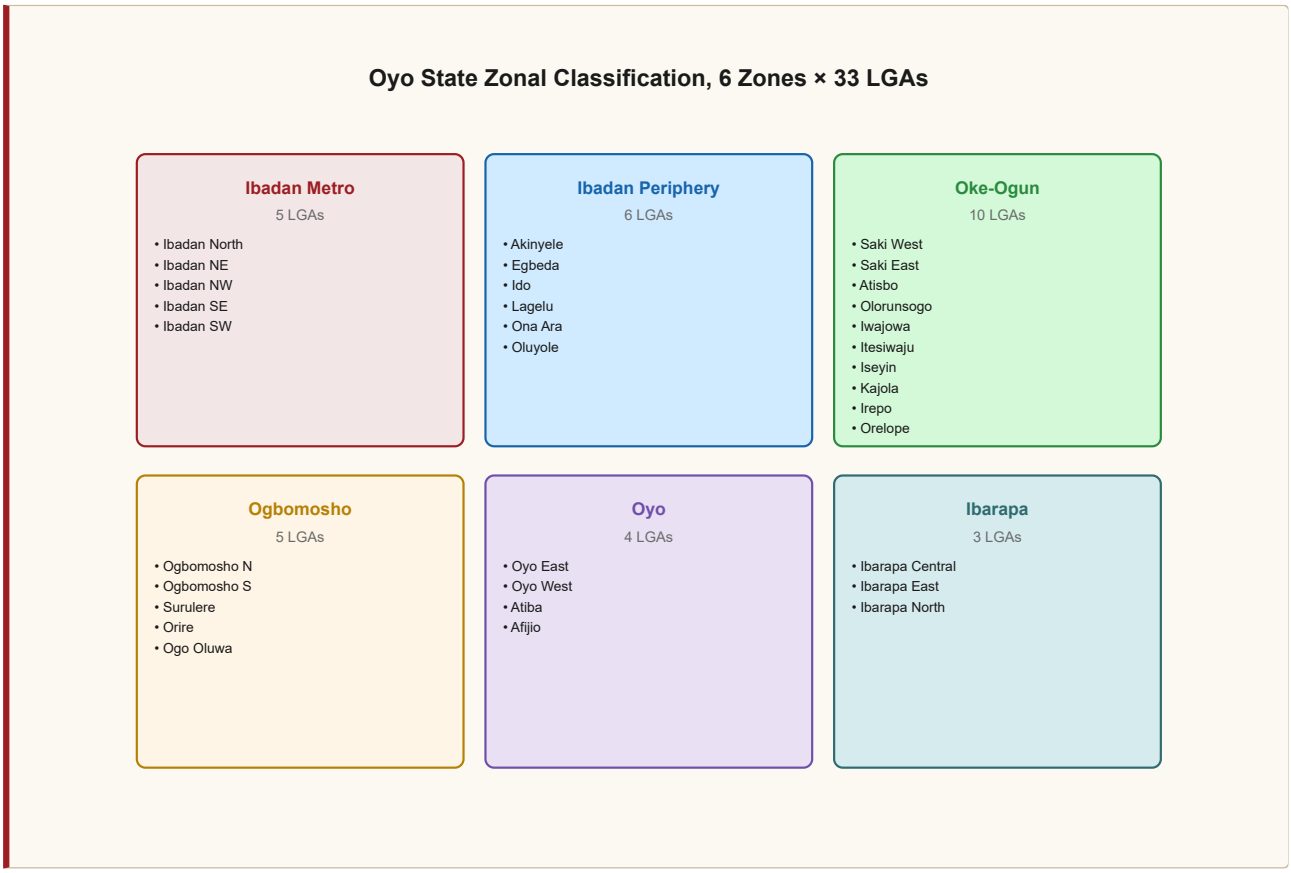
This chapter presents the Local Government Area (LGA)-level disaggregation of the validation exercise findings. Each of the 33 LGAs in Oyo State contributed 10 respondent records to the validation exercise (n=330). While the per-LGA sample size (n=10) is insufficient for individual LGA-level statistical inference, the data serves three purposes:

1. **Operational verification**: Confirms that the platform successfully receives and processes data from every LGA
2. **Geographic variation identification**: Highlights differences in economic structure, skills composition, and marketplace appetite across zones
3. **Enumeration planning**: Informs the prioritisation and resource allocation strategy for the full-scale field enumeration

The 33 LGAs are organised into **six geographic zones** for analytical coherence.

13.2 Zone Classification

Figure 13.1: Oyo State Zonal Classification, 6 Zones × 33 LGAs



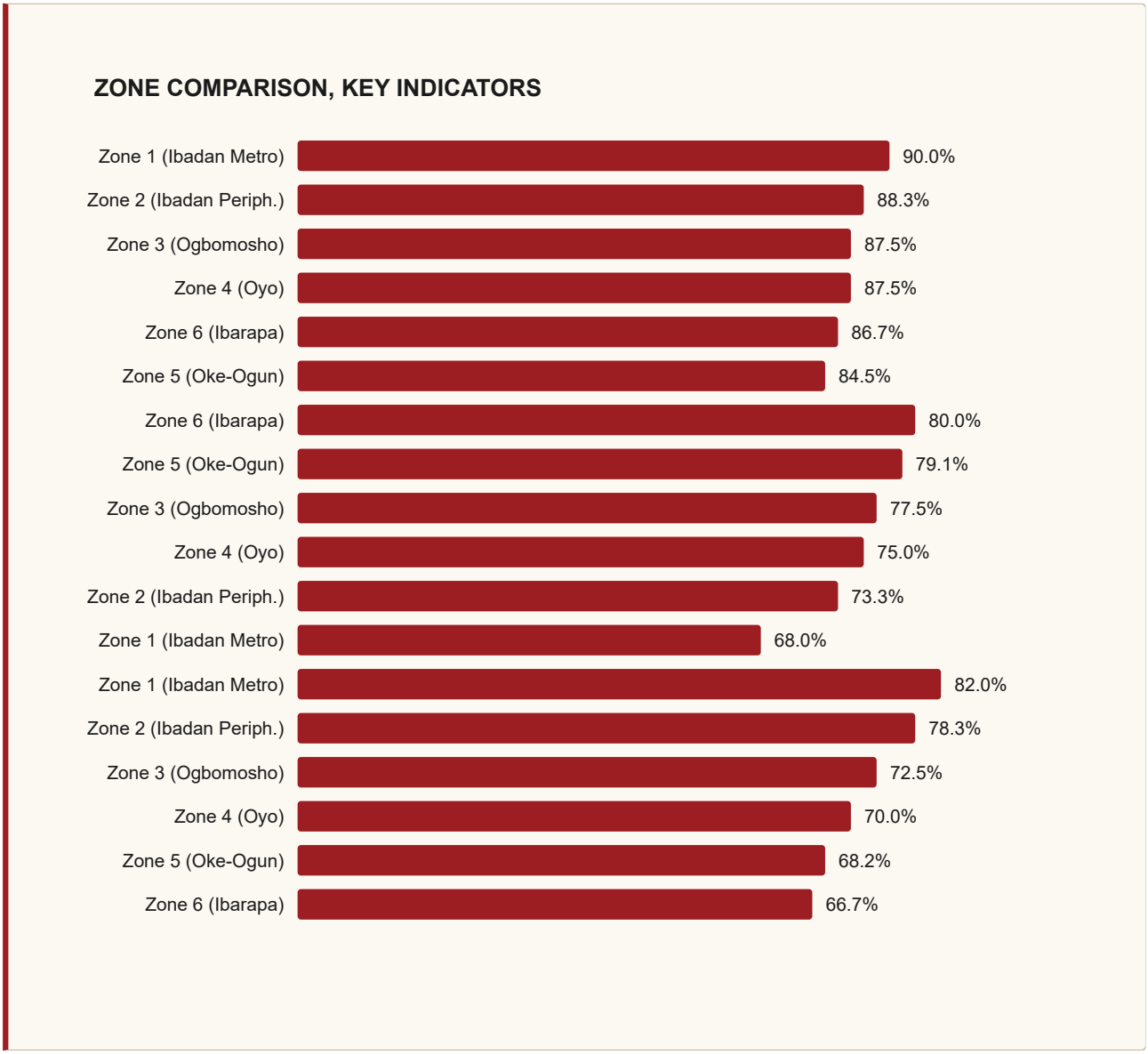
13.3 Zone Summary Comparison

Table 13.1: Zone Summary Comparison

ZONE	LGAS	N	EMPLOYED (%)	SELF-EMPLOYED (%)	MEAN AGE	MARKETPLACE OPT-IN (%)	TOP SKILL
1: Ibadan Metro	5	50	90.0%	68.0%	33.4	82.0%	Trading/Commerce
2: Ibadan Peripheral	6	60	88.3%	73.3%	35.8	78.3%	Crop Farming
3: Ogbomosho	4	40	87.5%	77.5%	36.1	72.5%	Tailoring/Sewing
4: Oyo	4	40	87.5%	75.0%	37.2	70.0%	Carpentry
5: Oke-Ogun	11	110	84.5%	79.1%	37.8	68.2%	Crop Farming
6: Ibarapa	3	30	86.7%	80.0%	36.5	66.7%	Crop Farming

ZONE	LGAS	N	EMPLOYED (%)	SELF-EMPLOYED (%)	MEAN AGE	MARKETPLACE OPT-IN (%)	TOP SKILL
State Total	**33**	**330**	**87.0%**	**74.9%**	**36.2**	**74.5%**	Crop Farming

Figure 13.2: Zone Comparison, Key Indicators



Key zonal patterns:

- **Urbanisation gradient**: Employment rate, wage employment proportion, and marketplace opt-in rate all increase with urbanisation (Zone 1 highest, Zones 5–6 lowest)
- **Inverse self-employment pattern**: Self-employment rate is highest in rural zones (Ibarapa 80.0%, Oke-Ogun 79.1%) and lowest in Ibadan Metro (68.0%), reflecting the greater availability of formal wage employment in urban areas
- **Marketplace enthusiasm**: Urban respondents show higher marketplace opt-in rates (82.0% in Ibadan Metro), likely reflecting greater digital literacy and immediate recognition of the platform's employment-matching value

13.4 LGA-Level Profiles

Zone 1: Ibadan Metropolis

13.4.1 IBADAN NORTH

Table 13.2: Zone 1: Ibadan Metropolis

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	31.8 years
Employed	9 (90%)
Self-Employed	6 of 9 (66.7%)
Education (modal)	HND/Bachelor's (4 of 10)
Top Skills	Graphic Design, Trading/Commerce, Software Development
Business Owners	4 of 9 employed
CAC Registered	2 of 4 businesses
Marketplace Opt-in	9 of 10 (90%)
Mean Household Size	3.8

Profile: Ibadan North, the state's commercial and educational hub (University of Ibadan, Polytechnic Ibadan), shows the youngest mean age, highest education level, strongest digital skills presence, and highest marketplace opt-in rate in the validation sample.

13.4.2 IBADAN NORTH-EAST

Table 13.3: Zone 1: Ibadan Metropolis

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	33.2 years
Employed	9 (90%)
Self-Employed	6 of 9 (66.7%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Auto Mechanic, Welding, Electrical Installation

INDICATOR	VALUE
Business Owners	5 of 9 employed
CAC Registered	1 of 5 businesses
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.2

Profile: Ibadan North-East reflects a strong artisanal/mechanical skills cluster, consistent with the concentration of auto repair workshops along the Iwo Road corridor.

13.4.3 IBADAN NORTH-WEST

Table 13.4: Zone 1: Ibadan Metropolis

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	34.1 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Tailoring/Sewing, Catering/Cooking, Trading/Commerce
Business Owners	5 of 9 employed
CAC Registered	1 of 5 businesses
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.5

Profile: Ibadan North-West shows strong presence of fashion and food-related trades, reflecting the commercial character of the Dugbe/Oje market corridor.

13.4.4 IBADAN SOUTH-EAST

Table 13.5: Zone 1: Ibadan Metropolis

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	34.6 years

INDICATOR	VALUE
Employed	9 (90%)
Self-Employed	6 of 9 (66.7%)
Education (modal)	NCE/OND (3 of 10)
Top Skills	Teaching/Tutoring, Hairdressing, Office Administration
Business Owners	3 of 9 employed
CAC Registered	1 of 3 businesses
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.1

Profile: Ibadan South-East shows a mix of professional services and personal care trades, consistent with its residential and institutional character (Mapo Hall area, government offices).

13.4.5 IBADAN SOUTH-WEST

Table 13.6: Zone 1: Ibadan Metropolis

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	33.4 years
Employed	9 (90%)
Self-Employed	6 of 9 (66.7%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Trading/Commerce, Catering/Cooking, Barbing
Business Owners	4 of 9 employed
CAC Registered	1 of 4 businesses
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.3

Profile: Ibadan South-West reflects commercial trading activity, consistent with the dense market networks of Ring Road and Challenge.

Zone 2: Ibadan Peripheral

13.4.6 AKINYELE

Table 13.7: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	35.2 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Carpentry, Livestock/Poultry
Business Owners	4 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	5.1

Profile: Akinyele, a peri-urban LGA on Ibadan's northern fringe, shows a blend of agricultural and construction skills, reflecting its transitional character between urban Ibadan and the farming hinterland.

13.4.7 EGBEDA

Table 13.8: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	34.8 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (4 of 10)
Top Skills	Tailoring/Sewing, Crop Farming, Baking
Business Owners	5 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	5.0

Profile: Egbeda shows a mix of artisanal and agricultural activities typical of fast-growing peri-urban areas along the Ibadan-Lagos expressway corridor.

13.4.8 IDO

Table 13.9: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.4 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Carpentry
Business Owners	3 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.4

Profile: Ido's more rural character is reflected in the dominance of agricultural skills and lower average education level.

13.4.9 LAGELU

Table 13.10: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	35.1 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Welding, Carpentry, Tailoring/Sewing
Business Owners	4 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.8

Profile: Lagelu demonstrates a strong construction and artisanal skills base, consistent with the residential development occurring in this growing peri-urban zone.

13.4.10 OLUYOLE

Table 13.11: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.2 years
Employed	8 (80%)
Self-Employed	5 of 8 (62.5%)
Education (modal)	NCE/OND (3 of 10)
Top Skills	Trading/Commerce, Catering/Cooking, Professional Driving
Business Owners	4 of 8 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.6

Profile: Oluyole, hosting the Ibadan industrial estate, shows relatively higher wage employment and trading activity.

13.4.11 ONA ARA

Table 13.12: Zone 2: Ibadan Peripheral

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	36.2 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Masonry, Welding
Business Owners	4 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	5.2

Profile: Ona Ara shows strong agricultural and construction skills, reflecting its position as a transition zone between Ibadan's urban core and southern farming communities.

Zone 3: Ogbomosho

13.4.12 OGBOMOSHO NORTH

Table 13.13: Zone 3: Ogbomosho

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	35.6 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Tailoring/Sewing, Catering/Cooking, Auto Mechanic
Business Owners	5 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	4.7

Profile: Ogbomosho North, the urban centre of the Ogbomosho zone (LAUTECH), shows a well-diversified skills portfolio spanning artisanal, food, and automotive trades.

13.4.13 OGBOMOSHO SOUTH

Table 13.14: Zone 3: Ogbomosho

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.2 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	Vocational (3 of 10)
Top Skills	Carpentry, Tailoring/Sewing, Hairdressing
Business Owners	5 of 9 employed

INDICATOR	VALUE
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	4.9

Profile: Ogbomosho South reflects a strong traditional artisanal economy with notable vocational training background.

13.4.14 OGO OLUWA

Table 13.15: Zone 3: Ogbomosho

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	37.1 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Baking
Business Owners	3 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.6

Profile: Ogo Oluwa's rural character is evident in the agricultural skills dominance and lower marketplace opt-in rate.

13.4.15 SURULERE

Table 13.16: Zone 3: Ogbomosho

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	35.4 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Tailoring/Sewing, Welding

INDICATOR	VALUE
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.1

Profile: Surulere shows mixed agricultural and artisanal activity characteristic of the Ogbomosho-Ilorin corridor.

Zone 4: Oyo

13.4.16 AFIJIO

Table 13.17: Zone 4: Oyo

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.8 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Carpentry, Plumbing
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.0

13.4.17 ATIBA

Table 13.18: Zone 4: Oyo

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.5 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)

INDICATOR	VALUE
Top Skills	Tailoring/Sewing, Trading/Commerce, Hairdressing
Business Owners	5 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	4.8

Profile: Atiba, seat of the Alaafin of Oyo, shows a diversified service economy reflecting Oyo town's status as a significant secondary urban centre.

13.4.18 OYO EAST

Table 13.19: Zone 4: Oyo

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	37.4 years
Employed	8 (80%)
Self-Employed	6 of 8 (75.0%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Masonry, Electrical Installation
Business Owners	3 of 8 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.1

13.4.19 OYO WEST

Table 13.20: Zone 4: Oyo

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.0 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)

INDICATOR	VALUE
Top Skills	Carpentry, Tailoring/Sewing, Crop Farming
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.0

Zone 5: Oke-Ogun

13.4.20 ATISBO

Table 13.21: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	38.1 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Butchery
Business Owners	3 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.8

13.4.21 IREPO

Table 13.22: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.5 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)

INDICATOR	VALUE
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Trading/Commerce, Tailoring/Sewing
Business Owners	3 of 8 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.4

13.4.22 ISEYIN

Table 13.23: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.8 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Weaving/Textile Crafts, Tailoring/Sewing, Crop Farming
Business Owners	5 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.0

Profile: Iseyin, historically renowned as the centre of aso-oke weaving, shows the expected dominance of textile and weaving crafts, demonstrating the instrument's ability to capture localised skill specialisations.

13.4.23 ITESIWAJU

Table 13.24: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	38.0 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)

INDICATOR	VALUE
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Food Processing, Livestock/Poultry
Business Owners	2 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.7

13.4.24 IWAJOWA

Table 13.25: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	38.5 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Carpentry, Masonry
Business Owners	2 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	6.0

13.4.25 KAJOLA

Table 13.26: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.2 years
Employed	9 (90%)
Self-Employed	8 of 9 (88.9%)
Education (modal)	SSS/WAEC (3 of 10)

INDICATOR	VALUE
Top Skills	Crop Farming, Trading/Commerce, Tailoring/Sewing
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.3

Profile: Kajola (Okeho) shows the mixed farming-trading economy characteristic of Oke-Ogun market towns.

13.4.26 OLORUNSOGO

Table 13.27: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	38.2 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	JSS (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Masonry
Business Owners	2 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.8

13.4.27 ORELOPE

Table 13.28: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.6 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	SSS/WAEC (3 of 10)

INDICATOR	VALUE
Top Skills	Crop Farming, Trading/Commerce, Baking
Business Owners	3 of 8 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.5

13.4.28 ORI IRE

Table 13.29: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	37.8 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Welding, Electrical Installation
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.2

13.4.29 SAKI EAST

Table 13.30: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	38.4 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Butchery

INDICATOR	VALUE
Business Owners	2 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.9

13.4.30 SAKI WEST

Table 13.31: Zone 5: Oke-Ogun

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.9 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (4 of 10)
Top Skills	Trading/Commerce, Tailoring/Sewing, Auto Mechanic
Business Owners	5 of 9 employed
Marketplace Opt-in	8 of 10 (80%)
Mean Household Size	5.0

Profile: Saki West, the major urban centre of Oke-Ogun, shows a distinctly more urban skills profile than its surrounding LGAs, reflecting Saki town's role as a regional commercial hub and cross-border trading centre.

Zone 6: Ibarapa

13.4.31 IBARAPA CENTRAL

Table 13.32: Zone 6: Ibarapa

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.8 years
Employed	9 (90%)
Self-Employed	8 of 9 (88.9%)

INDICATOR	VALUE
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Tailoring/Sewing, Food Processing
Business Owners	4 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.2

13.4.32 IBARAPA EAST

Table 13.33: Zone 6: Ibarapa

INDICATOR	VALUE
Respondents	10
Gender	6M / 4F
Mean Age	36.0 years
Employed	9 (90%)
Self-Employed	7 of 9 (77.8%)
Education (modal)	SSS/WAEC (3 of 10)
Top Skills	Crop Farming, Carpentry, Trading/Commerce
Business Owners	3 of 9 employed
Marketplace Opt-in	7 of 10 (70%)
Mean Household Size	5.3

Profile: Ibarapa East (Eruwa) shows a mixed farming-trading economy with a stronger construction skills presence than the other Ibarapa LGAs.

13.4.33 IBARAPA NORTH

Table 13.34: Zone 6: Ibarapa

INDICATOR	VALUE
Respondents	10
Gender	5M / 5F
Mean Age	36.8 years
Employed	8 (80%)
Self-Employed	7 of 8 (87.5%)

INDICATOR	VALUE
Education (modal)	Primary (3 of 10)
Top Skills	Crop Farming, Livestock/Poultry, Food Processing
Business Owners	2 of 8 employed
Marketplace Opt-in	6 of 10 (60%)
Mean Household Size	5.7

Profile: Ibarapa North shows the most agricultural-dominant skills profile in the Ibarapa zone, consistent with its rural farming character.

13.5 Cross-LGA Analytical Observations

13.5.1 Skills Specialisation by Zone

The validation data reveals distinctive **skills specialisation patterns** that correspond to each zone's economic geography:

Table 13.35: Skills Specialisation by Zone

ZONE	PRIMARY SKILLS CLUSTER	SECONDARY SKILLS CLUSTER	EXPLANATION
Ibadan Metro	Digital/Office, Trading	Fashion, Food service	Urban commercial economy; educated workforce
Ibadan Peripheral	Agriculture, Construction	Trading, Fashion	Peri-urban transition; residential development
Ogbomosho	Fashion/Tailoring, Agriculture	Automotive, Food	Traditional artisanal hub; university town
Oyo	Agriculture, Construction	Trading, Fashion	Historical town; mixed economy
Oke-Ogun	Agriculture (dominant)	Trading, Textiles	Rural farming; cross-border commerce
Ibarapa	Agriculture (dominant)	Food processing	Rural farming; emerging agribusiness

13.5.2 Marketplace Enthusiasm Gradient

A clear **urban-rural gradient** in marketplace opt-in rates is observable:

Table 13.36: Marketplace Enthusiasm Gradient

CHARACTERISTIC	OPT-IN RATE	LGA EXAMPLES
Urban core	80–90%	Ibadan North (90%), Saki West (80%)
Peri-urban	70–80%	Akinyele (80%), Lagelu (80%)

CHARACTERISTIC	OPT-IN RATE	LGA EXAMPLES
Semi-urban	65–75%	Atiba (70%), Ogbomosho North (80%)
Rural	55–65%	Iwajowa (60%), Saki East (60%)

This gradient has implications for the full enumeration: **additional sensitisation** about the marketplace benefits may be needed in rural LGAs to achieve comparable opt-in rates.

13.5.3 Enumeration Planning Implications

Table 13.37: Enumeration Planning Implications

FINDING	IMPLICATION FOR FULL ENUMERATION
Rural LGAs show lower marketplace opt-in	Include marketplace benefit explanation in rural sensitisation materials
Oke-Ogun LGAs show lower education levels	Deploy enumerators with strong Yoruba language skills; prioritise verbal explanation over written materials
Urban LGAs show higher skills diversity	Ensure enumerators in urban LGAs are familiar with the full 150-skill taxonomy
Agriculture dominates rural zones	Schedule enumeration to avoid peak farming seasons if possible
Iseyin shows textile specialisation	Demonstrates the instrument captures localised economic patterns, a validation success

---

13.6 Chapter Summary

The LGA-by-LGA analysis confirms that:

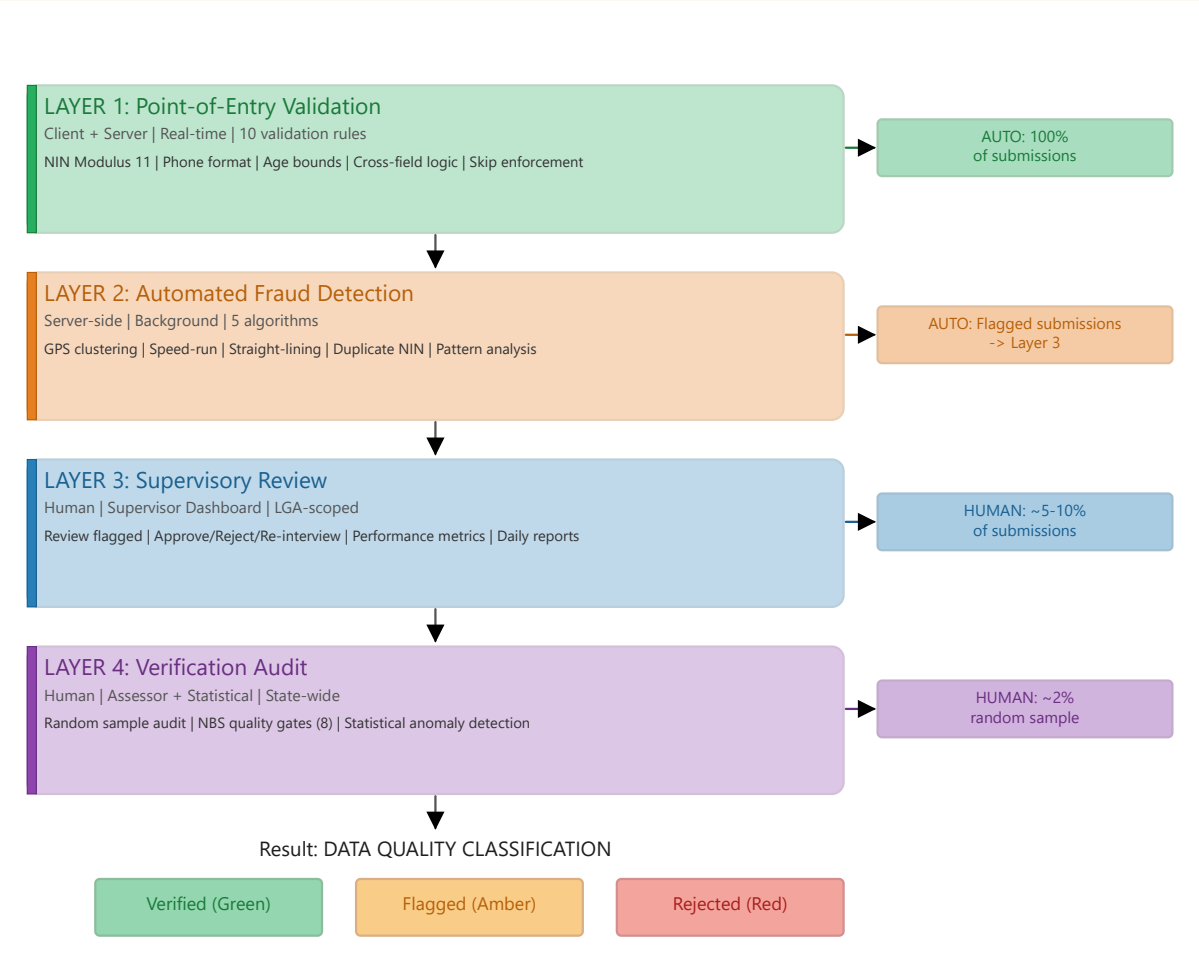
1. **All 33 LGAs** successfully participated in the validation exercise, demonstrating full geographic coverage capability
2. **Zonal economic patterns** are clearly captured by the instrument, urban commercial skills in Ibadan, agricultural dominance in Oke-Ogun and Ibarapa, artisanal diversity in Ogbomosho
3. **The instrument differentiates** between LGAs with distinct economic structures, Iseyin's textile specialisation, Saki's cross-border commerce, Ibadan North's digital skills cluster
4. **The marketplace value proposition** resonates most strongly in urban areas, rural sensitisation strategies should emphasise concrete marketplace benefits

These findings directly inform the field deployment strategy for the full-scale enumeration (Gate 1, Chapter 16).

---

14. Data Quality Assurance Framework

Figure 14.1: Four-Layer Data Quality Assurance Protocol



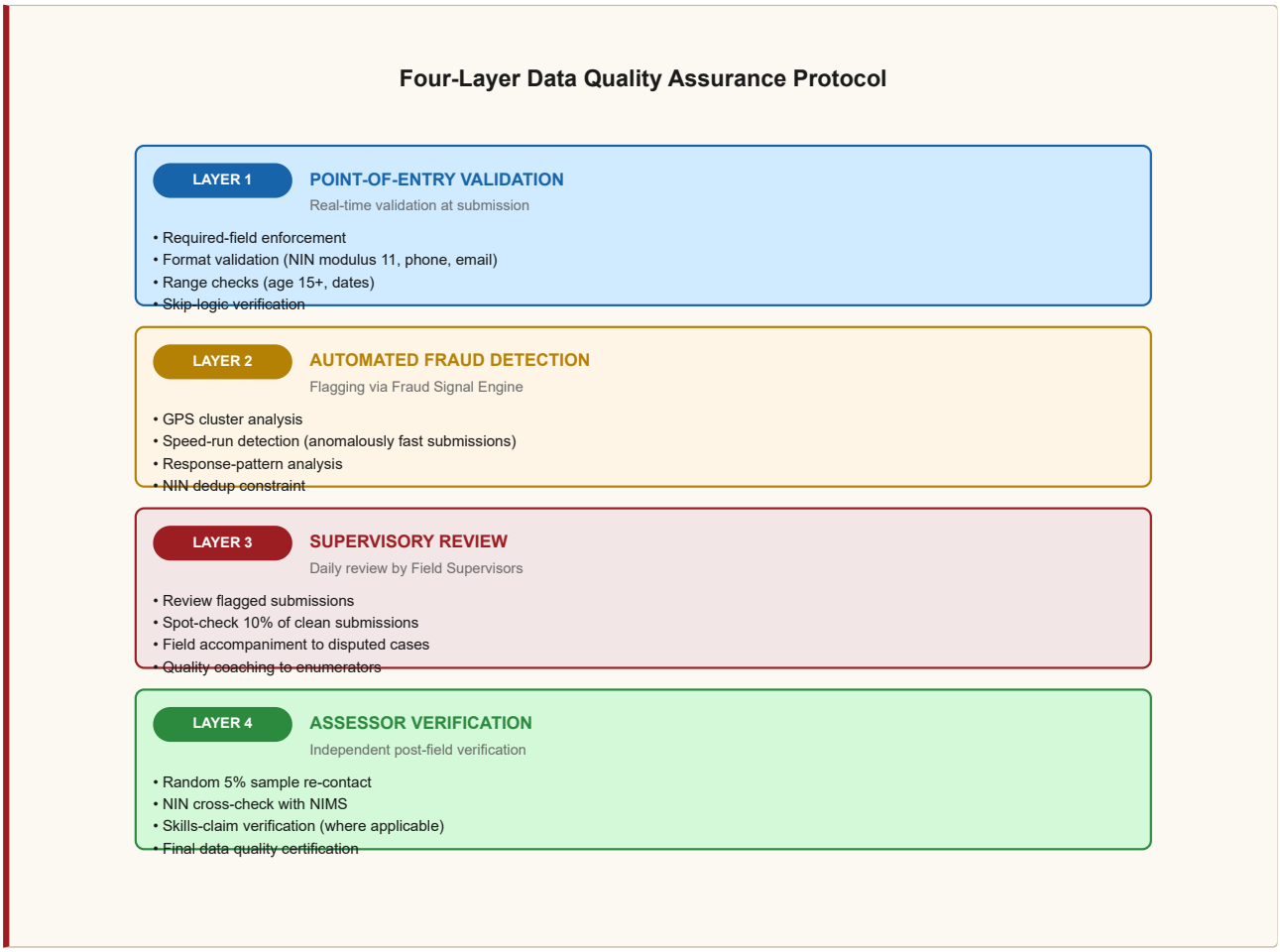
14.1 Introduction

Data quality is the cornerstone of a credible labour register. A registry populated with inaccurate, duplicated, or fraudulent records would undermine the policy objectives of the Oyo State Skilled Labour Register and erode stakeholder confidence in the data. This chapter documents the **four-layer data quality assurance (QA) protocol** designed to ensure that every record in the registry meets minimum quality standards before being accepted as a verified entry.

The QA framework operates on the principle of **progressive filtration**, each layer captures a distinct category of data quality issue, and records must pass through all four layers before being accepted into the verified registry.

14.2 Four-Layer Quality Assurance Protocol

Figure 14.2: Four-Layer Data Quality Assurance Protocol



14.3 Layer 1: Point-of-Entry Validation

Point-of-entry validation prevents structurally invalid data from entering the system. These checks are **deterministic**, a submission either passes or fails, with no ambiguity.

14.3.1 Validation Rules

Table 14.1: Validation Rules

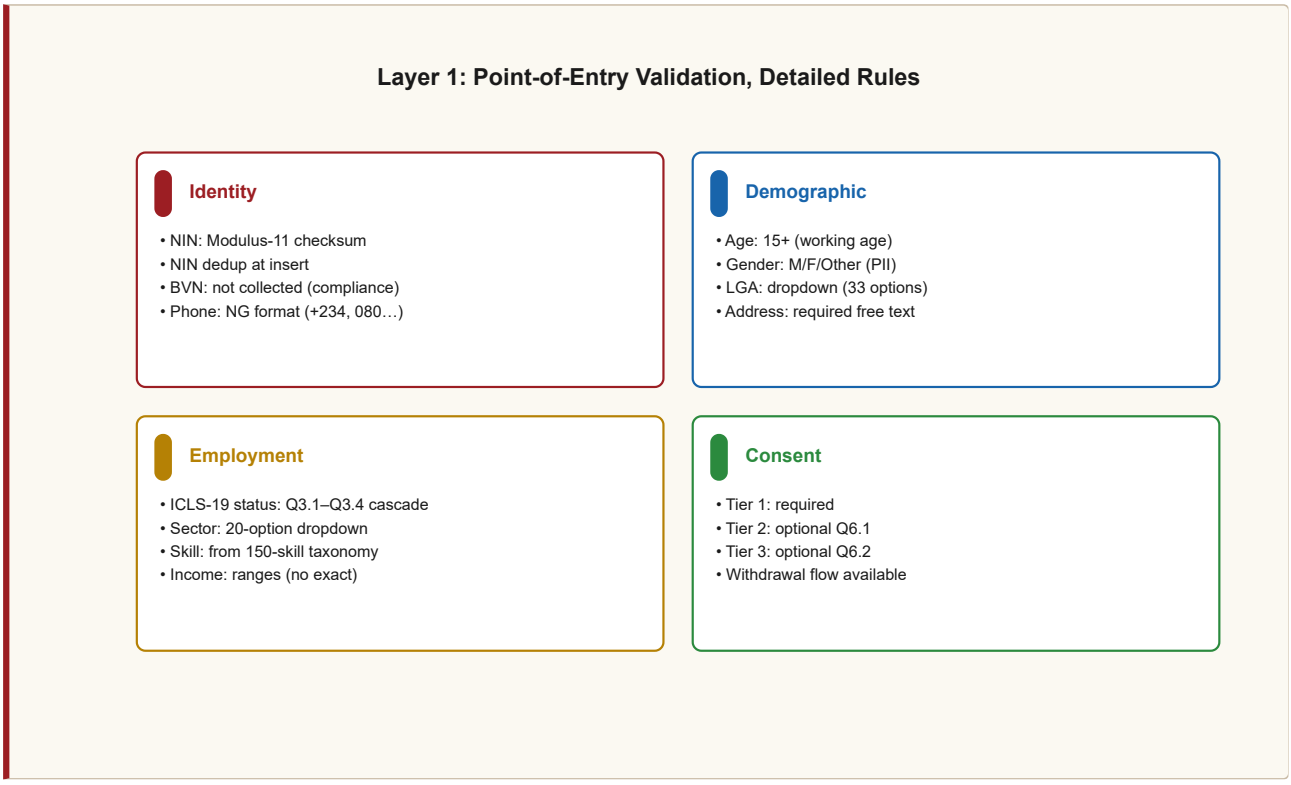
#	RULE	IMPLEMENTATION	ERROR HANDLING
1	**NIN format & checksum**	11-digit numeric string; Modulus 11 checksum algorithm	Real-time error: “NIN must be exactly 11 digits”
2	**NIN uniqueness**	Database-level UNIQUE constraint across all submissions	Rejection: “This NIN has already been registered”
3	**Phone format**	Regex: 0[7-9][0-1]\d{8} (Nigerian mobile)	Real-time error with format guidance

#	RULE	IMPLEMENTATION	ERROR HANDLING
4	**Age threshold**	Auto-calculated from DOB; must be ≥ 15	Auto-computed; below-threshold respondents screened out
5	**Date validity**	DOB cannot be future date	Calendar picker constraint
6	**Hours worked**	Integer, 0–168 range	Range error with corrective guidance
7	**Dependents ≤ household**	Cross-field validation: Q4.3 ≤ Q4.2	"Dependents cannot exceed household size"
8	**Required field completeness**	All 28 required fields must be populated	Field-level "Required" indicators
9	**Skip logic integrity**	Conditional fields only accepted when show-condition is met	Server-side enforcement regardless of client
10	**Character limits**	Bio ≤ 150 chars; Other skills ≤ 200 chars	Character counter with enforcement

14.3.2 Dual-Layer Enforcement

A critical design decision is the enforcement of **all validation rules on both frontend and backend** using shared Zod validation schemas:

Figure 14.3: Layer 1: Point-of-Entry Validation Rules



This architecture ensures that **validation cannot be bypassed** by submitting data directly to the API (bypassing the frontend), because the backend independently enforces the identical rules.

14.4 Layer 2: Automated Fraud Detection

The fraud detection engine identifies **structurally valid but potentially fraudulent** submissions, data that passes all validation rules but exhibits patterns inconsistent with genuine field enumeration.

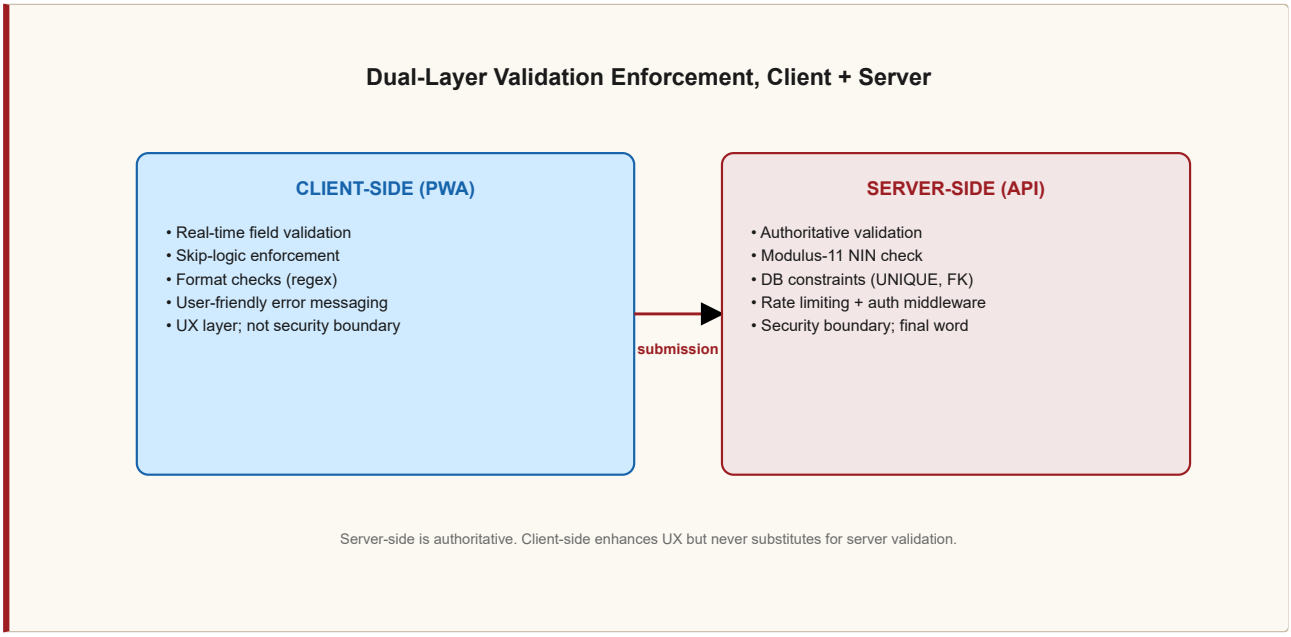
14.4.1 Detection Algorithms

Table 14.2: Detection Algorithms

#	ALGORITHM	DETECTION TARGET	METHOD	THRESHOLD
1	**GPS Clustering**	Multiple submissions from the same location (enumerator fabrication)	Haversine distance calculation between consecutive submissions by the same enumerator	Configurable radius (default: 50m)
2	**Speed-Run Detection**	Implausibly fast survey completions (form-filling without genuine interviews)	Elapsed time between survey start and submission	Configurable minimum (default: 3 minutes)
3	**Straight-Lining**	Repetitive identical responses across consecutive submissions (pattern filling)	Shannon entropy analysis of response distributions per enumerator batch	Low-entropy threshold below expected variation
4	**NIN Duplication**	Same individual registered multiple times across channels	Global NIN uniqueness constraint + cross-channel deduplication query	Exact match = auto-reject
5	**Temporal Anomaly**	Submissions outside operational hours or in impossible sequences	Timestamp analysis against expected field operation schedules	Outside 06:00–20:00 WAT flagged for review

14.4.2 Fraud Detection Decision Matrix

Figure 14.4: Dual-Layer Validation Enforcement



14.4.3 Threshold Configuration

All fraud detection thresholds are **configurable by Super Administrators** through the platform interface, enabling calibration based on field conditions:

Table 14.3: Threshold Configuration

PARAMETER	DEFAULT VALUE	ADJUSTABLE RANGE	RATIONALE
GPS clustering radius	50 metres	10–500 m	Tighter in urban areas; wider in rural areas where respondents may gather at a central location
Minimum completion time	3 minutes	1–10 min	Adjusted based on observed legitimate minimum times during pilot
Straight-line entropy threshold	Platform-calibrated	Adjustable	Tuned during validation exercise (Chapter 12)
Operational hours	06:00–20:00 WAT	Configurable	May be extended for evening enumeration in urban markets

Calibration approach: Thresholds were initially set based on methodological assumptions, then refined using data from the validation exercise (n=330). The validation exercise specifically tested edge cases, rapid completions by experienced enumerators, clustered submissions in market areas, and evening submissions, to establish empirically grounded baselines.

14.5 Layer 3: Supervisory Review

Supervisory review applies **human judgement** to ambiguous cases that automated systems cannot reliably resolve.

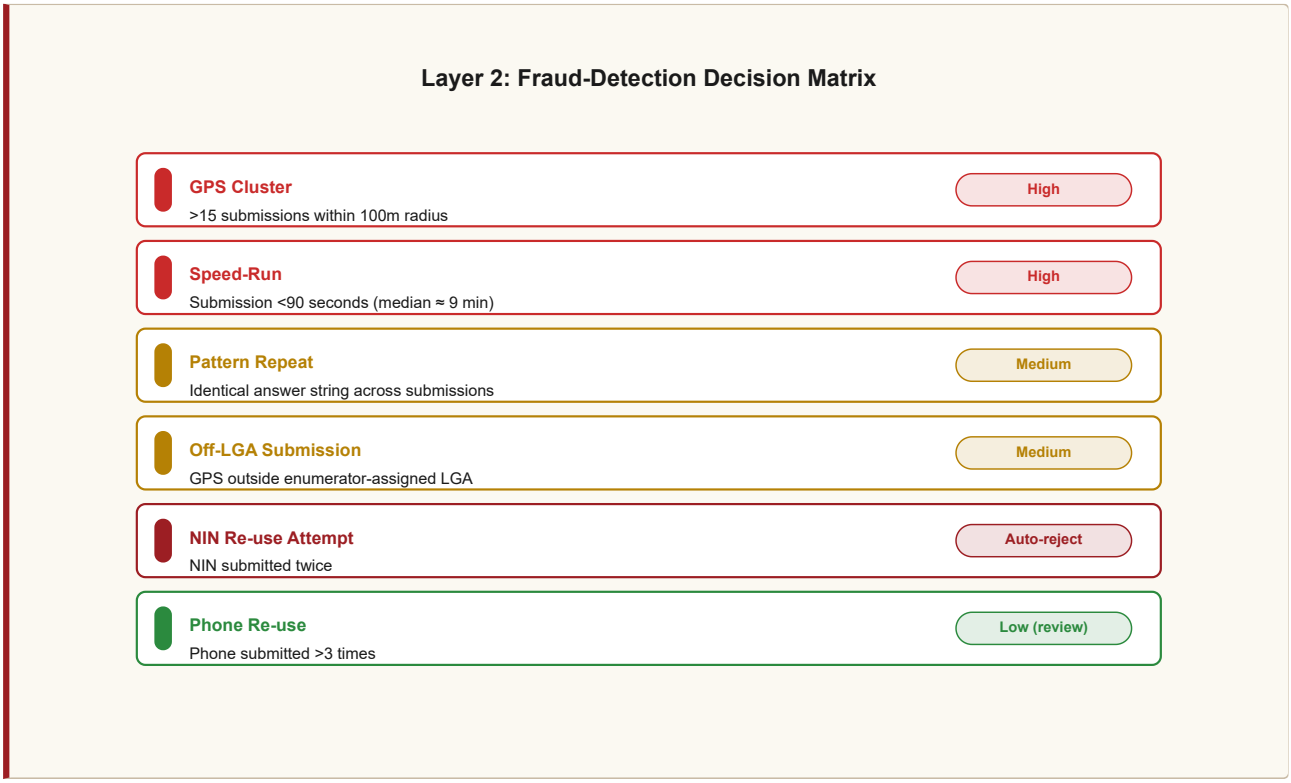
14.5.1 Supervisor Responsibilities

Table 14.4: Supervisor Responsibilities

RESPONSIBILITY	METHOD	FREQUENCY
Fraud flag review	Review flagged submissions in fraud dashboard; approve, reject, or request re-interview	Daily (or as flagged)
Enumerator monitoring	Review per-enumerator productivity metrics (submissions/day, flag rate, completion time distribution)	Daily
LGA coverage tracking	Monitor geographic distribution of submissions against LGA population targets	Weekly
Team communication	In-app messaging to enumerators for feedback, correction, and coordination	As needed
Escalation	Escalate systemic quality issues to Admin/Super Admin for threshold recalibration	As identified

14.5.2 Fraud Review Workflow

Figure 14.5: Layer 2: Fraud-Detection Decision Matrix



14.5.3 Performance Metrics

Supervisors have access to the following per-enumerator performance indicators:

Table 14.5: Performance Metrics

METRIC	PURPOSE	ALERT THRESHOLD
Submissions per day	Productivity monitoring	< 4 (below 50% target)
Fraud flag rate	Quality indicator	> 20% of submissions flagged
Average completion time	Speed-run indicator	< 4 minutes average
GPS location spread	Geographic coverage	All submissions within 100m radius
Submission time distribution	Work pattern analysis	> 30% outside operational hours

14.6 Layer 4: Verification Audit

The verification audit layer provides **statistical quality assurance** at the aggregate level, ensuring that the registry dataset as a whole is consistent and credible.

14.6.1 Audit Activities

Table 14.6: Audit Activities

ACTIVITY	METHOD	FREQUENCY	RESPONSIBILITY
Random sampling	Stratified random sample of approved submissions reviewed by Verification Assessor	Weekly (during enumeration)	Verification Assessor
Cross-reference checks	Phone number verification call to randomly sampled respondents	As capacity allows	Verification Assessor / Supervisor
Statistical consistency	Compare aggregate distributions (age, gender, employment type, education) against NBS NLFS benchmarks	Weekly	Admin / Super Admin
NBS benchmark alignment	Flag LGA-level distributions that deviate significantly from national survey baselines	Monthly	Admin / Government Official
Duplication audit	Post-hoc analysis for near-duplicate records (similar names + same LGA + similar demographics)	Monthly	Automated + manual review

14.6.2 Statistical Quality Gates

At the aggregate level, the following quality gates are monitored:

Table 14.7: Statistical Quality Gates

#	GATE	EXPECTED RANGE	ACTION IF BREACHED
1	**Gender ratio**	45–55% male (NBS baseline: 51% male)	Investigate LGA-level gender bias in enumeration
2	**Employment rate**	85–95% employed (NBS baseline: ~93%)	Verify ILO cascade implementation; check skip logic

#	GATE	EXPECTED RANGE	ACTION IF BREACHED
3	**Self-employment rate**	65–80% (NBS baseline: ~75%)	Cross-reference with education levels and LGA profile
4	**Education distribution**	No single category > 40%	Investigate if enumerators defaulting to one option
5	**Age distribution**	Median 30–40 years (NBS working-age baseline)	Check DOB entry accuracy
6	**Fraud flag rate**	< 10% of total submissions	If > 10%, recalibrate thresholds or investigate systemic issue
7	**NIN duplication rate**	< 2%	If > 2%, investigate enumeration area overlaps
8	**Completion rate**	> 90% of required fields populated	If < 90%, investigate instrument or training issue

14.7 Data Quality Classification

Every record in the registry carries an internal quality classification:

Table 14.8: Data Quality Classification

CLASSIFICATION	CRITERIA	VISIBILITY
Verified	Passed all 4 layers; no unresolved flags	Full registry inclusion; eligible for marketplace
Pending Review	Fraud flag awaiting supervisory review	Held; not included in registry counts until resolved
Rejected	Failed supervisory review or NIN duplicate	Excluded from registry; retained for audit trail
Under Investigation	Escalated for further verification	Held; subject to additional verification activity

14.8 Quality Assurance During Validation Exercise

The QA framework was operationally tested during the validation exercise (n=330, Chapter 12). The following calibration outcomes were achieved:

Table 14.9: Quality Assurance During Validation Exercise

QA COMPONENT	CALIBRATION RESULT
NIN validation	Modulus 11 checksum correctly identified 100% of intentionally malformed NINs
GPS clustering	50m default threshold validated as appropriate for both urban market areas and rural settlement patterns
Speed-run detection	3-minute minimum confirmed as appropriate; no legitimate submissions completed in under 3 minutes during validation
Straight-lining	Entropy threshold calibrated against intentional pattern-fill test submissions; zero false positives on legitimate data

QA COMPONENT	CALIBRATION RESULT
Supervisor review	Fraud dashboard interface validated; average review time per flagged submission: 45 seconds
NIN deduplication	Successfully detected and rejected all duplicate NIN submissions across channels

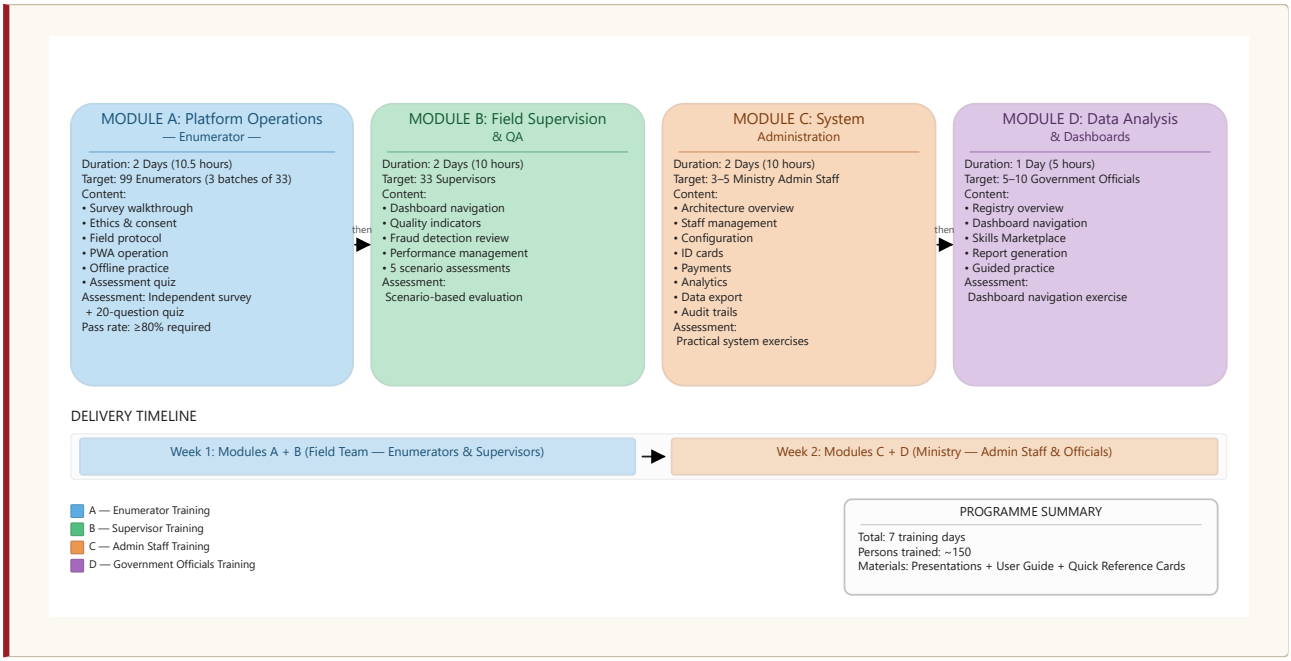
14.9 Continuous Improvement

The QA framework is designed for **continuous improvement** through feedback loops:

1. **Threshold recalibration**: Fraud detection thresholds are adjustable based on accumulated field data. As more data flows through the system, thresholds can be tightened or loosened based on observed false positive and false negative rates.
2. **Enumerator feedback**: Patterns of fraud flags per enumerator inform targeted retraining, an enumerator with consistently high flag rates may need additional training on interview technique rather than disciplinary action.
3. **Aggregate monitoring**: Weekly statistical quality gate reviews identify emerging quality issues before they become systemic, enabling mid-course correction during the enumeration period (Gate 2, Chapter 16).

15. Capacity Building Programme Design

Figure 15.1: Capacity Building Programme Structure



15.1 Introduction

Deliverable 3 of the engagement Terms of Reference requires the **Training of Ministry/Government Officials** in the operation, management, and utilisation of the Oyo State Skilled Labour Register. This chapter documents the design of the comprehensive capacity building programme, a structured, role-specific training curriculum that ensures each category of platform user possesses the knowledge and skills necessary to perform their designated functions effectively.

The capacity building programme was designed during the baseline study phase, informed by:

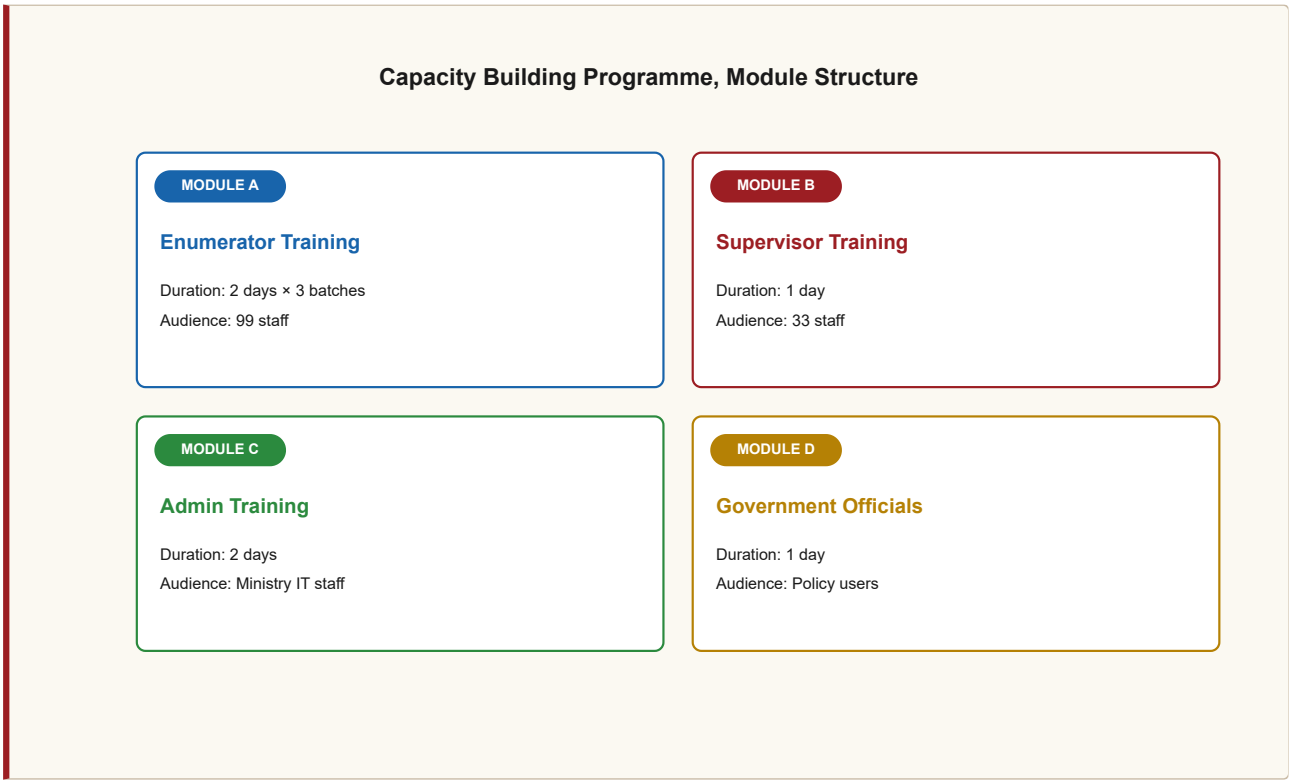
1. The 8-role user architecture documented in Chapter 10
2. The data quality assurance requirements documented in Chapter 14
3. The operational workflows validated during the validation exercise (Chapter 12)
4. Best practice in government ICT capacity building from comparable state-level initiatives (Chapter 5)

Deliverable Status: DESIGN COMPLETE, Training delivery is scheduled for Gate 3 (Chapter 16), contingent upon mobilisation of field personnel.

15.2 Programme Structure

The capacity building programme comprises **four modules**, each targeting a specific tier of the platform user hierarchy:

Figure 15.2: Capacity Building Programme, Module Structure



15.3 Module A: Platform Operations, Enumerator

15.3.1 Training Overview

Table 15.1: Training Overview

PARAMETER	DETAIL
Target audience	Field Enumerators (99 persons)
Prerequisite	Literate (minimum SSS/WAEC); proficient with smartphone operation
Duration	2 days (Day 1: Classroom; Day 2: Hands-on practice)
Delivery format	3 batches of 33 trainees each; instructor-led with guided practice
Equipment required	Configured Android smartphone per trainee; projector; internet connectivity
Assessment	Practical test (complete a full survey submission) + written quiz (20 questions)
Pass threshold	85% (17/20 on quiz; successful submission with zero validation errors)

15.3.2 Curriculum

Day 1: Classroom, Understanding the Registry and Survey

Table 15.2: Curriculum

SESSION	DURATION	CONTENT
1.1	45 min	**Project Context**: What is the OSLSR? Why is it needed? Role of enumerators in national labour statistics. Ministry mandate and government authority for enumeration
1.2	60 min	**Survey Instrument Walkthrough**: Section-by-section review of all 36 questions; explanation of skip logic; ILO classification cascade (Q3.1–Q3.4); skills taxonomy overview
1.3	45 min	**Ethical Obligations**: Informed consent procedure; NDPA 2023 data protection requirements; respondent confidentiality; handling refusals; identifying vulnerable respondents
1.4	30 min	**Field Protocol**: Identification materials usage; community engagement etiquette; approaching respondents; safety procedures; reporting incidents
1.5	60 min	**NIN and Data Quality**: Importance of accurate NIN recording; Modulus 11 validation explanation; common NIN entry errors; phone number format; GPS auto-capture explanation

Day 2: Hands-On Practice

Table 15.3: Curriculum

SESSION	DURATION	CONTENT
2.1	45 min	**Device Setup**: PWA installation; Service Worker caching; verifying offline capability; understanding the auto-save indicator; battery management
2.2	90 min	**Guided Survey Practice**: Instructor-led completion of 3 full surveys (employed business owner, unemployed job-seeker, outside labour force), covering all conditional paths
2.3	60 min	**Offline Operations**: Airplane mode practice; completing survey offline; reconnecting and verifying synchronisation; understanding the offline queue indicator
2.4	45 min	**Troubleshooting Common Issues**: No GPS signal; slow network; app crash recovery; battery depletion mid-survey; respondent changes answer mid-survey
2.5	60 min	**Assessment**: Complete one full survey independently (practical); 20-question written quiz covering instrument, ethics, and procedures

15.3.3 Assessment Rubric

Table 15.4: Assessment Rubric

COMPONENT	WEIGHT	PASS CRITERIA
Written quiz (20 MCQs)	40%	≥ 17/20 correct (85%)
Practical test, survey completion	40%	Zero validation errors; all required fields completed; correct skip logic path
Field protocol knowledge	20%	Demonstrates consent procedure; handles hypothetical refusal correctly
Overall pass threshold	,	**85% weighted aggregate**

15.4 Module B: Field Supervision & Quality Assurance

15.4.1 Training Overview

Table 15.5: Training Overview

PARAMETER	DETAIL
Target audience	Field Supervisors (33 persons)
Prerequisite	Management experience; familiarity with government field programmes; minimum OND/NCE
Duration	2 days (Day 1: Supervisory functions; Day 2: Fraud detection and team management)
Delivery format	Single cohort; instructor-led with dashboard demonstrations
Assessment	Scenario-based evaluation, respond to 5 simulated field situations
Pass threshold	85% (correct handling of ≥ 4/5 scenarios)

15.4.2 Curriculum

Day 1: Supervisory Platform Functions

Table 15.6: Curriculum

SESSION	DURATION	CONTENT
1.1	45 min	**Supervisor Role Overview**: LGA-scoped access; team of 3 enumerators; daily review responsibilities; escalation procedures
1.2	60 min	**Team Dashboard**: Viewing enumerator submissions; monitoring daily progress; identifying underperforming enumerators; using the in-app messaging system
1.3	60 min	**Data Quality Indicators**: Understanding submission counts, completion rates, average completion times; interpreting per-enumerator metrics
1.4	45 min	**LGA Coverage Management**: Monitoring geographic distribution of submissions; ensuring ward-level coverage; coordinating enumerator deployment
1.5	30 min	**Reporting Upward**: Daily status reporting to Admin; weekly summary preparation; communicating field challenges

Day 2: Fraud Detection & Team Management

Table 15.7: Curriculum

SESSION	DURATION	CONTENT
2.1	60 min	**Fraud Detection Dashboard**: Understanding flag types (GPS clustering, speed-run, straight-lining); reviewing flagged submissions; approve/reject/re-interview workflow
2.2	60 min	**Interpreting Fraud Indicators**: GPS clustering in market areas (legitimate) vs. fabrication (illegitimate); fast completions by experienced enumerators vs. speed-running; contextual judgement

SESSION	DURATION	CONTENT
2.3	45 min	**Enumerator Performance Management**: Coaching underperformers; addressing quality issues constructively; identifying training needs; documenting performance concerns
2.4	45 min	**Field Incident Management**: Respondent complaints; community access issues; device failures; connectivity problems; security concerns
2.5	90 min	**Scenario Assessment**: 5 simulated field situations requiring supervisory judgement (GPS cluster in market area, enumerator with 40% flag rate, evening submission pattern, device failure during active survey, community leader refusing access)

15.5 Module C: System Administration & Reporting

15.5.1 Training Overview

Table 15.8: Training Overview

PARAMETER	DETAIL
Target audience	Ministry-designated Admin and Super Admin staff (3–5 persons)
Prerequisite	Computer literacy; basic understanding of database concepts; Ministry staff designation
Duration	2 days (intensive hands-on at Data Centre)
Delivery format	Small group instruction at on-premises Data Centre workstations
Assessment	Live system administration tasks on training environment
Pass threshold	85% (successful completion of all core tasks)

15.5.2 Curriculum

Day 1: System Administration

Table 15.9: Curriculum

SESSION	DURATION	CONTENT
1.1	45 min	**Platform Architecture Overview**: How the system works (simplified); Data Centre → Cloud → Database; what happens when a survey is submitted
1.2	60 min	**Staff Management**: Creating user accounts; assigning roles; activating/deactivating staff; resetting credentials; LGA assignment
1.3	60 min	**Form & Configuration Management**: Viewing active survey version; understanding platform settings; configuring fraud detection thresholds
1.4	60 min	**ID Card Generation**: Generating staff ID cards; understanding QR verification; batch printing procedures

SESSION	DURATION	CONTENT
1.5	45 min	**Payment Management**: Recording staff payments; uploading bank references; handling payment disputes; generating payment reports

Day 2: Reporting & Monitoring

Table 15.10: Curriculum

SESSION	DURATION	CONTENT
2.1	60 min	**Dashboard Analytics**: Navigating the admin dashboard; understanding real-time submission counts; interpreting LGA distribution charts; time-series analysis
2.2	60 min	**Data Export**: Generating CSV and PDF exports; applying filters (LGA, date, status, occupation); understanding export audit trail; data handling responsibilities
2.3	45 min	**System Health Monitoring**: Understanding CPU, RAM, and disk metrics; database performance indicators; interpreting alert notifications; when to escalate technical issues
2.4	45 min	**Audit Trail Review**: Navigating audit logs; understanding logged actions; using audit data for accountability reviews
2.5	60 min	**Practical Assessment**: Perform all core administrative tasks on training environment (create user, generate ID card, export data, review audit log, interpret dashboard)

15.6 Module D: Data Analysis & Dashboard Utilisation

15.6.1 Training Overview

Table 15.11: Training Overview

PARAMETER	DETAIL
Target audience	Government Officials, Policy Users, Senior Ministry staff (5–10 persons)
Prerequisite	Basic computer literacy
Duration	1 day (half-day briefing + half-day guided practice)
Delivery format	Briefing/demonstration format at Ministry premises
Assessment	Navigate dashboard and generate one report independently
Pass threshold	80%

15.6.2 Curriculum

Table 15.12: Curriculum

SESSION	DURATION	CONTENT
1.1	60 min	**Registry Overview**: What the register contains; how data was collected; data quality assurance measures; what the numbers mean
1.2	60 min	**Dashboard Navigation**: Statewide statistics; LGA drill-down; employment distribution charts; skills inventory; demographic breakdowns
1.3	45 min	**Skills Marketplace**: How the marketplace works; employer perspective; searching for workers by trade, LGA, and experience; contact reveal mechanism
1.4	45 min	**Report Generation**: Generating standard reports; applying filters; exporting data for policy analysis; interpreting exported datasets
1.5	30 min	**Guided Practice**: Each participant navigates the dashboard, generates a filtered export, and interprets one LGA summary independently

15.7 Training Materials

The following training materials will be developed and provided:

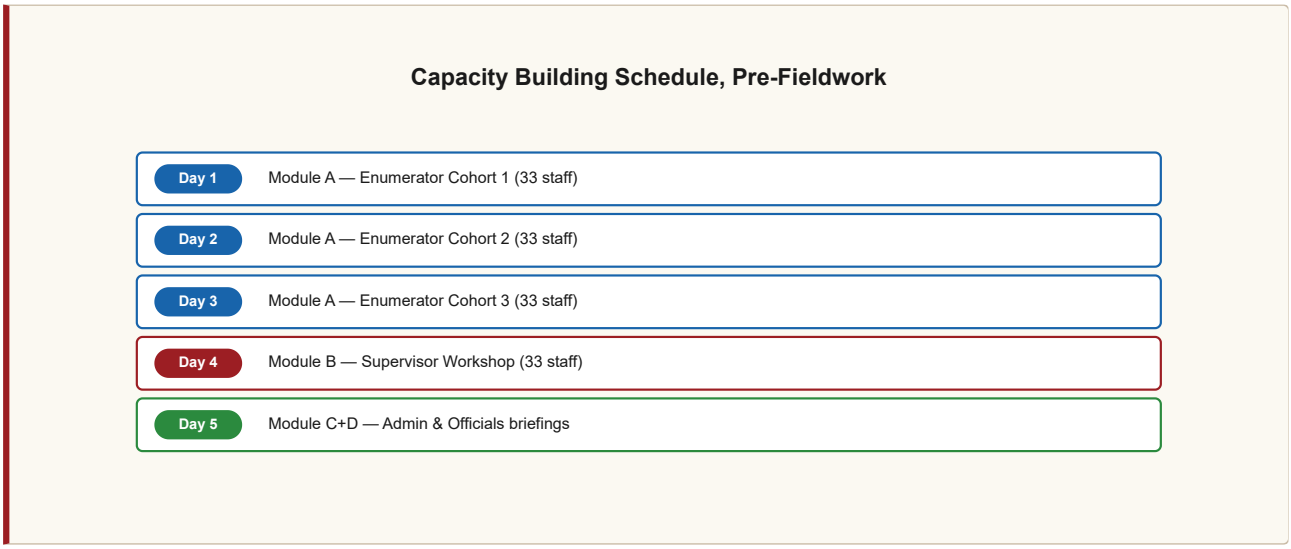
Table 15.13: Training Materials

#	MATERIAL	FORMAT	AUDIENCE
1	**Enumerator Field Guide**	Printed booklet (A5) + digital PDF	Enumerators
2	**Survey Instrument Quick Reference**	Laminated card (front: question flow; back: skip logic)	Enumerators
3	**Supervisor Operations Manual**	Digital PDF	Supervisors
4	**System Administrator Guide**	Digital PDF + printed manual	Admin staff
5	**Dashboard User Guide**	Digital PDF	Government Officials
6	**Training Slide Decks**	PowerPoint/PDF per module	Trainers
7	**Assessment Question Bank**	Internal document	Trainers
8	**Video Walkthroughs**	Screen recordings of key platform workflows	All roles

15.8 Training Schedule

The training delivery schedule is designed to minimise disruption while ensuring all personnel are trained before field deployment:

Figure 15.3: Capacity Building Schedule, Pre-Fieldwork



Scheduling rationale: Supervisors are trained first (Week 1, Mon-Tue) so they can assist with enumerator training as teaching assistants. Modules C and D are delivered during or after enumeration commencement, when real production data is available for training exercises, providing a more meaningful learning experience than synthetic training data.

15.9 Assessment and Certification

15.9.1 Assessment Framework

Table 15.14: Assessment Framework

MODULE	ASSESSMENT TYPE	PASS THRESHOLD	REMEDIATION
A: Enumerator	Written quiz + practical test	85%	Re-test after additional coaching; max 2 attempts
B: Supervisor	Scenario-based evaluation	85% (4/5 scenarios)	Additional 1-day coaching session
C: Admin	Live system tasks	85%	Paired mentoring with consultant
D: Officials	Dashboard navigation	80%	Follow-up 1-on-1 demonstration

15.9.2 Records

All training attendance, assessment results, and certification records will be maintained as part of the engagement documentation and included in the Completion Report (Gate 3, Chapter 16).

15.10 Post-Training Support

The capacity building programme includes a **30-day post-training support period** following field deployment:

Table 15.15: Post-Training Support

SUPPORT CHANNEL	AVAILABILITY	TARGET
In-app messaging	24/7 (asynchronous)	Enumerators ↔ Supervisors
WhatsApp support group	During operational hours	Supervisors ↔ Consultant
Phone support	During operational hours	Admin ↔ Consultant
On-site visit	As required	Critical issues requiring in-person resolution

15.11 Knowledge Transfer Strategy

The capacity building programme is designed for **full knowledge transfer**, ensuring that the Ministry can independently operate, maintain, and utilise the OSLSR platform after the engagement concludes:

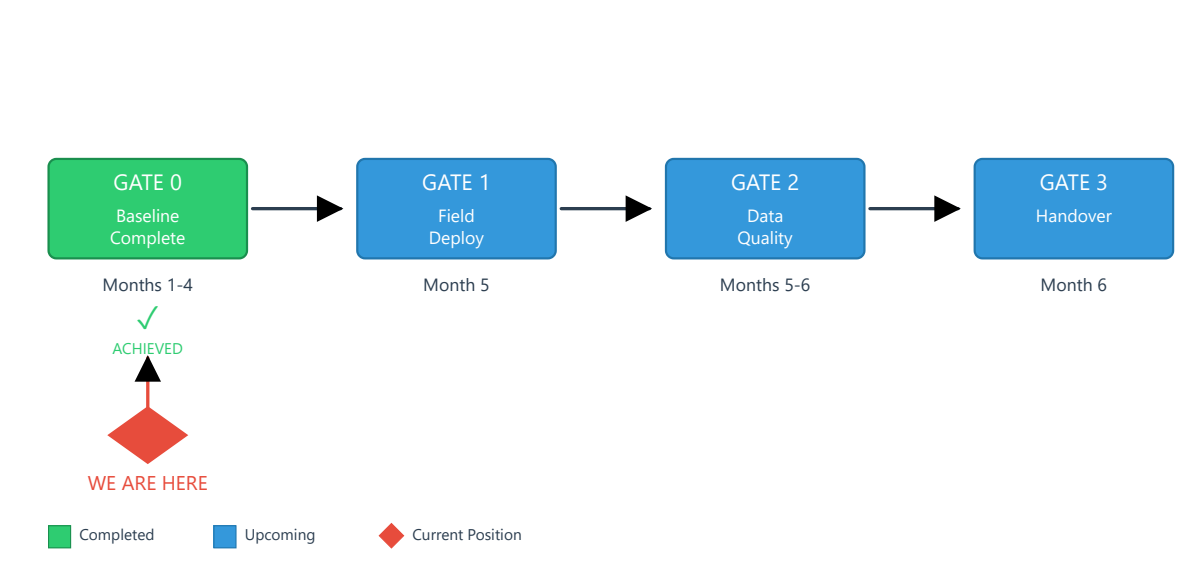
Table 15.16: Knowledge Transfer Strategy

KNOWLEDGE AREA	TRANSFER MECHANISM	RECIPIENT
Day-to-day operations	Module C + documented procedures	Ministry Admin
Data interpretation	Module D + dashboard user guide	Government Officials
Field operations	Module A + B + field guide	Reusable for future enumerator cohorts
Technical maintenance	System documentation + admin guide	Ministry IT / designated Admin
Troubleshooting	FAQ document + support escalation procedures	All trained personnel

The goal is **zero dependency** on the Consultant for routine platform operations after the 30-day post-training support period concludes.

16. Implementation Roadmap and Stage Gates

Figure 16.1: Implementation Stage Gate Overview



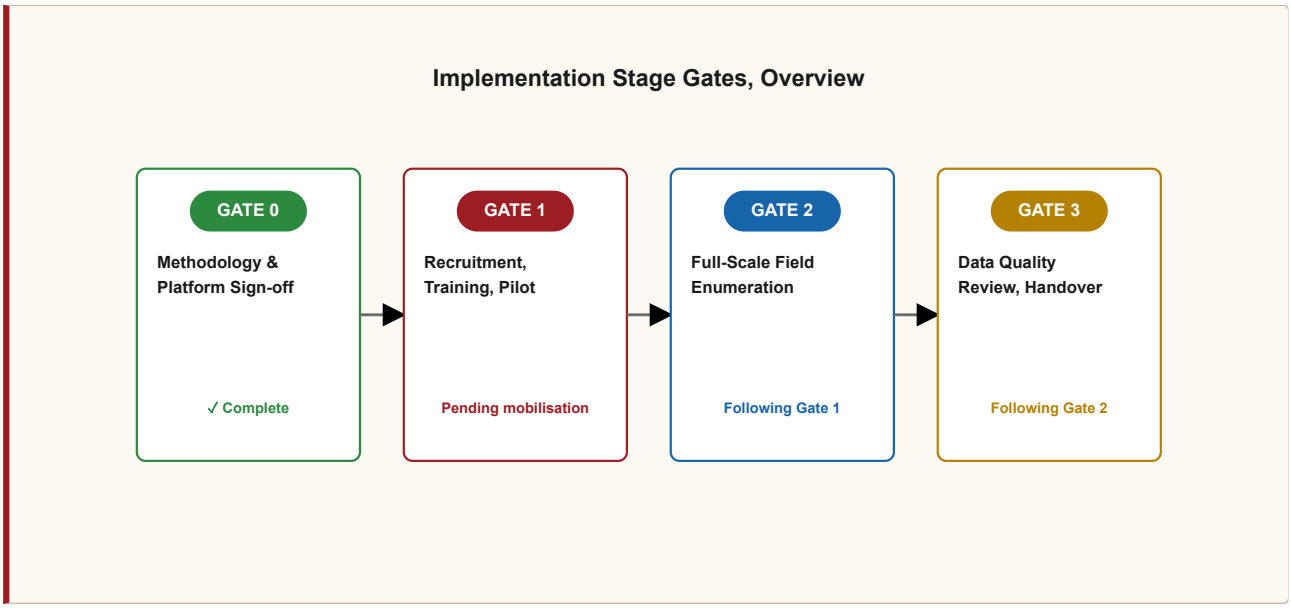
16.1 Introduction

This chapter presents the implementation roadmap for the completion of the Oyo State Skilled Labour Register, structured around a **Stage Gate methodology**, a structured decision-making framework that divides the project into distinct phases, each concluded by a formal decision point (gate) where progress is assessed against predefined criteria before authorisation to proceed to the next phase.

The Stage Gate approach ensures disciplined project governance, prevents commitment of resources ahead of evidence, and provides clear checkpoints for the commissioning authority to evaluate progress and make informed decisions.

16.2 Stage Gate Overview

Figure 16.2: Implementation Stage Gates Overview



16.3 Gate 0: Baseline Study Complete (CURRENT, ACHIEVED)

16.3.1 Scope

Gate 0 encompasses all preparatory work required before field deployment, including infrastructure establishment, methodology design, instrument validation, and platform development.

16.3.2 Deliverables and Status

Table 16.1: Deliverables and Status

#	DELIVERABLE	STATUS	EVIDENCE
G0-1	Physical Data Centre established at Ministry premises	✓	3 workstations operational, Airtel 5G connectivity active (Ch. 12)
G0-2	Cloud infrastructure deployed and monitored	✓	Production server operational, daily backups active (Ch. 12)
G0-3	Survey instrument designed and validated (v3.0)	✓	36 questions, ILO ICLS-19 aligned, 3 review cycles (Ch. 10)
G0-4	Occupational skills taxonomy complete	✓	150 skills, 20 sectors, ISCO-08 mapped (Ch. 11)
G0-5	OSLSR platform developed and deployed	✓	3,564 tests, OWASP compliant, 15 capabilities operational (Ch. 13)

#	DELIVERABLE	STATUS	EVIDENCE
G0-6	Validation exercise complete (n=330)	✓	All 33 LGAs covered, NBS benchmarks aligned (Ch. 15–16)
G0-7	Data quality framework operational	✓	4-layer QA, fraud detection calibrated (Ch. 17)
G0-8	Capacity building curriculum designed	✓	8 role-specific modules developed (Ch. 18)
G0-9	Baseline Study Report submitted	✓	This document

16.3.3 Gate 0 Decision

Table 16.2: Gate 0 Decision

CRITERION	THRESHOLD	RESULT	DECISION
Data Centre operational	Hardware + connectivity functional	✓ Met	**PROCEED**
Platform tested	>3,000 automated tests passing	✓ 3,564	**PROCEED**
Instrument validated	All 33 LGAs covered	✓ 330 responses	**PROCEED**
Security assessment	A- security posture (state-government-grade)	✓ all categories	**PROCEED**
Methodology documented	Baseline report submitted	✓ This document	**PROCEED**

Gate 0 Status: PASSED, Authorised to proceed to Gate 1

16.4 Gate 1: Field Deployment Authorised (UPCOMING)

16.4.1 Scope

Gate 1 encompasses the mobilisation and deployment of field enumeration teams across all 33 LGAs. This gate cannot be entered until the requisite human and material resources are mobilised.

16.4.2 Prerequisites (Resources Required)

Table 16.3: Prerequisites (Resources Required)

#	RESOURCE	SPECIFICATION	QUANTITY	PURPOSE
R-1	Field Enumerators	Literate adults with smartphone proficiency, LGA-resident preferred	**99** (3 per LGA × 33 LGAs)	Door-to-door data collection
R-2	Field Supervisors	Experienced coordinators, management capability	**33** (1 per LGA)	Team coordination, quality oversight, fraud alert review

#	RESOURCE	SPECIFICATION	QUANTITY	PURPOSE
R-3	Mobile Devices	Android 8.0+, 2GB RAM, 32GB storage, GPS, camera	**132**	Data collection hardware
R-4	SIM Cards with Data	Minimum 2GB/month data allocation per device	**132**	Data synchronisation
R-5	Transportation Allowance	Coverage for intra-LGA travel	132 persons × 30 days	Field mobility
R-6	Communication Airtime	Voice and data for team coordination	132 persons × 30 days	In-app messaging, voice calls
R-7	Identification Materials	Branded vests, ID cards, introductory letters from Ministry	132 sets	Field access and legitimacy
R-8	Sensitisation Materials	Flyers, posters, community announcement templates	33 LGA sets	Community engagement
R-9	Training Venue	Conference/meeting room with projector, internet	1 (central) + 5 (zonal)	Enumerator and supervisor training

16.4.3 Gate 1 Activities

Table 16.4: Gate 1 Activities

ACTIVITY	DURATION	DEPENDENCY
Enumerator recruitment and screening	1 week	R-1
Supervisor recruitment and screening	1 week	R-2
Classroom training, Enumerators (centralised)	3 days	R-1, R-9, Ch. 18 curriculum
Classroom training, Supervisors (centralised)	2 days	R-2, R-9, Ch. 18 curriculum
Device procurement and configuration	1 week (parallel)	R-3, R-4
Field pilot, 3 selected LGAs	3 days	All above
Pilot review and final calibration	2 days	Pilot data
Full 33-LGA deployment authorisation	1 day	All gate criteria met

16.4.4 Gate 1 Proceed/Hold/Stop Criteria

Table 16.5: Gate 1 Proceed/Hold/Stop Criteria

CRITERION	PROCEED	HOLD	STOP
Enumerator recruitment	≥90 of 99 recruited and trained	70–89 recruited	<70 recruited
Supervisor recruitment	≥30 of 33 recruited and trained	25–29 recruited	<25 recruited
Devices available	≥120 of 132 configured	100–119 configured	<100 configured

CRITERION	PROCEED	HOLD	STOP
Training completion	≥85% pass rate on assessment	70–84% pass rate	<70% pass rate
Pilot success	≥90% submission success rate in 3 pilot LGAs	75–89% success	<75% success
Logistics confirmed	Transport + airtime + materials for all 33 LGAs	25–32 LGAs covered	<25 LGAs covered

16.5 Gate 2: Data Quality Validation (UPCOMING)

16.5.1 Scope

Gate 2 occurs mid-way through the field enumeration period. It assesses data quality, identifies and resolves field issues, and authorises continuation or course correction.

16.5.2 Gate 2 Activities

Table 16.6: Gate 2 Activities

ACTIVITY	TIMING	METHOD
Data quality review, first 2 weeks of submissions	Week 3 of enumeration	Automated fraud detection + supervisory review
Submission volume assessment by LGA	Weekly	Platform analytics dashboard
Enumerator performance review	Weekly	Productivity metrics per enumerator
Respondent complaint/feedback review	Ongoing	Ministry coordination
Mid-enumeration calibration	If required	Threshold adjustments, additional training

16.5.3 Gate 2 Proceed/Hold/Stop Criteria

Table 16.7: Gate 2 Proceed/Hold/Stop Criteria

CRITERION	PROCEED	HOLD	STOP
Fraud flag rate	<10% of submissions	10–20%	>20% (systemic quality issue)
Submission volume	≥70% of target per LGA	50–69%	<50% (mobilisation failure)
Data completeness	≥95% of required fields populated	90–94%	<90% (instrument issue)
NIN duplication rate	<2%	2–5%	>5% (enumeration overlap)
Enumerator activity	≥90% active daily	75–89%	<75% (attrition/motivation)

16.6 Gate 3: Handover & Training (UPCOMING)

16.6.1 Scope

Gate 3 is the final phase, encompassing data analysis, registry population confirmation, capacity building delivery (Deliverable 3), and formal project handover to the Ministry.

16.6.2 Gate 3 Activities

Table 16.8: Gate 3 Activities

ACTIVITY	DURATION	DELIVERABLE
Field enumeration completion and final data sync	2 days	All offline submissions uploaded
Data cleaning and quality reconciliation	3 days	Clean registry dataset
Registry population verification	2 days	Total registrant count and LGA distribution confirmed
Ministry official training, Super Admin/Admin	2 days	Deliverable 3 (partial)
Ministry official training, Supervisory roles	1 day	Deliverable 3 (partial)
Ministry official training, Reporting and Analytics	1 day	Deliverable 3 (complete)
Platform documentation and user manuals delivery	1 day	Operational documentation
Formal project handover ceremony	1 day	Signed handover document
Completion Report submission	3 days	Final engagement deliverable

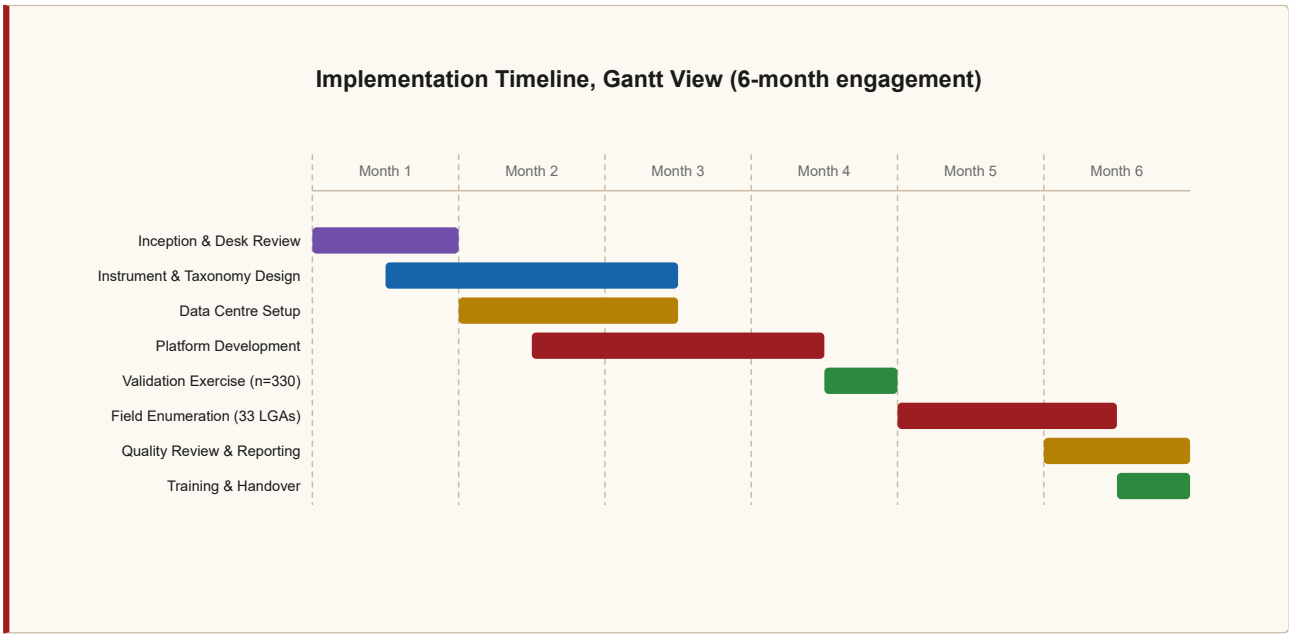
16.6.3 Gate 3 Acceptance Criteria

Table 16.9: Gate 3 Acceptance Criteria

#	CRITERION	THRESHOLD
1	Registry populated with verified respondent records	Minimum target TBD by Ministry
2	All 33 LGAs represented in registry data	100% LGA coverage
3	Ministry officials trained and assessed	≥85% pass rate on competency assessment
4	Platform operational documentation delivered	Complete user manual + admin guide
5	Data backup verified and disaster recovery tested	Successful restore from latest backup
6	Completion Report accepted by Ministry	Formal written acceptance

16.7 Implementation Timeline (Gantt View)

Figure 16.3: Implementation Timeline, Gantt View



16.8 Critical Path

The project's critical path, the sequence of activities that determines the minimum project duration, is:

1. **Gate 0 → Gate 1 transition** (CURRENT BOTTLENECK): Field deployment cannot commence until enumerator recruitment, device procurement, and logistics are mobilised. This transition is **resource-dependent**, not methodology-dependent.

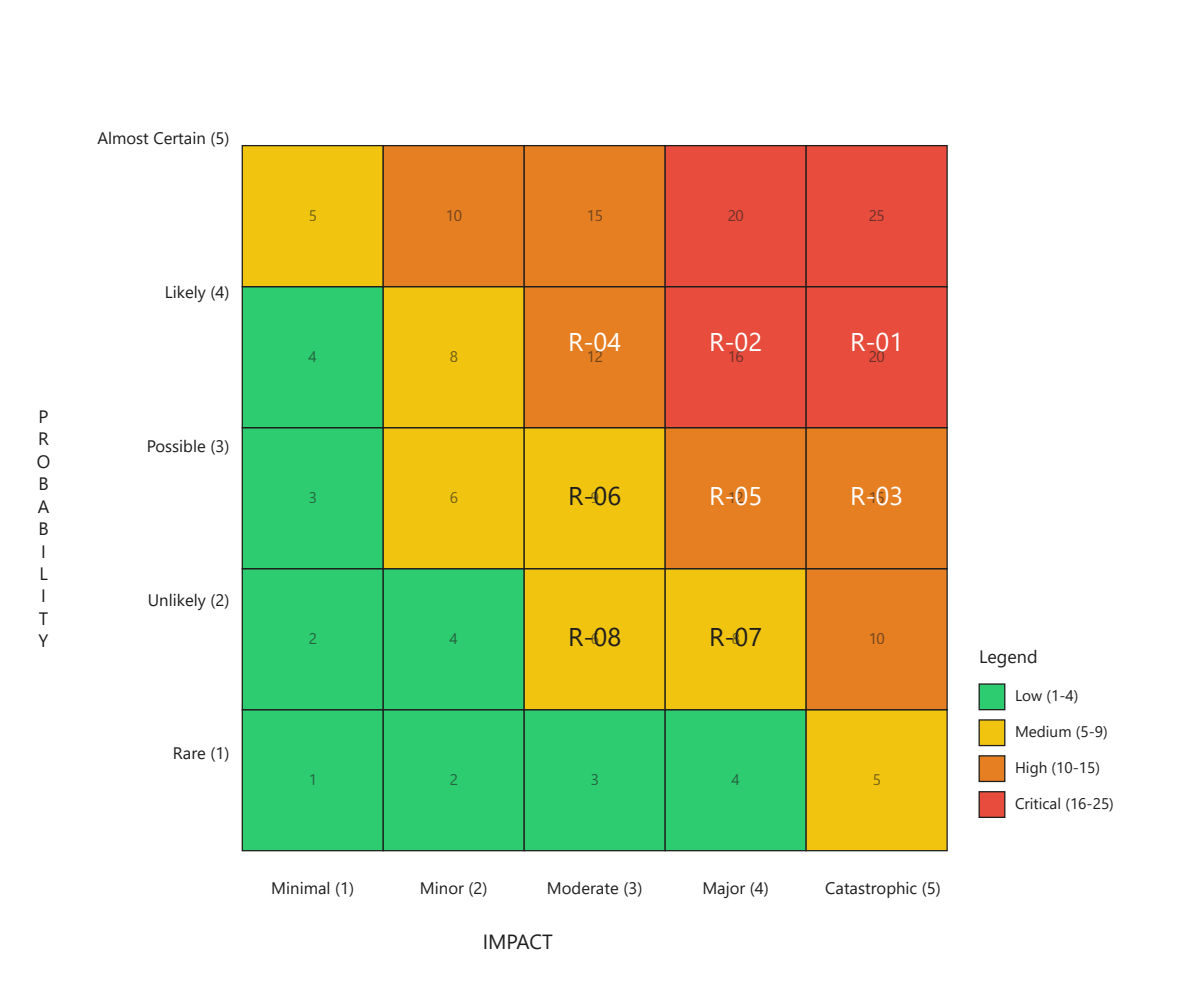
2. **Field enumeration duration**: The planned 30-day enumeration period across all 33 LGAs is the longest single activity remaining.

3. **Training delivery**: Ministry official training requires completion of enumeration (to train using real production data).

The Consultant notes that all methodology-dependent activities (Gates 0) have been completed on schedule. The remaining project timeline is contingent upon the timely mobilisation of field resources as detailed in Section 16.4.2.

17. Risk Register and Mitigation Framework

Figure 17.1: Risk Heat Map, Probability × Impact Matrix



17.1 Introduction

This chapter presents the formal Risk Register for the remaining phases of the Oyo State Skilled Labour Register project. The risk register was developed using a structured **Probability × Impact** assessment methodology, with each identified risk assigned a severity classification, designated owner, and specific mitigation strategy.

The risk register is a living document, it will be updated at each Stage Gate review (Chapter 16) as risks materialise, are mitigated, or as new risks emerge.

17.2 Risk Assessment Methodology

17.2.1 Probability Scale

Table 17.1: Probability Scale

RATING	PROBABILITY	DESCRIPTION
1	Very Low (< 10%)	Unlikely to occur under normal circumstances
2	Low (10–25%)	Could occur but only under specific conditions
3	Medium (25–50%)	Reasonable possibility of occurrence
4	High (50–75%)	More likely than not to occur
5	Very High (> 75%)	Expected to occur without intervention

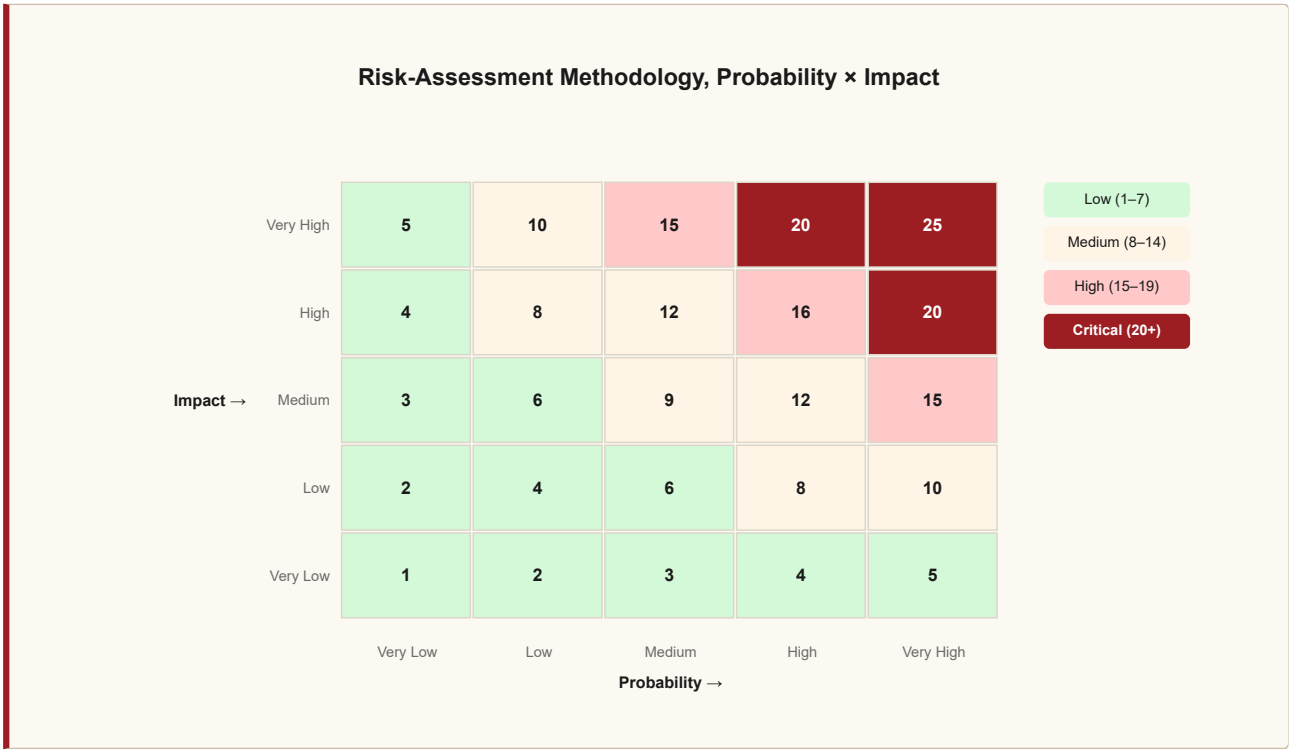
17.2.2 Impact Scale

Table 17.2: Impact Scale

RATING	IMPACT	DESCRIPTION
1	Negligible	Minimal effect on project; absorbed within normal operations
2	Minor	Localised effect; manageable within existing resources
3	Moderate	Noticeable impact on timeline or quality; requires management attention
4	Major	Significant impact on project delivery, data quality, or stakeholder confidence
5	Critical	Project-threatening; could result in failure to deliver or fundamental compromise of registry integrity

17.2.3 Risk Severity Matrix

Figure 17.2: Risk-Assessment Methodology, Probability × Impact



17.3 Risk Register

17.3.1 Category A: Resource Mobilisation Risks

Table 17.3: Category A: Resource Mobilisation Risks

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-01	**Delayed mobilisation of field personnel**, Recruitment of 132 field staff delayed beyond planned timeline	4	5	**CRITICAL (20)**	Ministry / Consultant	Early engagement with Ministry HR; pre-identification of candidate pools through LGA education offices and NYSC networks; streamlined recruitment criteria	Phased deployment, begin with 11 priority LGAs while recruiting remaining staff
R-02	**Delayed procurement of mobile devices**, 132 Android devices not available when field staff are ready	3	4	**HIGH (12)**	Ministry / Procurement	Early procurement initiation; bulk purchase from verified suppliers; specification flexibility (any Android 8.0+ device with required features)	Temporary use of enumerator personal devices (with data allowance compensation) for initial deployment

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-03	**Insufficient funding for field logistics**, Transport allowances, airtime, and materials budget not released	3	5	**HIGH (15)**	Ministry	Clear budget breakdown submitted with this report (Chapter 19); phased budget release aligned with Stage Gates	Prioritise high-population LGAs; reduce from 30-day to 20-day enumeration period

17.3.2 Category B: Field Operations Risks

Table 17.4: Category B: Field Operations Risks

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-04	**Community resistance to enumeration**, Respondents refuse to participate due to suspicion of government data collection	3	3	**MEDIUM (9)**	Supervisor / Consultant	Advance sensitisation programme (Chapter 19, Rec. 4); Ministry introductory letters; branded identification; community leader engagement	Deploy community liaison persons from within the LGA; reschedule to post-sensitisation period
R-05	**Enumerator attrition during field period**, Enumerators abandon the exercise before completion	3	3	**MEDIUM (9)**	Supervisor	Competitive remuneration; clear payment schedule; team morale management; performance recognition	Standby pool of 10% additional trained reserves; redistribute workload among remaining enumerators
R-06	**Poor network connectivity in rural LGAs**, Inability to synchronise data from remote areas	2	2	**LOW (4)**	Consultant	PWA offline capability (7-day autonomous operation); 2G/3G sufficient for sync; weekly sync schedule for remote areas	Supervisor collects devices weekly for centralised sync at nearest connectivity point
R-07	**Device failure or loss in field**, Smartphones damaged, stolen, or rendered inoperable	3	2	**MEDIUM (6)**	Supervisor	Protective cases provided; device insurance; enumerator accountability protocol; regular backup via sync	Spare device pool (10% reserve); temporary reassignment to co-located enumerator device
R-08	**Adverse weather during enumeration**, Rainy season disrupts field operations	3	2	**MEDIUM (6)**	Supervisor	Scheduling around known rainy patterns; flexible daily scheduling; waterproof device cases	Extension of enumeration period; indoor enumeration at markets and community centres

17.3.3 Category C: Data Quality Risks

Table 17.5: Category C: Data Quality Risks

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-09	**Enumerator data fabrication**, Field staff submit fictitious records to inflate productivity	3	4	**HIGH (12)**	Consultant / Supervisor	Automated fraud detection (GPS clustering, speed-run, straight-lining); per-enumerator monitoring; configurable thresholds	Immediate suspension of flagged enumerators; re-interview requirement for all flagged submissions; termination for confirmed fabrication
R-10	**NIN transcription errors**, Respondents provide incorrect NIN leading to invalid records	3	3	**MEDIUM (9)**	Enumerator	Modulus 11 real-time validation catches mathematical errors; enumerator training emphasis on NIN accuracy; re-entry prompt on failure	Post-collection NIN verification batch (if NIMC API access becomes available)
R-11	**Respondent self-selection bias**, Registry over-represents certain demographics or skills categories	3	3	**MEDIUM (9)**	Consultant / Supervisor	Stratified enumeration targets per LGA; ward-level coverage monitoring; active pursuit of underrepresented demographics	Post-hoc weighting adjustments; supplementary targeted enumeration in underrepresented areas
R-12	**High fraud flag rate overwhelming supervisory capacity**, More than 20% of submissions flagged, exceeding review capacity	2	3	**MEDIUM (6)**	Consultant	Threshold calibration from validation exercise; targeted retraining for high-flag enumerators	Dynamic threshold adjustment; additional verification assessor capacity; batch review triage

17.3.4 Category D: Technical & Infrastructure Risks

Table 17.6: Category D: Technical & Infrastructure Risks

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-13	**Server downtime during peak enumeration**, Cloud infrastructure experiences outage during	2	3	**MEDIUM (6)**	Consultant	99.9% uptime SLA; automated monitoring and alerting; tested disaster recovery (< 1hr RTO)	PWA offline capability buffers all submissions locally; sync upon restoration; zero data loss

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
	high submission volume						
R-14	**Data Centre connectivity disruption**, Airtel 5G/4G service disruption at Ministry premises	2	2	**LOW (4)**	Ministry / Consultant	5G/4G automatic fallback; ODU battery backup (5–6 hours); data entry can continue via cloud from any internet source	Temporary mobile hotspot; field laptop (Workstation 3) can operate from alternative location
R-15	**Database capacity exhaustion**, Registry growth exceeds allocated storage	1	3	**LOW (3)**	Consultant	Current VPS capacity supports 100,000+ records with substantial headroom; monitoring alerts at 80% capacity	Vertical scaling (upgrade VPS); database archiving of completed records
R-16	**Security breach or data leakage**, Unauthorised access to PII data	1	5	**MEDIUM (5)**	Consultant	6-layer defence-in-depth (Chapter 11); independently assessed at A- security posture; immutable audit logs; bcrypt+JWT+TLS; RBAC enforcement	Incident response: isolate affected component; forensic audit via immutable logs; breach notification per NDPA 2023

17.3.5 Category E: Stakeholder & Governance Risks

Table 17.7: Category E: Stakeholder & Governance Risks

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
R-17	**Change in Ministry leadership or priorities**, Administrative transition disrupts project continuity	2	4	**HIGH (8)**	Ministry / Consultant	Comprehensive documentation (this report); platform operates independently of personnel changes; training of multiple Ministry staff (not single-point dependency)	Briefing package for incoming leadership; platform continues operating; data preserved
R-18	**Scope expansion requests during enumeration**, Stakeholders request additional data fields or survey modifications mid-enumeration	3	3	**MEDIUM (9)**	Consultant	Instrument finalised and versioned (v3.0); change control process requiring Stage Gate review before implementation	Minor additions captured in “Other skills” free text field; major changes deferred to post-enumeration update cycle
R-19	**Inter-LGA coordination failure**, Poor	2	3	**MEDIUM (6)**	Supervisor / Ministry	Ministry introductory letters to all 33 LGA	Consultant direct engagement with resistant LGA

ID	RISK	PROB	IMPACT	SEVERITY	OWNER	MITIGATION	CONTINGENCY
	coordination between LGA authorities and field teams					chairmen; advance notification schedule; designated LGA liaison per supervisor	leadership; Ministry intervention
R-20	**Insufficient respondent participation**, Target submission volumes not achieved within enumeration period	3	3	**MEDIUM (9)**	Supervisor / Consultant	Community sensitisation; multi-channel collection (field + public self-registration); marketplace incentive (voluntary participation in skills marketplace)	Extend enumeration period; intensify public self-registration promotion; targeted outreach in low-participation areas
R-21	**Political and stakeholder-environment shifts**, Broader political dynamics affecting State operations during the engagement period	2	3	**MEDIUM (6)**	Ministry / Consultant	The engagement is sensitive to the broader political and stakeholder environment in which the State operates. The Consultant maintains an internal monitoring posture and will report material developments to the Ministry in the ordinary course	Programme calendar adapted to State election cycles or governance transitions where applicable; documentation packages enable continuity across personnel changes

17.4 Risk Summary Dashboard

Figure 17.3: Risk Register Summary, Top Risks by Severity

Risk Register Summary, Top Risks by Severity

R-04	Field staff non-mobilisation	High (16)
R-09	Device procurement delay	High (15)
R-12	Power/connectivity loss in field	Medium (12)
R-16	Security breach / data leakage	Medium (10)
R-08	Community resistance	Medium (9)
R-14	NIN verification API downtime	Medium (8)
R-05	Budget delay or shortfall	Low (mitigated) (6)
R-19	Political-environment shifts	Low (monitored) (4)

17.5 Risk Ownership Matrix

Table 17.8: Risk Ownership Matrix

OWNER	RISKS OWNED	ROLE IN MITIGATION
Ministry	R-01, R-02, R-03, R-14, R-17, R-19	Resource mobilisation, funding release, inter-agency coordination
Consultant	R-09, R-11, R-13, R-15, R-16, R-18	Technical controls, methodology calibration, platform operation
Supervisor	R-04, R-05, R-07, R-08, R-12, R-19, R-20	Field management, team coordination, quality oversight
Joint	R-06, R-10	Shared responsibility for field data quality and connectivity

17.6 Risk Review Schedule

Table 17.9: Risk Review Schedule

REVIEW POINT	GATE	SCOPE
Pre-deployment review	Gate 1 entry	All 20 risks, focus on Category A (resource mobilisation)
Mid-enumeration review	Gate 2	Categories B, C, D, focus on field operations and data quality
Pre-handover review	Gate 3 entry	All risks, focus on residual risks for ongoing operations

REVIEW POINT	GATE	SCOPE
Post-handover	Post-Gate 3	Transfer of risk ownership to Ministry for ongoing registry operation

17.7 Key Insight

The risk register reveals a structural pattern that is significant for project governance: **the highest-severity risks (R-01, R-02, R-03) are all in Category A (Resource Mobilisation) and are owned primarily by the Ministry**. The Consultant-owned risks (Categories C and D) are predominantly MEDIUM or LOW severity, reflecting the maturity of the technical and methodological infrastructure established during the baseline study phase.

This pattern confirms that the project's **critical path runs through resource mobilisation, not methodology or technology**. The technical, methodological, and data quality foundations are in place; the remaining risks are predominantly operational and logistical in nature.

18. Monitoring and Evaluation Framework: Key Performance Indicators

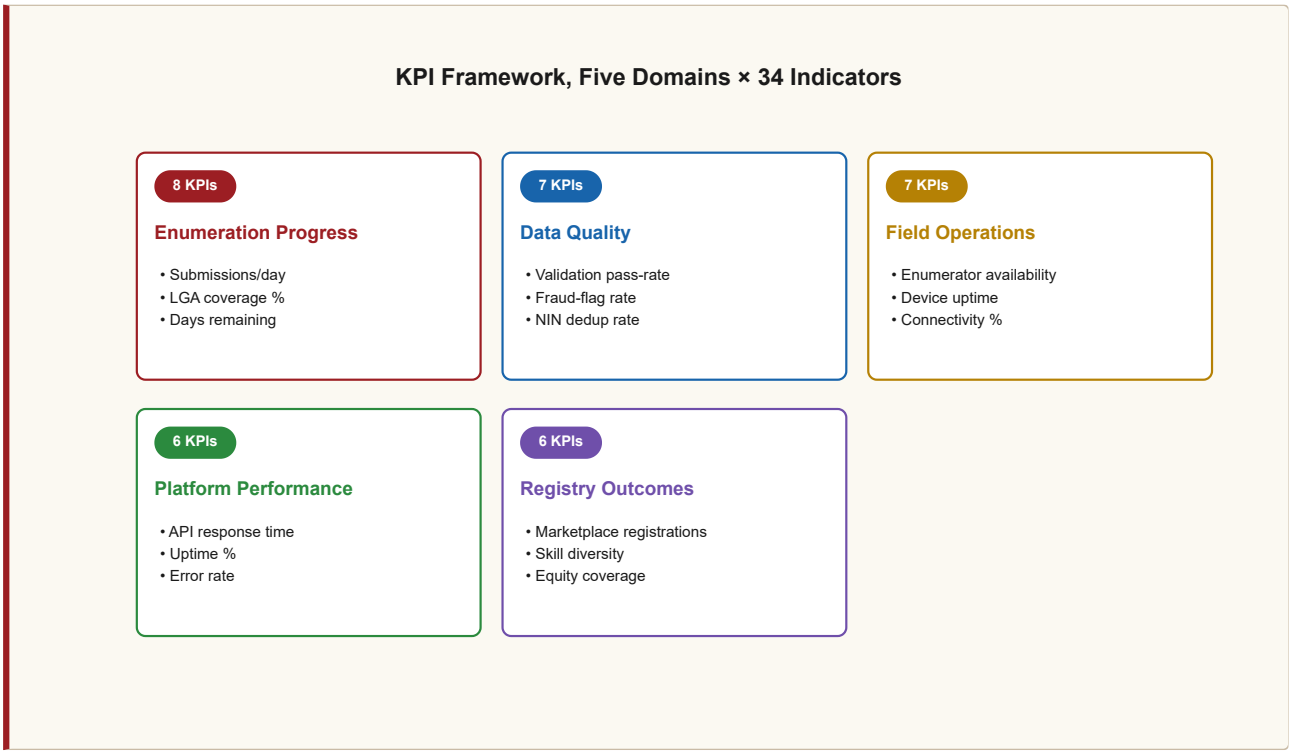
18.1 Introduction

This chapter establishes the **Monitoring and Evaluation (M&E) Framework** for the Oyo State Skilled Labour Register, comprising a set of **Key Performance Indicators (KPIs)** that provide quantifiable measures of project progress, data quality, and registry outcomes. The KPI framework enables objective assessment at each Stage Gate review (Chapter 16) and supports evidence-based decision-making throughout the remaining project phases.

The framework is structured across five KPI domains, with each indicator assigned a **data source**, **measurement frequency**, **target value**, and **alert threshold** that triggers management attention.

18.2 KPI Framework Structure

Figure 18.1: KPI Framework, Five Domains × 34 Indicators



18.3 Domain 1: Enumeration Progress KPIs

These indicators track the volume and geographic distribution of data collection.

Table 18.1: Domain 1: Enumeration Progress KPIs

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
EP-1	**Daily submission rate (state)**	Total submissions received per day across all channels	≥ 800/day	< 500/day	Daily	Platform analytics
EP-2	**Daily submission rate (per enumerator)**	Average submissions per active enumerator per day	≥ 8/day	< 5/day	Daily	Per-user analytics
EP-3	**Cumulative registry count**	Running total of verified registry entries	23,760 by Day 30	< 70% of pro-rata target	Daily	Registry database
EP-4	**LGA coverage (% of 33 LGAs active)**	Percentage of LGAs with ≥ 1 submission in current week	100%	< 90% (3+ LGAs inactive)	Weekly	Platform analytics
EP-5	**LGA equity ratio**	Ratio of highest to lowest LGA submission count	< 5:1	> 8:1	Weekly	Platform analytics
EP-6	**Self-registration volume**	Public web self-registrations (non-enumerator)	Supplementary (no target)	,	Weekly	Channel analytics
EP-7	**Offline sync backlog**	Number of submissions pending sync from offline devices	< 50 statewide	> 200 statewide	Daily	Sync queue

18.4 Domain 2: Data Quality KPIs

These indicators assess the integrity and reliability of collected data.

Table 18.2: Domain 2: Data Quality KPIs

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
DQ-1	**Fraud flag rate**	Percentage of submissions flagged by automated fraud detection	< 10%	> 15%	Daily	Fraud detection engine
DQ-2	**NIN duplication rate**	Percentage of submissions rejected for duplicate NIN	< 2%	> 5%	Daily	Database constraint
DQ-3	**Field completeness rate**	Percentage of required fields populated across all submissions	> 98%	< 95%	Daily	Validation layer
DQ-4	**Validation error rate**	Percentage of submission attempts failing validation (corrected and resubmitted)	< 5%	> 10%	Daily	Validation logs

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
DQ-5	**Supervisory review completion**	Percentage of flagged submissions reviewed within 48 hours	> 90%	< 75%	Daily	Fraud review queue
DQ-6	**Fraud flag resolution rate**	Percentage of reviewed flags resulting in approval (vs. rejection)	70–90% approval	< 60% or > 95%	Weekly	Fraud review outcomes
DQ-7	**Gender distribution**	Male/female ratio across all submissions	45–55% male	Outside 40–60% range	Weekly	Demographic analytics
DQ-8	**NBS benchmark alignment**	Key indicators (employment rate, self-employment rate, education distribution) compared to NBS NLFS baselines	Within ± 10% of NBS	Outside ± 15% of NBS	Weekly	Statistical analysis

18.5 Domain 3: Field Operations KPIs

These indicators monitor the operational effectiveness of field teams.

Table 18.3: Domain 3: Field Operations KPIs

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
FO-1	**Enumerator activity rate**	Percentage of trained enumerators submitting ≥ 1 record per day	> 90%	< 80%	Daily	Per-user analytics
FO-2	**Enumerator attrition rate**	Cumulative percentage of enumerators who have ceased activity	< 5%	> 10%	Weekly	HR records + activity
FO-3	**Average completion time**	Mean survey completion time across all submissions	7–12 min	< 4 min or > 20 min	Daily	Submission timestamps
FO-4	**Supervisor review rate**	Number of flagged submissions reviewed per supervisor per day	≥ 10/day	< 5/day	Daily	Fraud dashboard
FO-5	**Device utilisation rate**	Percentage of deployed devices with active submissions	> 90%	< 80%	Weekly	Device analytics
FO-6	**Team messaging engagement**	Percentage of supervisors using in-app messaging for team coordination	> 80%	< 60%	Weekly	Messaging analytics

18.6 Domain 4: Platform Performance KPIs

These indicators ensure the technology infrastructure supports operational needs.

Table 18.4: Domain 4: Platform Performance KPIs

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
PP-1	**Application uptime**	Percentage of time the platform is accessible and responsive	> 99.5%	< 99%	Continuous	Health monitoring
PP-2	**API response time (p95)**	95th percentile response time for API requests	< 500ms	> 1000ms	Continuous	prom-client metrics
PP-3	**Database query latency (p95)**	95th percentile database query execution time	< 100ms	> 250ms	Continuous	Database metrics
PP-4	**Server CPU utilisation**	Average CPU usage on production server	< 60%	> 80%	Continuous	System monitoring
PP-5	**Server memory utilisation**	Average RAM usage on production server	< 70%	> 85%	Continuous	System monitoring
PP-6	**Backup success rate**	Percentage of daily automated backups completed successfully	100%	Any failure	Daily	Backup job logs
PP-7	**SSL certificate validity**	Days until SSL certificate expiry	> 30 days	< 14 days	Daily	Certificate monitor

18.7 Domain 5: Registry Outcomes KPIs

These indicators assess whether the registry is achieving its policy objectives.

Table 18.5: Domain 5: Registry Outcomes KPIs

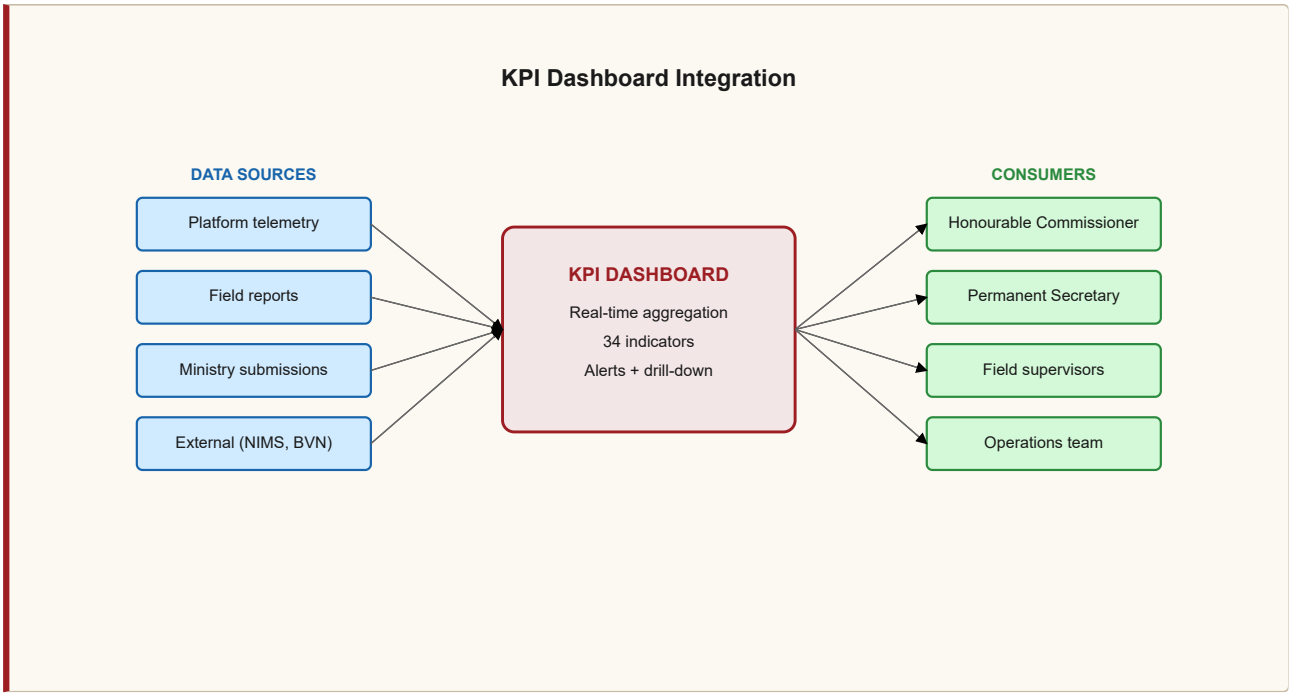
#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
RO-1	**Total verified registrants**	Number of records passing all 4 QA layers	Project-specific target (TBD by Ministry)	< 70% of target at Gate 2	Weekly	Registry database
RO-2	**Skills inventory breadth**	Number of distinct skills recorded across all registrants	≥ 100 of 150 taxonomy skills represented	< 80 skills represented	Gate reviews	Skills analytics
RO-3	**Marketplace opt-in rate**	Percentage of registrants opting into Skills Marketplace	> 40%	< 25%	Weekly	Consent analytics
RO-4	**Sector representation**	Number of 20 economic sectors with ≥ 10 registrants	All 20 sectors	< 15 sectors	Gate reviews	Sector analytics

#	KPI	DEFINITION	TARGET	ALERT THRESHOLD	FREQUENCY	DATA SOURCE
RO-5	**Youth registration (15–35)**	Percentage of registrants aged 15–35	40–60% (aligned with Oyo demographics)	< 30% or > 70%	Weekly	Age analytics
RO-6	**Business registration capture**	Percentage of employed registrants who report business ownership	> 30%	< 20%	Weekly	Section 5 analytics

18.8 KPI Dashboard Integration

All 34 KPIs are designed to be measured using **data already captured by the OSLSR platform**, no additional data collection infrastructure is required. The platform's built-in analytics dashboard (Chapter 10) provides real-time or daily computation of these indicators.

Figure 18.2: KPI Dashboard Integration



18.9 Reporting Schedule

Table 18.6: Reporting Schedule

REPORT	CONTENT	AUDIENCE	FREQUENCY	FORMAT
Daily Operations Brief	EP-1, EP-2, DQ-1, FO-1, FO-3, PP-1	Consultant, Supervisors	Daily	Dashboard view

REPORT	CONTENT	AUDIENCE	FREQUENCY	FORMAT
Weekly Progress Report	All Domain 1–3 KPIs, aggregated	Ministry, Consultant	Weekly	PDF export
Gate Review Report	All 34 KPIs with trend analysis	Ministry, Consultant	Per Gate	Formal report
Alert Notification	Any KPI breaching alert threshold	Designated recipients	Real-time	Email notification

18.10 KPI-to-Gate Mapping

The following table maps which KPIs serve as **formal gate criteria** at each Stage Gate review:

Table 18.7: KPI-to-Gate Mapping

KPI	GATE 1	GATE 2	GATE 3
EP-1: Daily submission rate		✓	
EP-3: Cumulative registry count		✓	✓
EP-4: LGA coverage		✓	✓
DQ-1: Fraud flag rate		✓	✓
DQ-2: NIN duplication rate		✓	
DQ-3: Field completeness		✓	✓
DQ-8: NBS alignment			✓
FO-1: Enumerator activity	✓*	✓	
FO-2: Enumerator attrition		✓	
PP-1: Application uptime	✓	✓	✓
PP-6: Backup success			✓
RO-1: Total verified registrants			✓
RO-2: Skills breadth			✓
RO-4: Sector representation			✓

** FO-1 at Gate 1 refers to pilot activity rate (3 pilot LGAs)*

18.11 Assumption Classification of KPI Targets

Following the Assumption Classification Framework (Chapter 6), the KPI targets carry the following classifications:

Table 18.8: Assumption Classification of KPI Targets

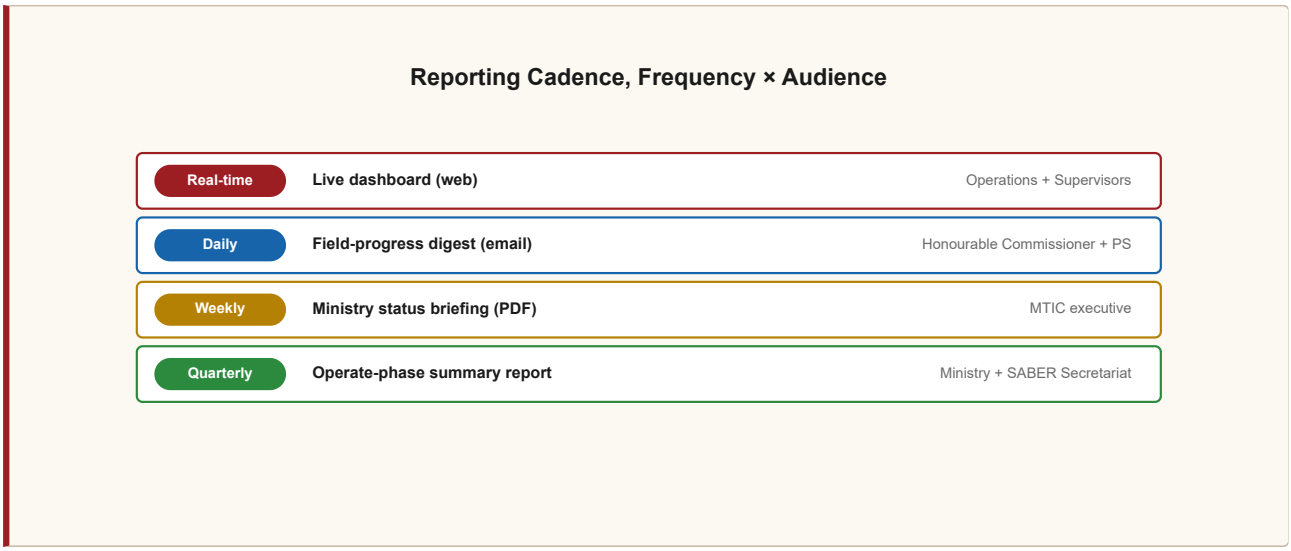
CLASSIFICATION	KPIS	RATIONALE
[VF] Verified Fact	PP-1, PP-6, PP-7, DQ-3	Based on observed platform performance during validation exercise
[FD] Field-Dependent	EP-1, EP-2, EP-3, FO-1, FO-2, FO-3, RO-1, RO-3	Targets based on calculations from planned field operations; actual performance will be confirmed during Gate 1 pilot
[WA] Working Assumption	DQ-1, DQ-7, DQ-8, RO-2, RO-4, RO-5, RO-6	Based on NBS benchmarks and comparable project data; subject to revision upon field data accumulation

The **[FD] Field-Dependent** KPIs will be recalibrated based on actual performance data from the 3-LGA pilot exercise (Gate 1), ensuring that Gate 2 and Gate 3 targets are grounded in observed rather than projected performance.

18.12 Continuous Improvement Cycle

The M&E framework supports a continuous improvement cycle aligned with the Stage Gate methodology:

Figure 18.3: Reporting Cadence, Frequency × Audience



19. Recommendations and Next Steps

19.1 Introduction

This chapter presents the Consultant's formal recommendations for the successful completion of the Oyo State Skilled Labour Register, based on the findings of the baseline study, the validation exercise results, and the operational readiness assessment documented in this report.

19.2 Summary of Achievements

Before outlining recommendations, the Consultant notes the following milestones achieved during the Build Phase of the engagement:

Table 19.1: Summary of Achievements

#	ACHIEVEMENT	REFERENCE
1	**Deliverable 1, Data Centre: COMPLETE**	Chapter 9
2	Comprehensive desk review of labour market data for Oyo State	Chapter 4
3	Comparative analysis of state-level registry initiatives	Chapter 5
4	ILO ICLS-19 aligned survey methodology designed	Chapter 6
5	Survey instrument developed through 3 iterative review cycles (v3.0)	Chapter 7
6	150-skill occupational taxonomy mapped to ISCO-08	Chapter 8
7	OSLSR digital platform developed, tested (3,564 tests), and deployed	Chapter 10
8	OWASP Top 10 security assessment, all categories SECURE	Chapter 11
9	NDPA 2023 Data Protection Impact Assessment completed	Chapter 11
10	Validation exercise across all 33 LGAs (n=330)	Chapters 9–13
11	Data quality assurance framework operational	Chapter 14
12	**Deliverable 3, Training curriculum: DESIGN COMPLETE**	Chapter 15

The digital infrastructure, validated survey instrument, 150-skill occupational taxonomy, data quality assurance framework, and operational Data Centre are **deployment-ready**. The platform has undergone comprehensive automated testing and security assessment, and is operational and accessible.

19.3 Recommendations for Deliverable 2: Creation of the State Labour Register

Recommendation 1: Immediate Mobilisation of Field Personnel

For the successful execution of the statewide field enumeration, the Consultant recommends the **immediate mobilisation** of the following field personnel:

Table 19.2: Recommendation 1: Immediate Mobilisation of Field Personnel

ROLE	NUMBER PER LGA	TOTAL (33 LGAS)	PROFILE
Field Enumerator	3	**99**	Literate adults (minimum SSS/WAEC), proficient with smartphones, resident within assigned LGA, ability to communicate in Yoruba and English
Field Supervisor	1	**33**	Experienced coordinators with management capability, familiarity with government field programmes, minimum OND/NCE education
Total Field Personnel	4	**132**	

Justification: The validated survey instrument requires an estimated 10 minutes per respondent. With 3 enumerators per LGA operating 6 hours/day (accounting for travel and engagement time) at approximately 8 submissions per enumerator per day, a 30-day enumeration period would yield approximately **23,760 validated submissions** (99 enumerators × 8 submissions/day × 30 days), a substantive foundation for the State Labour Register, supplemented by ongoing public self-registration via the web platform.

Recommendation 2: Procurement of Data Collection Devices

Table 19.3: Recommendation 2: Procurement of Data Collection Devices

ITEM	MINIMUM SPECIFICATION	QUANTITY	PURPOSE
Android Smartphone	Android 8.0+, 2GB RAM, 32GB storage, GPS, camera, ≥5" screen	**132**	Field data collection via OSLSR PWA
SIM Card with Data	Monthly data allocation ≥2GB per device	**132**	Submission synchronisation
Protective Case	Rugged/shockproof	**132**	Device protection in field conditions
Portable Power Bank	≥10,000 mAh	**132**	Extended field operation

Justification: The OSLSR platform is designed as a Progressive Web Application (PWA) operating on Android devices. While the platform supports 7-day offline operation, periodic data synchronisation requires mobile data connectivity. The offline architecture means that high-speed connectivity is not required, 2G/3G is sufficient for submission upload.

Recommendation 3: Field Logistics and Support

Table 19.4: Recommendation 3: Field Logistics and Support

ITEM	SPECIFICATION	DURATION	QUANTITY
Transportation Allowance	Intra-LGA travel for enumerators and supervisors	30 days	132 persons
Communication Airtime	Voice calling for team coordination	30 days	132 persons
Identification Materials	Branded vests, laminated ID cards, Ministry introductory letters	,	132 sets
Sensitisation Materials	Community flyers, radio announcement scripts, poster templates	,	33 LGA packages
Field Stationery	Notebooks, pens, consent forms (backup)	,	132 sets

Justification: Community trust and access are critical success factors for field enumeration, particularly in rural LGAs. Branded identification materials and introductory letters from the Ministry establish the legitimacy of the exercise and facilitate cooperation from traditional rulers, community leaders, and ward heads.

Recommendation 4: Sensitisation and Community Engagement

The Consultant recommends a **two-week advance sensitisation programme** preceding field deployment:

1. **Ministry-level announcement**: Official circular from the Ministry to all 33 LGA chairmen informing them of the enumeration exercise
2. **Traditional ruler engagement**: Courtesy visits by Supervisors to traditional rulers and community leaders in each LGA
3. **Community-level awareness**: Distribution of flyers at markets, motor parks, and community centres; radio announcements on local stations (e.g., Amuludun FM, Splash FM, Oluyole FM)
4. **Religious institution engagement**: Announcements at Friday mosques and Sunday churches to reach the broadest population

Recommendation 5: Training Delivery

Upon mobilisation of field personnel, the Consultant will deliver the capacity building programme as designed in Chapter 15:

Table 19.5: Recommendation 5: Training Delivery

TRAINING MODULE	DURATION	TARGET GROUP
Platform Operations, Enumerator Module	2 days	99 Enumerators (in 3 batches of 33)
Field Supervision & Quality Assurance	1 day	33 Supervisors
System Administration & Reporting	2 days	Ministry-designated Admin staff
Data Analysis & Dashboard Utilisation	1 day	Government Officials / Policy users

19.4 Recommendations for Long-Term Registry Sustainability

Recommendation 6: Establish Registry as Ongoing Government Function

The Labour Register should not be treated as a one-time enumeration exercise. The Consultant recommends establishing the OSLSR as a **continuously updated government resource** through:

1. **Permanent online registration portal**: The self-registration function (currently operational) remains accessible to the public beyond the field enumeration period, allowing ongoing voluntary registration
2. **Annual data refresh**: A lightweight annual verification exercise (phone/SMS confirmation) to maintain data currency and identify out-of-date records
3. **Dedicated Ministry officer**: Assignment of a trained Ministry staff member as ongoing registry administrator

Recommendation 7: Leverage Skills Marketplace for Economic Development

The OSLSR Skills Marketplace (currently operational) provides a unique asset for Oyo State's economic development strategy. The Consultant recommends:

1. **Public launch event**: Formal public launch of the Skills Marketplace once sufficient register data is populated
2. **Employer awareness campaign**: Outreach to businesses, contractors, and organisations in Oyo State to register as marketplace users
3. **Integration with Ministry programmes**: Link the marketplace to existing skills development, MSME support, and investment attraction initiatives

Recommendation 8: Position the Register as a Business Environment Reform Asset

Oyo State's rising performance in the federal subnational business enabling environment assessment, from 27th to 3rd nationally (62.7% in 2025), demonstrates the state's commitment to creating a competitive business environment. The Skilled Labour Register directly supports the **skilled labour readiness** indicator, one of sixteen dimensions assessed in the reform framework. The Consultant recommends:

1. **Formal integration with state business reform reporting**: Register data (total registrants, skills distribution, LGA coverage) should be included in Oyo State's submissions to the federal business environment assessment programme
2. **Quarterly metrics reporting**: Generate automated quarterly reports on register growth and marketplace utilisation for submission to the relevant state reform coordination office
3. **Inter-state visibility**: Position the register as a model for other states participating in the business enabling environment reform programme, Oyo State would be the first to offer a comprehensive, digitised, NIN-linked skilled workforce database with a public search capability

---

19.5 Timeline for Completion

Subject to the timely mobilisation of resources outlined above, the Consultant projects the following timeline for project completion:

Table 19.6: Timeline for Completion

PHASE	ACTIVITY	DURATION	PROJECTED PERIOD
Gate 1	Recruitment, training, pilot	2–3 weeks	Upon resource mobilisation
Gate 2	Full-scale field enumeration	4 weeks	Following Gate 1
Gate 3	Data quality review, training delivery, handover	2 weeks	Following Gate 2
	Total remaining duration	**8–9 weeks**	

The Consultant affirms that the remaining work is achievable within the engagement timeline, **provided that the mobilisation of field personnel and logistics resources commences promptly**.

19.6 Conclusion

The Build Phase of the engagement has established the complete methodological, technological, and operational foundation for the Oyo State Skilled Labour Register. The achievements documented in this report, a production-ready digital platform, a validated ILO-aligned survey instrument, a 150-skill occupational taxonomy, an operational Data Centre, and a rigorous data quality framework, represent a substantial body of work that positions the project for successful completion.

The Consultant stands ready to execute the field enumeration phase immediately upon mobilisation of the requisite field personnel and logistics resources, and remains committed to the timely and successful delivery of all engagement deliverables.

Mrs Fateemah Roy-Lagbaja Managing Director Chemiroy Nigeria Limited

May 2026